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Master in the Political Economy of Capitalism

Beyond Imbalances: A Competition-Theoretic Account of the U.S.-China Rivalry

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Abstract

This thesis investigates whether the recent shift in U.S. foreign economic policy – across-the-board tariffs, profit-sharing investment agreements and a turn towards industrial policy – reflects underlying pressures on the profitability of U.S.-American corporations. To make this question tractable, it traces out the relationship between trade and profitability. Adopting an absolute cost framework – where imbalances are understood as a normal outcome of international competition – it shows that the relationship between the trade deficit (or surplus) and profitability is ambiguous, depending on the composition of the traded goods and the nature of international competition. Drawing on Crotty (2017b)’s theory of shifting competitive regimes, it hypothesises that rising Chinese technological capabilities and price competition in core industries – the ‘*China Shock 2.0*’ – are triggering a shift from *corespective* to *coercive competition*, threatening the profit rates and markups of U.S.-American firms. This hypothesis is tested on a panel of large public U.S. corporations over 2000–2024, using a novel firm-level indicator of exposure to Chinese import penetration. They confirm that the association between exposure and profitability has reversed sign over the investigated period and that the profit-rate dynamics have shifted towards faster mean-reversion; suggesting that inter-capitalist competition has become more intense, especially in sectors where Chinese firms are active. The sensitivity of the regime of investment to profitability has significantly declined, suggesting a transformation of the competitive regime consistent with the hypothesised situation. The doctrinal shift of U.S. policy is then read as a response negotiated on behalf of U.S. transnational capital to its deteriorating conditions of profitability.

Keywords: U.S.-China rivalry, trade imbalances, transnational corporations, profit rate, competitive regimes, China Shock.

JEL Codes: B51, E12, F14, F23, L52

AI usage transparency: artificial intelligence tools were used in the drafting of this work, namely the coding assistant *opencode* together with the model *deepseek-v4-flash* as an assistant for writing the code and for proofreading of grammar and syntax.

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Chapter 1

Introduction

Trump's second term in office has been characterised by a significant increase in tariffs, an increasingly unilateral approach to foreign policy and a tendency towards abandoning the *laissez-faire* doctrine associated with neoliberalism in favour of a more interventionist or (neo-)mercantilist (Wolf 2026b) approach to economic policy¹. On the world stage, the Trump 2 administration appears to be divorcing itself from the international agenda of liberalisation of trade and capital flows, which the United States had been championing internationally since the 1980s. Instead it is adopting a much more aggressive foreign policy, one of *permanent predation* (Bürbaumer 2025) marked by intense interventionism on selected fronts (especially in Latin America, as well as in the Middle East), by the intensification of the trade war against China and by an increasingly unilateral and aggressive approach to international relations. The so-called 'Liberation Day' tariffs – the most substantial rise in trade barriers since World War II – and the following political battle which the administration has waged to keep trade barriers high (Williams 2026) mark a complete break with previous U.S. trade policy. They are presented as an attempt to *rebalance* the massive trade deficit of the United States and protect U.S.-American manufacturing jobs. However, they raise their own set of contradictions, both regarding the preservation of the United States' role as the provider of the hegemonic currency of the global market (Ali 2025) and in terms of their stated goal of protecting jobs in the

¹It should be mentioned that during the so-called *neoliberal* period, the United States never was fully committed to the *laissez-faire* doctrine and maintained a high level of state intervention, especially through the command economy of the defence sector. See Block, Keller, and Negoita (2024) on that matter.

U.S. domestic manufacturing industry (Grothe and Nikiforos 2025). On the domestic front, while initially the new administration turned its back on the more interventionist policies of the Biden era, this was quickly overturned. In the context of the reheating of the U.S.-China trade war, as the U.S. government was attempting to restrict the access of Chinese companies to advanced chips, it took a stake of 10 per cent in Intel. A few months later, as China retaliated by restricting U.S. access to rare earths, Treasury Secretary Scott Bessent announced an extremely comprehensive industrial policy package with price floors, buffer stocks of critical minerals and a curtailment of stock buybacks for U.S. companies (Rappeport and Swanson 2025). In parallel, the new administration tried (initially with limited success) to overturn the founding neoliberal principle of central bank independence, another step away from what had been understood to be a core pillar of the neoliberal economic policy package (Ali 2026).

This drastic doctrinal shift led by the new administration has, of course, given rise to a variety of interpretations. While some commentators argue that the administration's decisions are fundamentally incoherent and cannot be interpreted through any other lens than a psychological one, some commentators see an intelligible, renewed policy package in the moves made by the new administration². As was remarked by Burbaumer (2024), while much ink has been spilt regarding Trump's personal responsibility in this turnaround, a careful observer would have picked up on a deeper trend. Indeed, the U.S.-China rivalry and the profound reconfiguration of U.S. foreign economic policy find roots before (and for that matter between) Trump's two mandates. The renewed commitment to a more interventionist policy, directly favouring domestic industries, of the Trump administration of course, presents itself as a continuation of the Biden era policies of the IRA and CHIPS Act. What is perhaps more surprising is that the central importance given to the rebalancing of trade, while often associated with Stephen Miran's 2024 policy proposal, also found an advocate as early as 2010 in the person of Janet Yellen. At the time Yellen (2010), who would be Biden's treasury secretary and was then Fed vice chair, was already arguing that *rebalancing* was a key imperative for the United States³. This view is symptomatic of the understanding that the U.S. is '*burdened by having to absorb*

²See Blanchard and Mateos y Lago (2025) for an example of both points of view expressed by broadly speaking neoclassical economists.

³although she was not committed to rebalancing by way of tariffs in the same way that the second Trump administration would be.

the world's exports' (Ali 2025), which has become prevalent far beyond the Trump camp in Washington.

At the centre of this reconfiguration stands China. It is against Chinese competition that the trade war is being waged, and it is the rise of China which, more than any other factor, has unsettled the consensus in favour of the liberalised world market in the United States (Burbaumer 2024). The first '*China Shock*' (Autor, Dorn, and Hanson 2016) – the irruption of Chinese manufactures on the world market in the decades following China's opening – displaced U.S.-American labour and capital in low-wage, labour-intensive sectors, but left the high-value-added segments and the firms dominating them, largely untouched. The current debate concerns a shock of a different kind. As argued by Autor and Hanson (2025), who coined the phrase, a '*China Shock 2.0*' is underway: China is exiting the low-value sectors it had occupied since its opening and establishing export dominance in high-value-added industries – industrial machinery, electronics, transportation equipment, renewables – which had until recently remained the preserve of U.S.-American, European and advanced Asian corporations. One central feature distinguishes it from its predecessor: the emergence of Chinese corporations, headquartered in China and capable of competing head to head with the strongest U.S.-American, European and advanced Asian corporations. The material basis of this catching up – technological upgrading, economies of scale of unprecedented magnitude and immense productivity gains – will be examined later in this thesis, but what interests us most here is the repercussions it is having on the U.S.-American economy, and by extension polity.

While the discussion regarding the Trump 2 doctrinal shift has mostly been focused either on its effects on the preservation of the status of the dollar as the world's hegemonic currency (Boz et al. 2025; Pforr, Pape, and Petry 2025; Murau 2026), or on the effects of the tariffs on world trade and on their inflationary effects (Ignatenko et al. 2025; Amiti et al. 2026; Yale Budget Lab 2026), little consideration has been given to the context of evolving U.S.-American corporate profits in which the new policy package is being adopted. Yet, there are reasons to believe that the question of preserving the profit margins of U.S.-American corporations is central to the decision-making process of the new administration. This is perhaps most clear in the policy of enforcing profit-sharing foreign direct investment agreements, whereby the U.S. government monetises access to

its domestic market, against the agreement by foreign countries to invest in the United States, while abandoning a share of the profits to (or securing market access for) U.S.-American companies. Two striking examples here are the agreement by Japan to invest \$550bn on U.S. soil while abandoning 90 per cent of the resulting profits to the United States (Williams and Lewis 2025) and the setting up of a ‘reconstruction investment fund’ giving U.S.-American firms guaranteed access to the Ukrainian critical minerals and natural resources. The general spirit of these policies is translated into the rather explicit formulation of the 2025 National Security Strategy issued by the White House, which states that ‘[a]ll our embassies must be aware of major business opportunities in their country, especially major government contracts. Every U.S. Government official [...] should understand that part of their job is to help American companies compete and succeed’ (The White House 2025, p.18). It appears that direct intervention in favour of U.S.-American firms abroad is acquiring greater and greater importance in U.S. foreign policy – or, at the very least, it is being pursued more and more unabashedly. I argue that this can be read as a weak signal that ‘something is up’ with the profitability of the U.S.’s transnational corporations. This constitutes a starting point for the investigation undertaken in this thesis.

In discussing the shift away from the *Keynesian compromise* world of the post-war period towards the *neoliberal* world that followed, Gérard Duménil and Dominique Lévy wrote ‘the economic and social transformations of recent years were not driven by the interests of the masses, but by those of privileged minorities who have actually benefited from them [...] As soon as we shift our focus from the masses to these minorities, the common-sense analytical framework, which ranges from the emergence of difficulties to the implementation of solutions, regains its full relevance.’⁴ (Duménil and Lévy 2000, p.11). In doing so, they were arguing in favour of a renewal of the analysis of crises as solutions proposed by certain class fractions – and negotiated in society at large – to certain problems posed, not to humanity in general, but to a given social group. What followed in Duménil and Lévy’s book was an analysis of *neoliberalism* as a response to the fall of the profit rate in the United States in the 1960s–1970s. It is in an attempt to adhere to a similar guiding principle which drives this thesis’ investigation. If we take seriously the thesis of Panitch

⁴This citation, like all other citations of non-English works cited in this text, is translated by the author.

and Gindin (2013) and Burbaumer (2024) according to which the current organisation of the world market was instated specifically to allow U.S. corporations to raise their profit rates after the period of low profit rates in the 1970s, then the current policy shift begs the question of whether profitability also plays a role in changes currently at hand. In other words, is globalisation becoming less favourable for U.S. capital? And if so, in what way?

This thesis argues that the question can be made tractable by shifting the analytical focus away from the imbalances, which is where the Trump administration and most of its detractors are drawing it, and towards the profitability of U.S.-American (mostly transnational) corporations that organise international production. Trade deficits, I will show, are neither an anomaly that trade theory must explain away as the result of exogenous factors, nor an unambiguously burdensome weight on firms in the deficit country, and consequently on that country's ruling class. They are an expected outcome of international competition governed by absolute costs, and their impact on firm profitability depends on *what* is traded and under which conditions of inter-capitalist competition. Building on Crotty (1993)'s Post-Keynesian/Marxian theory of inter-capitalist competition, and on the distinction between *corespective* and *coercive regimes of competition* which he introduces. I hypothesise that the rapid technological catch-up of Chinese firms and the intense competitive pressure they bring in the industries they are entering are eroding the conditions which had enabled U.S.-American *lead firms* to secure high markups and profit rates. This threatens to trigger a shift from the *corespective* regime which had prevailed for leading U.S.-American firms until recently, towards a *coercive* one. The doctrinal shift of U.S. foreign economic policy can then be read, in the spirit of Duménil and Lévy (2000) and Burbaumer (2024), as an attempt to respond to a particular problem (a threatened position in the world market) faced by a particular social group (the dominant fractions of U.S.-American capital). This response is to be understood, not as an optimal plan guaranteed to restore the position of U.S.-American firms on the world market, but rather as part of a historical process of *institutional bricolage* where U.S.-American capitalism is in search of solutions to its problems.

The thesis is structured as follows. Chapter 2 lays the foundations. After reviewing balance-of-payments accounting, and confronting comparative cost theory with the em-

empirical persistence of trade imbalances, reviewing the exogenous explanations usually advanced for that persistence, it argues for the alternative framework of absolute cost theory. There, imbalances are considered a normal outcome of competition, reflecting the unequal competitiveness of trading partners, or equivalently inequality in unit labour costs across borders. Chapter 3 turns from *how much* to *what* is being traded. It develops the categories needed to relate the profitability of transnational corporations, trade imbalances and the practice of offshoring. Through a simple model of offshoring, it identifies under which conditions a trade deficit can be profitable for the deficit country's firms, a situation which we coin that of a *profitable trade deficit*. Chapter 4 introduces the theory of competitive regimes: grounding the firm's investment and pricing decisions in fundamental uncertainty and the irrecoverability of sunk costs. In that context it discusses how part of the tech sector can be identified with the notion of a *core industry*. It contrasts, after Crotty (2017b), *corespective* with *coercive* competition and identifies the entry of efficient foreign competitors as a credible path from the former to the latter. Chapter 5 brings us back to the very timely debate surrounding the ongoing '*China Shock 2.0*'. We discuss the different dimensions of China's catching up in high-value-added sectors and of its export dominance. We leave the chapter with a clear narrative of how China's catch up may trigger a shift in competitive regime for U.S.-American firms. Chapter 6 puts this hypothesis to the test, constructing a novel firm-level indicator of exposure to Chinese import competition for a panel of large listed U.S. firms over 2000–2024 and estimating profit-rate dynamics and investment functions across three successive time windows. Chapter 7 concludes and discusses further avenues of research.

Chapter 2

Explaining Trade Imbalances

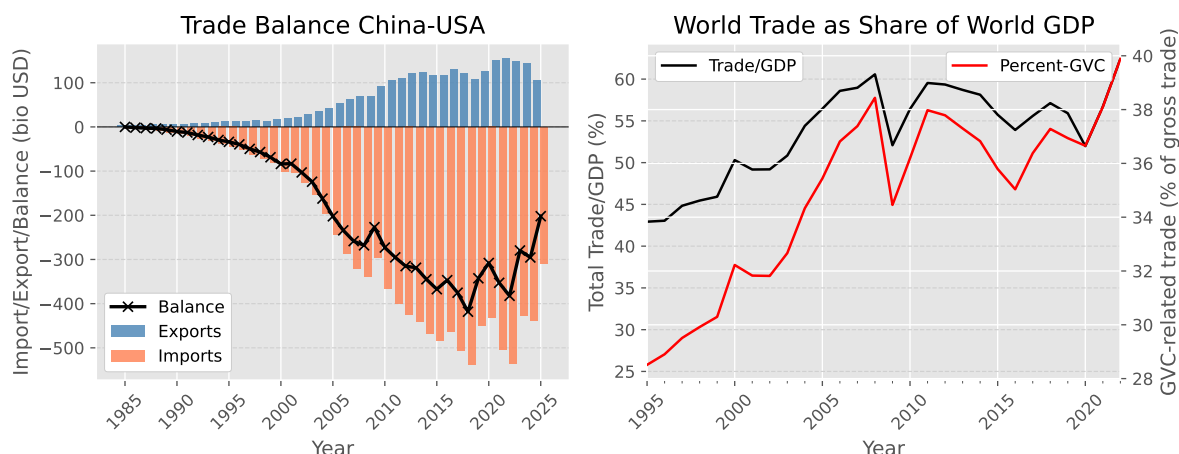


Figure 2.1: **Left:** the evolution of the nominal (non-inflation-adjusted) trade imbalances (gross exports and imports as well as the trade balance) between China and the United States (U.S. Census Bureau 2026). **Right:** in black, relative importance of trade in world GDP (World Bank 2026b), in red share of world trade attributed to GVC trade (in other words to fragmented production) computed from the TiVA database (Borin, Mancini, and Taglioni 2025).

We start this section on the theory of trade by considering a few stylised facts. These are currently at the centre of the discussion of the economic relation between China and the U.S., but they are also relevant to the more general discussion of trade under ‘*globalisation*’. The first of these stylised facts is the observation that trade between the two countries is deeply *imbalanced*, in the sense that China imports far more into the U.S. than the U.S. exports to China. We say that the U.S. runs a deep *trade deficit with China* (see Figure 2.1, right). We will see that the way this is explained is a central matter in

debates around the theory of trade. The second stylised fact is the increasing volume of trade across the world, relative to economic activity (see Figure 2.1) and of international investments which rapidly happened from the mid 1980s to the 2007–2008 Great Financial Crisis (GFC). Third, and perhaps most significantly, there is a tendency for the splintering of international production (Baldwin 2006; Milberg and Winkler 2013). An increasing share of products are now manufactured in disaggregated production networks which allocate different production steps to different geographies; to the point that a ‘*made in the world*’ label would better describe the location of production to the consumer than the famous ‘*made in*’ labels we are familiar with. These production networks, which the literature has come to refer to as *global value chains* (GVCs), have challenged economists’ understanding of trade flows. Lastly, since the GFC, the increase in the integration of the world’s production has stalled or reached a plateau or a turning point. This has led to discussions of a hypothetical *deglobalisation* (Milberg, Liess, and Tedesco 2024) – although, as Antràs (2020) notes, the current degree of international economic integration remains extremely high by historical standards – or of *slowbalisation*, reflecting the slowdown of the world’s integration dynamics (Antràs 2020, p.10). Concerning the relationship of the United States and China, several authors (García-Herrero and Tan 2020; Milberg and Johnston 2025; Jiang and L. Xing 2025) discuss the hypothesis that both economies are decoupling. At this point, talk of decoupling must be taken with a grain of salt as the two economies are still highly integrated (Meng, Gao, et al. 2022), but it is true that the integration between both economies stalled and, since 2018, has partially reversed (Milberg and Johnston 2025).

These four stylised facts will serve to guide us in the discussion of trade imbalances which follows. The rest of the chapter is structured as follows: first, we introduce elementary notions of balance-of-payment accounting which we require for clarity later on. Second, we discuss the mainstream theory of trade, which is mostly structured around the notions of comparative advantage and comparative costs, and discuss its limitations. Finally, we introduce an alternative theory of trade, which we argue allows for overcoming the limitations of comparative-advantage theories.

2.1 The Balance of Payments

Before we dive into our discussion of imbalances, let us quickly introduce the national accounting framework which enables their measurement: that of the balance of payments (Lavoie 2022, Ch.7). Let us start from the well-known ‘three balances’ equation, which states that – because to every credit there must correspond a matching debit – the sum of the private sector surplus ($S - I$), the public sector deficit ($T - G$), and the foreign sector deficit (which is the negative of the current account balance, $(-CAB)$) must necessarily be equal to 0 (Lavoie 2022, Ch.7):

$$(S - I) + (T - G) - CAB \equiv 0. \quad (2.1)$$

This is an accounting identity, and on its own, it does not carry a causal interpretation. While (2.1) is a flow-consistency equation, the significance of trade surpluses and deficits cannot be fully comprehended without also introducing an equation for stock-flow consistency (flows accumulate to stocks). In the case of foreign trade, the stock-flow consistency equation is the balance of payments:

$$CAB = AFAB. \quad (2.2)$$

where the current account balance (CAB), which measures the net payments incoming to the country, matches the accounting financial balance ($AFAB$) which measures the net assets acquired by the country abroad (be it in terms of deposits, foreign assets, etc.). Let us decompose both sides of (2.2). The current account balance itself can be broken down into three components:

$$CAB = (X - M) + (FYI - FYO) + SIB. \quad (2.3)$$

First, the balance of trade, exports less imports, $(X - M)$, which may or may not be decomposed in terms of goods and services. Second, the primary income balance of incoming less outgoing payments $(FYI - FYO)$, which corresponds to the balance of wages, return on investments or taxes. Finally, the secondary income balance, which by convention corresponds to the balance of payments not directly associated with economic

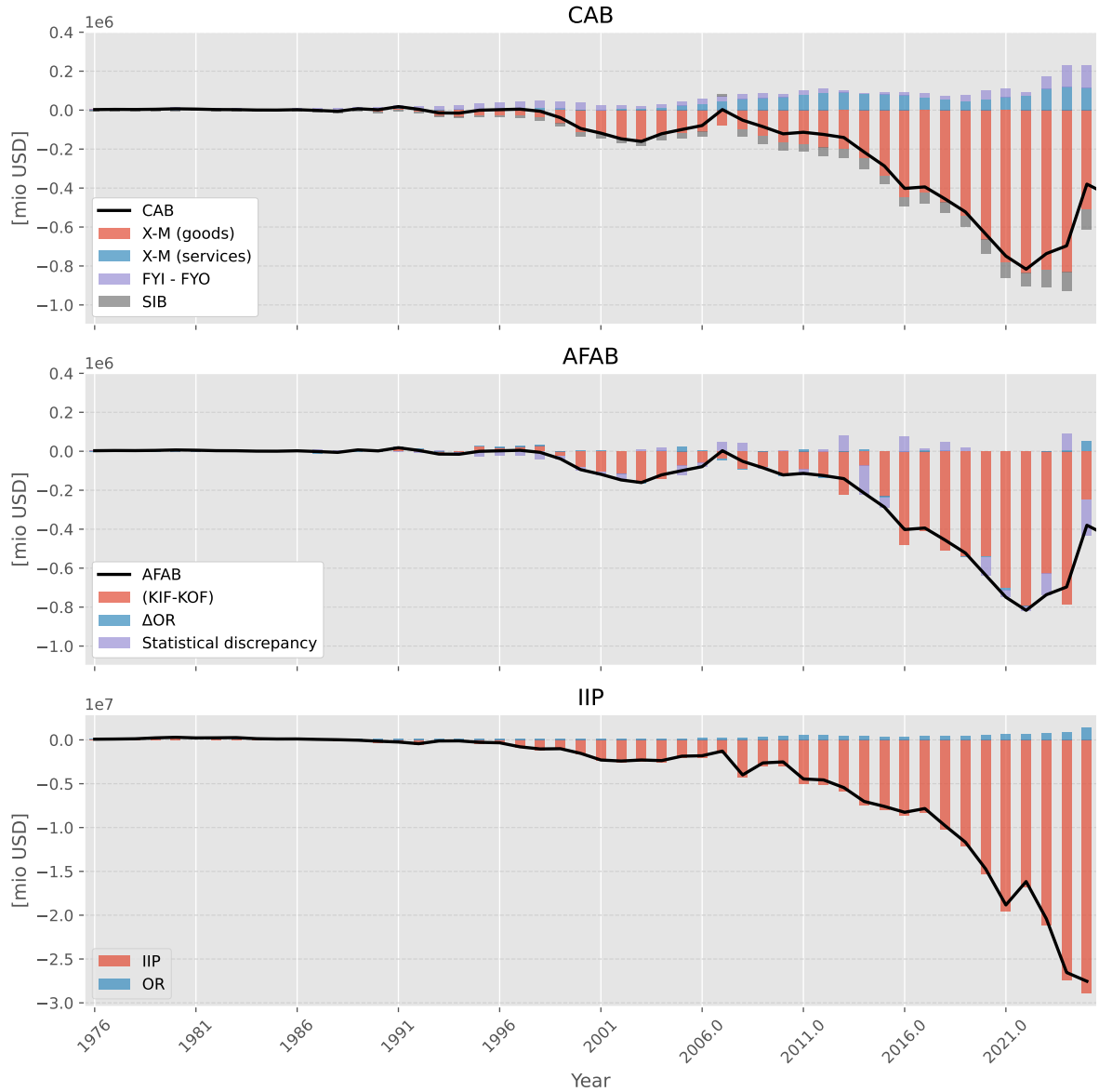


Figure 2.2: Current account balance (CAB), accounting financial account balance (AFAB) and net international investment position (NFK_t) for the United States. The plot clearly shows that the current account deficit leads to a build up of liabilities, which accumulates to a negative international investment position. Data from U.S. Bureau of Economic Analysis (2026a); U.S. Bureau of Economic Analysis (2026b). Note that there is a statistical discrepancy between the AFAB and the CAB data.

activities, for instance remittances or pensions (*SIB*). On the other side of (2.2) we find the equation for the accounting financial account balance. It decomposes into two terms:

$$AFAB = (KOF - KIF) + \Delta OR. \quad (2.4)$$

The first term designates the net outflow of capital ($KOF - KIF$), in other words foreign investments into domestic assets less domestic investments into foreign assets. The second concerns the changes in official international reserves (ΔOR), the reserves of foreign currency held by the central bank. The two terms of (2.4) accumulate over time, respectively to the official international reserves and to the net international investment position:

$$NFK_t = NFK_{t-1} + KOF_{t-1}, \quad (2.5)$$

$$OR_t = \Delta OR_{t-1} + OR_{t-1}, \quad (2.6)$$

where NFK_t designates the net international investment position and OR_t the official international reserves. This is clearly visible in Figure 2.2.

2.2 Comparative Cost and Advantage

Almost all of mainstream trade theory is articulated through comparative advantage models, which are themselves underpinned theoretically by the theory of comparative costs (CCT) (Shaikh 2016, Ch.11). We will start by formally introducing CCT, its fundamental premises and conclusions, before turning our attention to how it is operationalised in models of comparative advantages. Our goal in this section is not only to provide an introduction to these theories but also to discuss how comparative advantage theories deal with the three main stylised facts which we just introduced: the U.S. running a deep trade deficit with China, the volume of trade in the world having increased relative to the world's GDP and production having incrementally become more fragmented.

2.2.1 Comparative Cost Theory

CCT was first proposed by David Ricardo [1817] (1992) as a challenge to the prevailing theories of his time, which were constructed on a notion of absolute cost advantage. Although in its original presentation Ricardo's theory would appear quite heterodox nowadays (if only because Ricardo's theory is built on a version of the labour theory of value), the central principles of the theory as Ricardo presented them still form the foundation of contemporary trade theory. There are three such central premises.

- P.1. labour is internationally immobile** (there is neither equalisation of wages nor movement of qualified labour across borders).
- P.2. In the absence of *exogenous* forces, capital is approximately internationally immobile** ($KOF - KIF \approx 0$).
- P.3. The balance of payments is 0** ($CAB - AFAB = 0$) since capital is expected to be internationally immobile, this implies that the current account must be balanced.

The central point is that the combination of premises P.2 and P.3 implies that the current account must be roughly balanced: if most of the time, capital is approximately internationally immobile ($KOF - KIF \approx 0$), then the balance of payment constraint $CAB = AFAB$ becomes $CAB \approx \Delta OR$. In other words, the current account (and the trade balance) become constrained by the stock of international reserves, which is by definition finite. This means that the trade balance must be subject to a *long-run intertemporal budget constraint*, which leads to the conclusion that the current account balance must follow stationary dynamics (Coakley, Kulasi, and R. Smith 1996). Thus, any theory based on comparative cost must find a way to explain how the current account balance '*rights itself*'. Since the largest component of the current account balance is trade, it means that CCT hinges on the existence of a mechanism which ensures that trade flows have a tendency to balance themselves.

Since the original proposition of Ricardo [1817] (1992), multiple versions of such a mechanism have been proposed, but the core structure of these mechanisms has remained largely unchanged. The idea is that there exists a mechanism through which the real

exchange rate¹ improves (or worsens) for a country which is running a trade deficit (surplus), making it easier for that country to import (export) and to balance out the deficit (surplus). For clarity of the argument, let us write down the China-U.S. bilateral real exchange rate:

$$e_r = \frac{P_{rX}^* \cdot e}{P_{rX}}. \quad (2.7)$$

Where P_{rX} is the real export (from the U.S. to China) price level for the domestic (U.S.) production, P_{rX}^* is the real foreign export (from China to U.S.) price level and e is the nominal exchange rate (in the direct quote form, \$/元). For this indicator to be interpretable, the two price levels must be computed for equivalent consumption baskets.

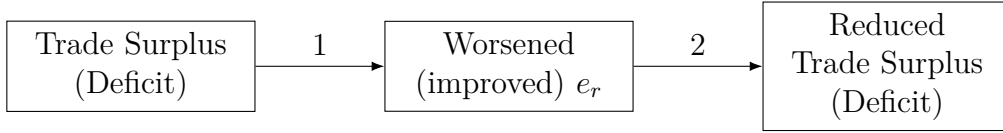


Figure 2.3: Causal chain of the ‘self-righting’ mechanism underlying CCT. The causal arrow 1 describes the notion that a trade surplus (deficit) worsens (improves) the real exchange rate, while arrow 2 describes the notion that a worsened (improved) real exchange rate leads to a reduced trade surplus (deficit).

In the formulation which Ricardo originally gave, the mechanism which ensured that trade was balanced was grounded in the quantity theory of money. His argument went as follows (Figure 2.3): suppose a country is running a trade surplus, this means that there is an influx of additional foreign money in the national economy (ΔOR increases). This influx of new money, by the quantity theory of money² leads to domestic inflation (i.e., the price level P_{rX} increases), which worsens the terms of trade: the real exchange rate e_r decreases. The prices of national goods are now higher relative to goods from the rest of the world. The final part of the argument states that a worsened real exchange rate leads to a reduction in the volume of exports, which balances out trade (CAB decreases, restoring the equilibrium). Ricardo’s argument works for fixed exchange rates, while more contemporary formulations are constructed to hold in a world of floating exchange rates, without relying on the quantity theory of money. In these arguments, a current

¹Which measures ‘the terms of trade’ for a country, in other words how advantageous it is to import a commodity from abroad compared to buying it at home.

² ΔOR is a component of the money supply M , and the quantity theory of money asserts that $MV = PY$, assuming a constant velocity of money V and output level Y , it implies that $M \propto P$ and thus that $\Delta OR \propto P$.

account surplus of the domestic country causes the nominal value of the domestic currency to appreciate, through the action of foreign exchange markets. The real exchange rate (e_r) is understood to track the nominal exchange rate (e), decreasing as well. In this understanding of CCT, the central aspect lies in the exchange rate dynamics and how they relate to current account imbalances. Regardless of which mechanism is posited, the assumption that some mechanism ensures balanced trade ($X \approx M$) holds.

2.2.2 Comparative Advantage Models

Models of trade based on comparative costs are called comparative advantage models. They are usually built on two main premises: they assume the principle of comparative costs (CCT), and some version of Say's law (Shaikh 2016, Ch.11). From these two key assumptions comparative advantage theory reaches the following conclusion. In a setting of unregulated trade, domestic prices will converge to world market prices (though the action of the *law of one price*). Therefore, producers will tend to assign productive forces under their control to sectors in which they face lower comparative costs. What is key here is that what matters is the advantage relative to other sectors within the country, not relative to the same sector in other countries.

country	sector	exchange rate	unit cost (local-currency)	normal world-market unit price	markup
U.S.	steel	$\$/\$ = 1$	$UNC_s = \$8$	\$10	$\mu_s = 0.2$
U.S.	wood	$\$/\$ = 1$	$UNC_w = \$4.5$	\$5	$\mu_w = 0.1$
CN	steel	$\$/\bar{\text{元}} = 0.2$	$UNC_s^* = \bar{\text{元}}30$	\$10	$\mu_s^* = 0.4$
CN	wood	$\$/\bar{\text{元}} = 0.2$	$UNC_w^* = \bar{\text{元}}10$	\$5	$\mu_w^* = 0.6$

Table 2.1: Numerical example of the comparative advantage logic: each country specialises in the sectors in which it enjoys a *comparative* advantage, that is, the sectors where its markup is higher relative to its other sectors.

Let's take an example to make the argument clearer. Say that in the United States, it is cheaper to produce steel than wood, relative to world market prices. If this is the case, then steel has a cost advantage relative to wood and the margins which a steel-making company will make are higher than a wood-making one. In this example, comparative advantage theory makes two key predictions. First, it predicts that the United States will

specialise in steel (see Table 2.1), because within the national borders, it is the sector in which firms get the highest markup³ even if in China, the markup on the same product is higher. This is important because this is why what matters is the *comparative* (relative to other national sectors) advantage, not *absolute* advantages (relative to all sectors producing equivalent goods in the world market). The second prediction is that since Say’s law applies, all productive forces in the country will be allocated some productive activity, even if that country is less productive than its competitors *in every possible sector*. This is the case in the example of Table 2.1, where China outcompetes (faces lower unit normal costs than) the U.S. in both sectors.

One point which should be underlined here is that in the comparative advantage model, the reason that a low productivity country is able to successfully sell its production on the world market is because of an adjustment of the terms of trade, which necessarily entails a reduction of the domestic real wages⁴. Hence, trade liberalisation is not expected, in the long run, to lead to job losses and underutilisation of productive capacity.

In that argument, the immobility (or the low mobility) of capital internationally is key. Indeed, it is required for the comparative advantage argument to hold that domestic capitalists do not react to lower absolute costs abroad by reallocating their productive activities wherever costs are lowest. This was a point recognised by Ricardo [1817] (1992), who argued that international capital mobility was, in practice, kept in check by ‘*the natural disinclination which every man has to quit the country of his birth*’.

By introducing an adjustment mechanism that ensures that $X \approx M$ through an adjustment of the real exchange rate, the CCT assumptions specify that it is always worth it to specialise domestically, even if the sector in which specialisation happens is a sector which is outcompeted abroad. Thus, according to CCT, it is thus unambiguously beneficial for a country to lower its trade barriers. Without the assumptions from CCT, the unambiguity of these conclusions breaks down. A world without CCT is one where there can exist situations of persistent trade deficit or surplus ($X \neq M$); a world without Say’s

³Which is the same as saying it is the sector in which firms face the lowest unit normal costs relative to world market prices.

⁴When adopting a neoclassical theoretical framework – which is what most modern comparative advantage models do – where real wages are set by the marginal productivity of labour, this is a trivial remark.

law is one there may be a situation of persistent unemployment of labour and capital alike (if they are both immobile), or labour exclusively (if capital is more mobile than labour). If both of these assumptions are relaxed, then the interaction of both trade deficits and unemployment of labour and capital becomes an expected outcome of trade.

What we have discussed so far does not constitute a model *per se*. Our specification of what gives a specific sector its comparative advantage (lower unit normal cost⁵ relative to the world market prices) is too vague, and different approaches to tackling this question lead to different models of comparative advantages. Modern acceptations of comparative advantage, such as the Heckscher-Ohlin-Samuelson (HOS) model, explain cost differences by inequality of factor endowments (Feenstra 2016, Ch.2). Early versions of the HOS model only considered abundance of the two usual neoclassical factors (labour and capital), but in the face of the observation that capital-abundant countries import capital-intensive goods (this is known as the ‘Leontief Paradox’), have been reformulated to consider the abundance of ‘skilled’ versus ‘unskilled’ labour (this is known as the Heckscher-Ohlin-Wood framework). The HOS framework is an example of the kind of refinements that contemporary comparative advantage models adopt. There are other such models, but the two key premises which we introduced (CCT and Say’s law) are never abandoned (Milberg and Winkler 2013; Shaikh 2016; Kaczmarczyk 2023).

The conclusions of comparative advantage are often presented through a story about the benefits of specialisation. A country would benefit from reducing the amount of trade regulation it imposes, because once trade regulations are relaxed and industry protection is dropped, the forces of competition drive domestic producers to specialise in the commodities in which they enjoy a comparative cost advantage. Since Say’s law is assumed, this process of specialisation is not expected to happen at the expense of domestic employment; if it does, it will only be a transitory process which will dissipate on its own. On the other hand, since, by CCT, trade is assumed to be balanced, the exports of the goods in which the country is comparatively advantaged yield the highest possible revenue (X takes the maximum possible value), with which the maximum level of imports can be brought into the country (since $X \approx M$). In other words, the maximum

⁵By unit normal cost we refer to the cost of producing one unit of output (for instance one commodity, or a fixed quantity of an uncountable commodity in some cases) accounting for overhead costs and capital depreciation at the normal rate of capacity utilisation (Lavoie 2022, Ch.3).

level of social welfare is attained in the setup without trade regulations. There are, of course, rigorous demonstrations of this claim by social welfare economists (Feenstra 2016, Ch.7). Once trade is deregulated across the world, what happens is a reconfiguration of the division of labour across the world in which each country specialises, leading to productivity gains and faster growth across the planet (Feenstra 2016, Ch.11).

Comparative advantages provide a natural interpretation for two of the four stylised facts which we introduced at the beginning of this chapter: the increased importance of trade in world GDP and the increased fragmentation of production across global production chains.

The narrative goes as follows: if, under conditions of unregulated trade, there is a tendency for countries to specialise in sectors in which they enjoy comparative advantage (and are relatively more productive), then trade liberalisation leads to a situation in which different countries specialise in different sectors, increasing their interdependence. This naturally explains the fragmentation of production as each country specialises in a production step. The increase in the volume of trade is then a corollary of specialisation and fragmentation of production, as intermediaries and final goods alike now need to be shipped across locations to join the production steps together and bring the final goods to their consumers. The increased volume of trade and fragmentation of production are thus coherent with the predictions of the comparative advantage framework and models.

The same cannot be said about the persistence of trade imbalances, or the stalling of international economic integration.

2.3 The Problem of Imbalances

As we just saw, most of the mainstream theory of trade is constructed on the assumption that there is a mechanism (which is assumed to work according to the blueprint of Figure 2.3) that brings trade back in balance, ensuring that $X \approx M$ holds. When mathematically formulating a comparative advantage model, it is indeed often simply assumed that $X = M$. But this initial assumption clashes with what appears to be the reality of international trade: most countries run an imbalanced current account most of

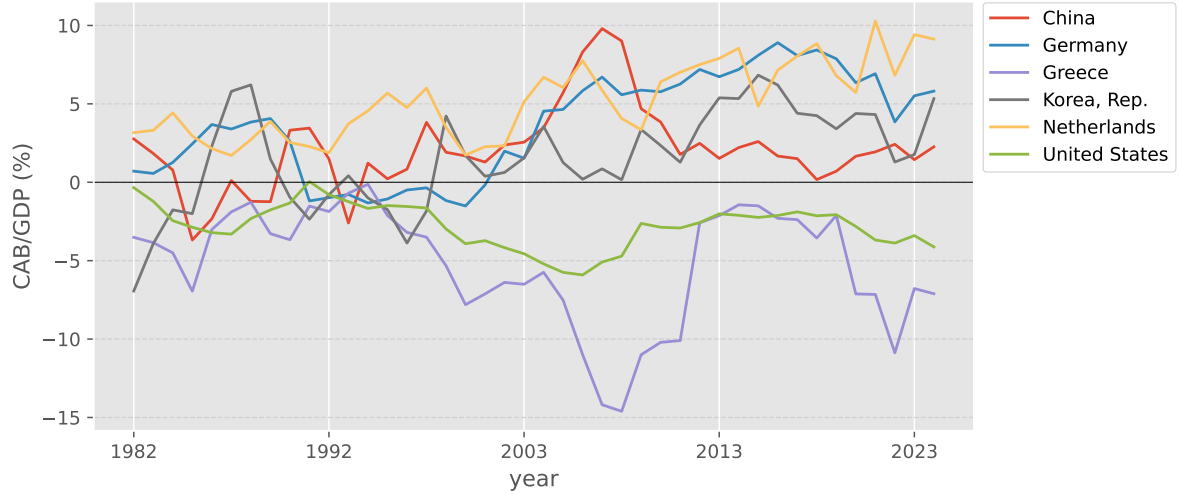


Figure 2.4: Illustration of the evolution of current account imbalances across multiple countries (measured relative to GDP), data from World Bank (2026a).

the time (see Figure 2.4). Balanced trade is a historical exception rather than the rule. Naturally, by the balance of payment identity $CAB = AFAB$, these persistent current account imbalances are matched not only by official reserve changes, but also by capital flows, which presents a major challenge for comparative advantage theory.

As we previously discussed, the normal assumption of CCT is not that trade is always strictly in balance, but that it is subject to a *long-run intertemporal budget constraint*, and thus that the current account balance must follow stationary dynamics (Coakley, Kulasi, and R. Smith 1996). Here we run into an issue: empirical economists tend to have trouble rejecting the hypothesis that the current account balance (or the trade balance) is a non-stationary time-series⁶, especially if a relatively short time-frame is considered (less than fifty years) (Froot and Rogoff 1994; Herwartz and F. Xu 2008). While this could be interpreted as raising a fundamental issue with the assumption of approximately balanced trade, the conclusion most mainstream economists come to is that reversion to balanced trade is subject to *slow dynamics*. For instance, Froot and Rogoff (1994) argue that the mean-reversion process has a four-year half-life, and thus that the amount of data required to test the stationarity of a time-series is around seventy-five years. Implicitly, it means that the time-frame over which the comparative advantage mechanism should

⁶A stationary series (we write $I(0)$) is a time-series which ‘gravitates’ in a pattern around a specific value. In the case of the trade balance, if there were a mechanism that ensured balanced trade, we should expect the current account time-series to be stationary and to gravitate around 0, hence it should be an $I(0)$ time-series. See Hamilton (1994) for proper definitions.

be expected to work is slow as well. Specifically, it should be expected to work as slowly as the mean-reversion process works. Thus, it implies that policy prescriptions derived from comparative advantages should not be expected to come to fruition over the short term, defined here as a little less than half a century.

This gives us a first element to answer one of the main questions which we asked ourselves coming into this section, that is, how the mainstream (comparative advantage) theories of trade deal with the question of imbalances. The answer is that they do not. The question of imbalances is necessarily approached through an ad-hoc lens, a late addition to a theoretical framework constructed on the assumption that trade imbalances, most of the time, do not (or should not) exist. Later in this text (Section 2.4), we will see that adopting an *absolute cost* framework addresses this issue: imbalances become expected as a result of trade. But until we get there, let us discuss how the mainstream deals with the stubborn fact that trade imbalances are a fixture of national accounting data.

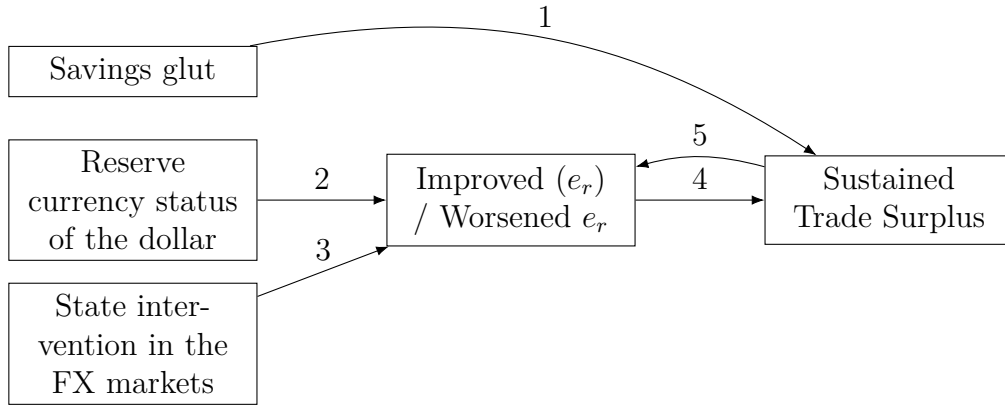


Figure 2.5: Causal chain of the three exogenous explanations for U.S.-China trade imbalances. The causal arrow 1 describes the effect of (relatively to the United States) overabundant savings in China flowing into the United States and directly driving the trade surplus. Causal arrows 2 and 3 describe the exchange rate channels, which are understood to affect the real exchange rate, leading indirectly to 4, sustained trade imbalances. Causal arrow 5 describes the slow ‘self-righting’ mechanism, which is still understood to be acting while the exogenous forces 1-2-3-4 throw the current account out of balance.

As we briefly introduced, mainstream economists argue that the combination of slow mean-reversion dynamics and fast perturbations explains the persistently imbalanced nature of trade. This raises a certain number of issues. First, regardless of whether or not there exist mean-reversion dynamics, since the predominant theory of trade assumes no imbalances, economists make policy recommendations constructed on a no-imbalance

model, in a world where trade stays imbalanced for periods longer than fifty years. Second, the claim that reversion to the mean is occurring (albeit slowly) does not constitute a theory for explaining actually existing imbalances. This thus leaves the mainstream in a situation where they must provide an explanation for the actually existing imbalances, but where they lack a systematic approach for doing so. Instead, mainstream economists thus construct eclectic, multifactorial explanations which are structured as follows (Figure 2.5). First, they recognise that the mechanism which restores the balance of trade is a very slow one, and second, they introduce *exogenous* mechanisms, which throw off the balance of trade. Because of how the CCT premises are structured, these explanations assume that exogenous forces act on the accounting financial balance, which in turn impacts the current account. In what follows we explain three such *exogenous forces*, which are mobilised as explanations for U.S.-China trade imbalances.

2.3.1 Exogenous explanations for the persistence of trade imbalances

There are three main hypotheses: the savings-glut, imbalances as a consequence of the dollar's reserve currency status and imbalances as a consequence of exchange rate manipulation by the Chinese government. All of these narratives assume an essentially identical underlying structure, which we presented in Figure 2.5.

The savings' glut hypothesis First put forth by Ben Bernanke (2005), the hypothesis is constructed on the sectoral accounting identity: $(S - I) + (T - G) = CAB$. Assuming a constant public deficit, the identity states that any lowering of private savings $\Delta(S - I) < 0$ must be matched with a corresponding current account deficit increase $\Delta CAB = \Delta(S - I)$. Of course, an accounting identity by itself does not imply any particular direction of causality; the savings' glut hypothesis assigns causality from the savings rate, to savings, to financial account and finally to current account balance.

In the words of Bernanke (2007) '*[d]omestic investment not funded by domestic saving must be financed by capital flows from abroad, and, indeed, the large increase in the U.S. current account deficit was matched by a similar expansion of net capital inflows.*' Of

course, to any deficit there must correspond a surplus. The savings glut hypothesis thus requires other countries, in which the savings rate is ‘too high’ to fund this disequilibrium. Bernanke’s hypothesis is that China runs a higher than normal savings rate, which leads to its trade surplus with the U.S. We find a textbook example of this point being made in a very recent Martin Wolf (2026a) opinion piece published in the *Financial Times*:

If a country has an extremely low share of household consumption in GDP (40 per cent in China’s case), it will also have an enormously high share of savings in GDP. Surplus savings then have to be absorbed abroad, especially after the alternative of a domestic property boom (as in Japan in the 1980s and China in the 2010s) collapses. If a country is to export surplus savings, it needs to produce a surplus of tradeables. In China’s case, that means manufactures. In sum, China’s trade surplus of \$1.2tn last year is not just a product of competitiveness, but also of its macroeconomic imbalances. Moreover, if China has huge savings surpluses, others must have offsetting deficits. The U.S. is the world’s most creditworthy country and so naturally its safest borrower. (Wolf 2026a)

Notice how to Wolf, it is clear that the financial imbalances (financial flows from China directed towards the rest of the world) drive the trade imbalances, not the other way around. This is central to the savings glut argument, as it is for all exogenous explanations discussed here.

While the accounting identity $(S - I) + (T - G) = CAB$ must necessarily hold, on its own, it says nothing about the direction of causality. Bernanke (2005)’s assertion – which is essentially the same as Wolf’s – is that there is an autonomous (and thus causal) decrease in the savings rate (of households) in the U.S. Since the savings rate reduction is unmatched by an increase in the investment rate, a persistent trade deficit appears. In the case of China, the argument that the country’s weak social safety net has led to a high domestic savings rate, which in turn has driven the imbalance, has gained traction (Y. Xu et al. 2025; Bayoumi and Gagnon 2026). Another explanation is demographic; the one-child policy is understood to have left Chinese households with fewer reasons to consume, increasing their savings (Dieppe et al. 2018).

The reserve currency hypothesis Another hypothesis is that the dollar’s position as the reserve currency generates ‘exogenous’ capital flows into the U.S. as foreigners try to acquire dollar-denominated assets, because the U.S. is the most creditworthy country. This is understood to push the exchange rate up (because it stimulates the demand for dollars), leading to an overvalued currency, and therefore, as a result of a deteriorated real exchange rate, to a persistent trade deficit (McKinnon 2001). There is no doubt that the dollar holds a unique position in the world’s economy and that there is high demand for dollar-denominated assets. On this issue, Weber and Shaikh (2021) remark that the dollar was also the world’s currency at a time when the United States ran a surplus (before the 1970s), contradicting the notion that demand for dollar-denominated assets necessarily drives the trade deficit.

The currency manipulation hypothesis The proposition according to which China’s trade-surplus *vis à vis* the U.S. is best interpreted as the result of exchange rate manipulation on the part of China has come to the centre of the policy debate. It is a position which has successively been held by Barack Obama (Weaver 2010), by Trump’s Director of Trade and Industrial Policy (Navarro and Autry 2016) and by the U.S. Treasury U.S. Department of the Treasury (2019). The argument goes as follows: it assumes that China, by intervening in foreign exchange markets and accumulating dollar-denominated assets, artificially pushes the dollar-to-renminbi exchange rate above its equilibrium level, leading to a persistent trade surplus for China and a corresponding deficit for the U.S.

This hypothesis in particular has led to the production of a dense literature concerned with estimating the degree of *over-* or *undervaluation* of the renminbi. The idea according to which a currency can be *over-* or *undervalued* hinges on the notion that, assuming the absence of *market distortions*, exchange rates should converge to a certain *equilibrium level*, itself determined by *macroeconomic fundamentals* (Cheung 2012). This, of course, supposes that whatever level of exchange rate is measured is not the (distortionless) equilibrium level and thus requires the production of an estimation of said equilibrium level. A good criticism of this argument is provided by Weber and Shaikh (2021), who go through measurements of the *undervaluation* of the renminbi (the Chinese currency) in

detail and show that there is no consensus about the devaluation of the renminbi across estimates by different researchers. This is because there is no consensus about what the equilibrium level of the renminbi to dollar exchange rate should be. As we will soon discuss in greater detail, exchange rate determination theories fail to reach any kind of scientific consensus, greatly undermining the credibility of this argument.

The industrial subsidies argument A fourth narrative locates the source of the imbalance in Chinese industrial policy itself: by subsidising its manufacturers, China is alleged to have engineered excess capacity and export dominance, producing its surplus and, correspondingly, deficits abroad. This allegation has recently received apparently systematic empirical support on part of the OECD in the publication of the MAGIC database which tracks the subsidies received by the world’s largest manufacturing groups over 2005–2024 (**oecd2026b**). According to this database, Chinese firms received on average three to eight times more government support than firms based in the OECD. These findings were widely relayed in the financial press as evidence of a subsidy-driven *China Shock* – (Foster and Leahy 2026; Tett 2026).

Three remarks are in order. First, the headline comparison is partly an artefact of presentation: China is the only economy which the report displays as a single country, while every other economy is pooled into revenue-weighted averages of continental blocs (‘OECD – North America’, ‘OECD – Europe’, ‘OECD – Asia Pacific’, ‘Rest of the World’), a grouping the report justifies only by the confidentiality of firm-level data. Averaging mechanically dampens the values of the other large subsidisers – the report itself notes, for instance, that firms based in the United States were the largest beneficiaries of semiconductor subsidies in absolute terms, but they appear only as one component of the OECD – North America average – thereby inflating China’s apparent exceptionality relative to values that have been diluted by construction. Second, the measurement itself is contested: Chinese economists have accused the OECD of miscalculating below-market borrowings, and narrower definitions of subsidies. Third, and most tellingly from the perspective of the present work, the OECD’s own estimates indicate that while subsidies appear to have caused an increase in firms’ market shares, they did not lead to any significant increase in their productivity or profitability (OECD 2026). This is a point we will

come back to: in Chapter 5, as it may relate to the character of competition in China.

2.4 Absolute Cost and Competitive Advantage

Having gone through *exogenous* explanations for trade imbalances, let us now turn to an alternative explanation, that of the theory *absolute cost advantage* (ACT). This line of argumentation finds its roots in Adam Smith [1776] (2014)'s work on trade and was formalised later on by Harrod (1957) and Shaikh (2016). As CCT served as a theoretical foundation to comparative advantage models, ACT serves as a theoretical foundation to *competitive advantage* models. In the following section we introduce the former through a discussion of the criticism levelled at CCT by Shaikh (2016), Ch.11, before turning to the alternative conclusions brought upon by adopting ACT.

2.4.1 Absolute Cost Theory

The argument in favour of ACT (and against CCT) begins by questioning the very existence of the '*strong equilibrating forces*' (Krugman 1991), which, in CCT, are expected to bring back trade to a balanced state. This is equivalent to a rejection of the second premise of CCT (P.2, see Section 2.2.1) since, in the absence of a mechanism that balances out the current account, the balance-of-payment equilibrium accounting identity $AFAB = CAB$ implies that capital cannot be assumed to be internationally immobile anymore. As we will see, this leads to a theory where trade imbalances become an expected outcome and contradictory financial flows move to adjust the balance of payments.

Let us start our discussion by considering again the causal mechanism that is expected, in CCT, to lead to the outcome of the 'self-righting' of the current account, which we schematically represented in Figure 2.3 of Section 2.2.1. In this criticism, we focus on the '*modern*' version of the mechanism (the exchange rate channel) rather than the '*classical*' version (the quantity of money channel) because we argue that the adoption of the quantity theory of money is sufficient to discredit the '*classical*' version. There are two causal mechanisms at play.

1. Trade surpluses or deficits have an impact on the real exchange rate, through the nominal exchange rate channel. Specifically, trade surpluses lead to appreciation of the surplus country's currency and deficits to a devaluation.
2. If a country's currency depreciates (appreciates), its trade balance will improve (worsen). This is known as the Marshall-Lerner condition.

The core of the issue lies in the first proposition. In more modern versions of comparative advantage models, it is understood that trade surpluses lead to increased demand for the currency of the country on FX markets, and thus affect the nominal exchange rate, which in turn affects the real exchange rate. So, to be empirically valid, this first proposition would require that there exists a statistically significant relationship between the exchange rate and either the balance of trade or the current account balance. The issue here is that economists have failed to produce models of floating exchange rate dynamics that outperform a naive random walk specification (Meese and Rogoff 1983; Cheung, Chinn, and Pascual 2005). In other words, the discipline's best (empirically validated) guess about nominal exchange rate dynamics is that they are fully random. This, of course, shines a very different light on the difficulties which econometricians face when trying to show that *CAB* showcases stationary dynamics (Froot and Rogoff 1994; Herwartz and F. Xu 2008). It may be that the dynamics are not slow, but absent. In his survey of orthodox theories of the exchange rate, Harvey (1996) simply writes:

‘Over the same period, neoclassical economists have expressed increasing frustration over their failure to explain exchange rate movements [...] Despite the fact that this is one of the most well-researched fields in the discipline, not a single model or theory has tested well. The results have been so dismal that mainstream economists readily admit their failure’ (Harvey 1996, p.567)

The alternative to these theories is to abandon the hypothesis that the real exchange rate adjusts in such a way that the current account becomes balanced, but to accept that the current account is floating ($CAB \neq 0$) and to take the real effective exchange rate to simply reflect the difference in currency-adjusted production price levels (hence, unit normal costs) across countries.

This is the alternative mechanism (Figure 2.6) proposed by Milberg (1994); Shaikh and

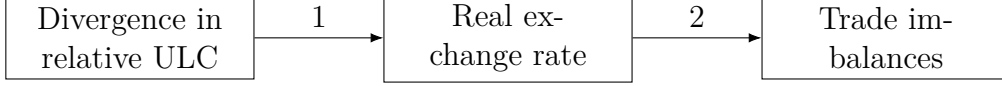


Figure 2.6: Causal chain of the mechanism underlying ACT. The causal arrow 1 reflects the notion that the terms of trade reflect the relative unit labour costs divergences. The causal arrow 2 reflects the notion that the terms of trade are reflected in a trade surplus or deficit. Unlike in the CCT example there is no self-righting mechanism.

Antonopoulos (2013) who argue that the real effective exchange rate of a country is regulated by vertically integrated unit labour cost (adjusted by the tradable to non-tradable goods ratio), and that the nominal exchange rate is floating (cannot be predicted). Vertically integrated unit labour costs (v) are defined as direct labour costs plus the direct labour cost of every intermediate input required for producing a commodity, and every intermediate required for producing the intermediates and so on. This is a quantity which is very close to unit normal cost UNC (especially when taking an average across sectors), as when decomposing the labour costs of a unit of output vertically we necessarily find that every cost is split into a remuneration of labour (wages) or of capital (profits) (Shaikh 2016, Ch.9). They can be computed using input-output tables (Pasinetti 1977). The relationship hypothesised by Shaikh and Antonopoulos (2013) is expressed as follows:

$$e_r = \frac{P_{rX}^* \cdot e}{P_{rX}} \approx \frac{v_r^*}{v_r} \cdot \frac{\tau^*}{\tau}. \quad (2.8)$$

Where $v_r = v/P_c$, $v_r^* = v^*/P_c^*$ are real vertically integrated unit labour costs (weighted according to the same basket as our price index) computed from the vertically integrated labour costs v , v^* (which can be computed using input-output data) and the price indices P_c , P_c^* which need to be computed using equivalent consumption baskets. The coefficients $\tau = P_c/P_{cT}$ and $\tau^* = P_c^*/P_{cT}^*$ are ratios between the price coefficients of tradable versus non-tradable goods. Assuming that trade is relatively open, which implies that $P_c \approx P_{cT}$ and hence that $\tau \approx 1$ for both countries, we can drop the tradable/nontradable term τ/τ^* . Since the vertically integrated unit labour costs are correlated with the direct unit normal costs, the main claim of ACT can be approximated as:

$$e_r \approx \frac{UNC^* \cdot e}{UNC}. \quad (2.9)$$

This leaves us with the main conclusion of ACT: the real exchange rate (i.e. the terms of

trade) is a function of the divergence of overall *competitiveness* across borders; itself being determined by how high wages are and how productive labour is, in a given location or country.

We are left with a completely different theoretical framework. The terms of trade become a reflection of the divergence in *competitiveness* of different countries, and so do the imbalances which are now understood to result from a competitive process akin to the one which occurs within national borders (albeit with some additional costs caused by trade barriers). On that matter, Shaikh and Antonopoulos (2013) remark that:

‘As far as internal competition is concerned, virtually all theories agree that competition within a given country is driven by the law of absolute costs, that is to say, firms with lower unit costs of production enjoy an absolute competitive advantage. From this point of view, within any one country, high-cost regions would suffer from a competitive disadvantage. If unprotected from competition, firms in such a region would tend to have declining shares in the national market.’ (Shaikh and Antonopoulos 2013, p.207)

The main claim of ACT: *that the real exchange rate is proportional to the divergence in competitiveness, proxied by direct unit labour costs* is one which can easily be gauged using a database such as the Penn World Tables (Feenstra, Inklaar, and Timmer 2015). We commit ourselves to that exercise by producing plots of the real bilateral exchange rate and relative unit labour cost between the U.S. and China (Figure 2.7, left). The real exchange rate (left axis) and the relative unit labour cost (right axis) are drawn on different scales. Using the same dataset, as well as Comtrade data for computing international trade volume we compute real effective exchange rate and unit labour costs for both countries (Figure 2.7, right). The qualifier *effective* indicates that these are geometric averages of the bilateral exchange rates, weighted by each partner’s share in the country’s trade. Here the series track each other less well. Which may be the result of the omission of important trade partners of either China or the United States. We provide details on the computation of both series in Appendix B. While Figure 2.7 does not constitute a rigorous check of ACT – we only consider two countries, do not perform a co-integration test or robustness checks – it does give it some empirical credibility.

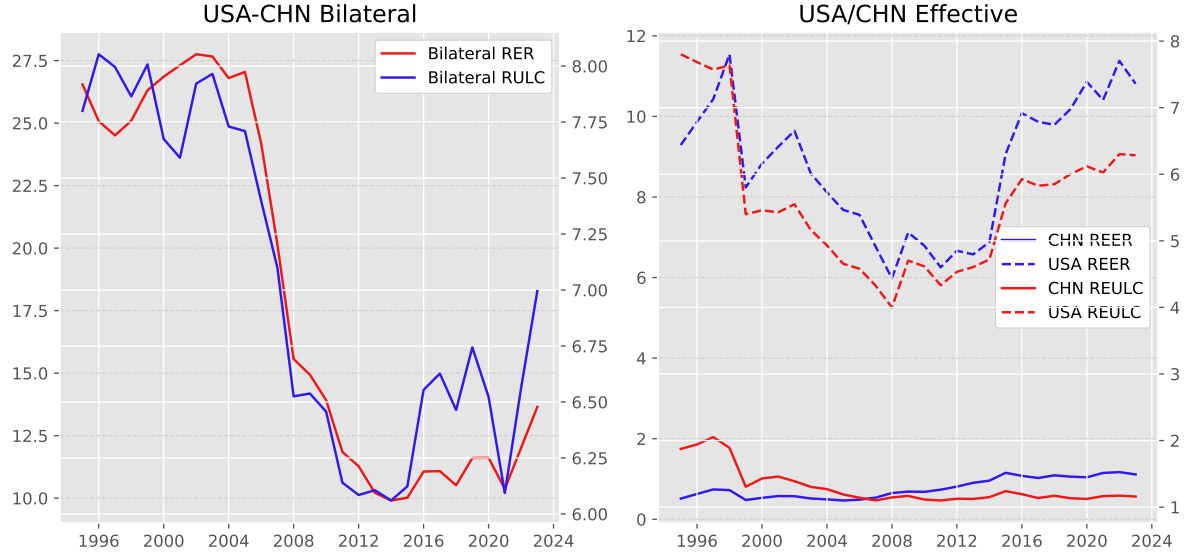


Figure 2.7: **Left:** bilateral real exchange rate between the USA and China, computed from price levels and exchange rates from the Penn World Table 11 (Feenstra, Inklaar, and Timmer 2015), together with bilateral relative unit labour costs in manufacturing, proxied by relative labour shares from the same dataset. **Right:** real effective exchange rates for the USA and China, computed from the same datasets, with United Nations (2026) used to weight the relative importance of trade partners. Details of the computations are given in Appendix B.

2.4.2 Competitive Advantage

In the same way that CCT was the basic building block for the comparative advantage model, ACT is the basic building block for the theory of competitive advantage. It starts by rejecting the notion that there exists a ‘self-righting’ mechanism in the current account balance; it is thus expected that current accounts are imbalanced and that they follow a non-stationary process. Without this self-righting assumption, there is no expectation anymore that a sector which is absolutely disadvantaged but comparatively advantaged (like the U.S.-American steel sector was in the example from Subsection 2.2.2) will export its production nonetheless. Instead, it is likely that all production concentrates where absolute advantages lie; so, in the same example as Subsection 2.2.2, *China would dominate the production of both steel and wood* (see Table 2.2).

But the accounting constraint of the balance of payment (2.2) does not disappear, which means that any current account deficit must be compensated by a corresponding financial account deficit (since $AFAB = CAB$). Thus there subsists the question of how can

	domestic	foreign		domestic	foreign
steel	$\mu_s = 0.2$	$e \cdot \mu_s^* = 0.4$	steel	$\mu_s = 0.2$	$e \cdot \mu_s^* = 0.4$
wood	$\mu_w = 0.1$	$e \cdot \mu_w^* = 0.6$	wood	$\mu_w = 0.1$	$e \cdot \mu_w^* = 0.6$

Table 2.2: Numerical 2-sector, 2-countries example showcasing the logics of comparative advantage (left, the sectors with comparative advantage in each country are coloured green) and absolute cost theory (right, the sectors with absolute cost theory across all countries are coloured green).

persistent imbalances be financed, and what role do interest rates play? Two distinct answers circulate in the literature. We keep them carefully separate, because they imply different causal roles for the rate of interest.

The first answer holds that the interest rate is itself the channel through which the balance of payments is closed – that is, the mechanism by which the financing of the external deficit is elicited. The argument goes as follows: assume there is an increase in trade imbalances; the first-round effect is an appreciation of the nominal interest rate. But in the medium to long run, what happens is that the trade deficit reduces liquidity in the deficit country, raising the interest rate, whereas it will increase liquidity in the trade surplus country, lowering the interest rate. This interest rate differential is expected to drive financial flows from the surplus to the deficit country, balancing out the balance of payments Shaikh (2016), Ch.11. However, authors like Porcherot and Féliz (2024) object that this closure ‘*leaves the door open to the full comeback of loanable funds theory and money supply exogeneity, as the central bank does not seem to have any sort of control on the interest rate*’ (p.23). This raises an issue if one adopts, as we do, a contemporary (endogenous money-compatible) Marxian/Post-Keynesian view on finance. Because we do not wish to reintroduce a theory in which interest rates clear a market for savings, we do not adopt this closure.

An alternative mechanism, reconciling ACT with a theory of endogenous money is proposed by Harvey (2019) who argues that precisely the assumption of endogenous finance removes the contradiction between persistent current-account imbalances and the balance-of-payments identity. In his account, imports are not paid for out of a fixed stock of foreign currency but by the exporters and financial institutions extending trade credit to importers. Trade finance is thus endogenously created for importers, since this credit is ‘*brand new currency*’, the supply of currency generally rises together with the demand

generated by trade; autonomous changes in trade flows therefore have little direct effect on the exchange rate Harvey (2019), p.17. This resolution nevertheless requires two qualifications. First, the segmentation between trade finance and speculative finance is not absolute: securitised trade credit can affect the speculative capital market Harvey (2019), p.19. Second, endogenous finance does not mean that every desired import is automatically financed, since banks may ration credit and prevent a trade flow from taking place Harvey (2019), p.18. Subject to these qualifications, his argument shows that persistent current-account imbalances are compatible with both the balance-of-payments identity and an endogenous-money theory of finance.

Regardless of which mechanism is posited, the conclusion of ACT is that a country's competitiveness will drive a current account deficit (surplus) which will be compensated by an inflow (outflow) of foreign capital (Figure 2.6). In other words, trade deficits are financed by the acquisition of debt, equity and other property titles by foreigners in the country running the deficit. This scheme accurately describes what the United States has been going through in the last half century, as it has run a negative current account balance, accumulating liabilities *vis à vis* the rest of the world, leading to a deeply negative international investment position (Figure 2.2).

This is a very different outlook from the view of mainstream trade theory according to which international capital flows are understood to be exogenous from trade flows (all three of the mainstream explanations which we discussed for trade imbalances started from exogenous international capital flows). Here, it is understood that it is an expected result of international competition that trade and capital flow across borders. The kind of imbalances which characterize the U.S.-China economic relationship is not a puzzle to be answered by the addition of exogenous factors, but a result of the difference in competitive advantage between the two countries. This is not to say that the exogenous mechanisms identified in Subsection 2.3.1 cannot possibly have an effect; some of them may have influence. But they are in no way required to explain the imbalances, and the difference in real unit normal costs between China and the U.S. provides an explanation which is sufficient by itself.

Unlike most comparative advantage models, competitive advantage theory does not require an implicit acceptance of Say's law. This means that while indeed trade liberali-

	CCT	ACT
labour	Immobile	Immobile
Capital	Immobile (or mobile for exogenous reasons)	Mobile (reacts to absolute cost differences with trade)
BoP	$= 0$ (accounting identity)	$= 0$ (accounting identity)
CAB	≈ 0 , subject to intertemporal budget constraints	unconstrained, reacts to ReR
ReR	determined by CAB imbalances	determined by unit normal cost divergences

Table 2.3: Summary of the differences between the CCT and ACT assumptions.

sation will tend to create a pattern of specialisation, there is no guarantee that it leads to an efficient reallocation of resources. Since specialisation happens through a process where domestic businesses either risk being driven to bankruptcy by their foreign competitors or are in a position to grow by appropriating the market share of their foreign competitors, if one does not assume some version of Say’s law, there is no guarantee that the resources and labour used by the outcompeted businesses will be reallocated to other activities. Therefore, we should expect some regions to specialise while others become derelict. ACT theory predicts a pattern of regional specialisation (and, consequently, of increased trade) which matches the stylised facts of fragmented production and increased trade. The rejection of both premises of comparative advantage theory leads to the conclusion that trade liberalisation has a fundamentally ambiguous effect. As Shaikh (2007), p.58 puts it: ‘*what-you-see is what-you-get: free trade does what it is supposed to do, namely reward the strong*’. This means that exposure to intense foreign competition is now understood as a likely cause for mass unemployment both of labour and capital, which is consistent with the history of what is known as the first ‘*China Shock*’, an event of mass unemployment in the United States, caused by intense foreign competition, mainly from China, following its opening to trade (Autor, Dorn, and Hanson 2016).

2.5 Conclusion

At this point we have reached the following conclusions. The mainstream theory of trade (CCT) faces a fundamental issue when attempting to explain the reality of trade imbalances. Because it assumes the existence of a ‘self-righting’ mechanism balancing out

trade, large trade imbalances such as the U.S.-China imbalance require the introduction of *exogenous explanations*. We reviewed the three main strands of such *exogenous explanations*. Some are credible, some are less, but we have a more fundamental disagreement with them. We argue that the basic premises of CCT do not hold, specifically, the mechanism which brings about balanced trade does not exist. Instead, we argue in favour of ACT theory, which does not posit the existence of such a mechanism. ACT predicts that trade should be expected to be imbalanced as a result of disparities of competitiveness and expects that the balance of payments should be equilibrated through international flows of capital. This tracks well with what is currently happening in the relationship between the USA and China.

This chapter started by stating four basic stylised facts about trade under ‘*globalisation*’: (1) U.S.-China trade is deeply imbalanced; (2) the volume of trade and international investment across the world has increased fast (relative to World GDP) in the period between the mid-1980s and the 2010s; (3) over that period of time production has become more and more fragmented; (4) since the 2010s the increase in the volume of trade and international investment has been stalling. We have shown how the first of these stylised facts raises a fundamental issue for all theories constructed on the assumptions of CCT. The second and third stylised facts are consistent with both theoretical frameworks. The fourth is explained by neither and we will come back to it later. We have also shown that ACT provides an alternative foundation for a theory of trade which overcomes this difficulty.

We thus have very good reasons to adopt ACT instead of CCT as the foundations of our upcoming discussion of trade imbalances and international competition. But the adoption of ACT alone merely explains the trade deficit, and does not suffice to construct a solid understanding of the question at hand. This is because trade relations cannot be properly understood when considering competitive advantage at the aggregate level. Knowing *how much* is being traded (the trade deficit) is not enough; we also need to look at *what* is being traded. This is the object of the next chapter, which discusses why some configurations of trade push profitability up while others push it down.

Chapter 3

Trade Imbalances, Profitability and Offshoring

At this point, we have an explanation for why persistent trade imbalances exist: because of disparities in competitiveness (proxied by unit normal costs) of firms across borders. But this does not resolve the issue which we first touched upon in the introduction, namely that the trade deficit with China was successively welcomed and opposed by representatives of U.S.-American businesses. We argue that this is likely the case because the nature of Chinese and U.S.-American imports and exports has changed; because the imports coming from China went from supporting U.S.-American profits to threatening them. To understand how this is possible, let us move beyond trade imbalances and look into the relationship between trade and domestic firm-level profitability.

The upcoming chapter is constructed as follows: first, we briefly discuss the internationalised character of production in the modern economy. Second, we go over some basic accounting definitions of different measures of profitability and discuss why the profit-rate is such a central variable for capitalist firms. Finally, we turn our attention to the key question at hand, that of relating firm profitability to international trade and transnational activities of corporations (namely offshoring and exports) by constructing a simple model which illustrates the main effects that interest us.

3.1 International Production

	Task performed within the firm	Task outsourced to another firm
Domestic	Domestic in-house production (Q1)	Domestic outsourcing (Q2)
Foreign	Foreign direct investment (Q3)	Foreign outsourcing (Q4)

“Outsourcing”

“Offshoring”

Figure 3.1: The location and integration decision from the perspective of the firm. Plot by the author.

In Chapter 2, we have touched upon the fact that contemporary production is increasingly fragmented in complex production networks (this was our third stylised fact: the geographical splintering of production), which organise the production process across borders. The *transnational* corporation (TNC) is the organisational form of enterprises which co-ordinate such production networks. In her textbook dedicated to the matter, Grazia Ietto-Gillies (2019) insists that TNCs¹ are not only companies which operate in multiple countries, but companies which co-ordinate production across borders. They do so through diverse modalities, which, following Dunning (2000), can roughly be categorized along two axes (see Figure 3.1). First, firms make an *integration decision*: should the firm perform a given task within the firm itself (perform the task in-house) or should it *outsource* it to a subcontractor? Second, they make a *location decision*: answering the question of whether a task should be performed domestically (in the same country as the

¹She refers to *transnational corporations* instead of *multinational corporations* for that reason, as the prefix *trans* puts emphasis on ‘*their ability to manage, control and develop strategies across and above national frontiers*’ (Ietto-Gillies 2019, p.13), which is what distinguishes TNCs from other firms.

firm’s headquarters) or abroad; in the latter case it is said that the task is offshored. In practice, modern TNCs use the full range of configurations.

The central question of transnational production lies in constructing an understanding of the decision process underlying the firm’s growth and extension across borders. Expanding across borders has to be conceptualised as a decision taken in the process of firm expansion, instead of a decision to expand or diversify domestically (Kay 2006). This decision is taken under constraints brought about by the firm’s organisational growth limits and by its capacity to secure financing, and is directed by the *specific advantages* which the firm holds. Such *specific advantages* can be constituted of *intangible assets* (technological or organisational capabilities, intellectual property, branding) or can be directly related to the localisation of its new activities (geographical or institutional context facilitating firm expansion) (Horstmann and Markusen 1989). For instance, early investments in China after the country opened its doors to foreign investments were dominated by companies owned or managed by members of the Chinese diaspora, these companies enjoyed the *specific advantage* of having managers with a good mastery of the Chinese language and an already existing social network in China. This enabled them to circumvent regulations and navigate the burgeoning Chinese market (Burbaumer 2024, p.89).

In the subsections which follow, we first discuss foreign direct investments (Q3, in Figure 3.1) before turning to the practice of outsourcing abroad (Q4, in Figure 3.1), and domestically (Q2, in Figure 3.1).

3.1.1 Foreign Investment

The very possibility of offshoring ‘in-house’ is conditioned by ownership of capital abroad, which takes the form of foreign investments. There are two types of foreign investments: foreign direct investments (FDIs) and foreign portfolio investments (FPIs). The distinguishing factor between both types, according to the IMF’s definition, is that FDIs correspond to a 10 per cent or more equity stake being taken in a foreign company, whereas FPIs correspond to a smaller stake, or to a non-equity investment, for instance bonds. The rationale behind distinguishing foreign investments according to equity stakes is to

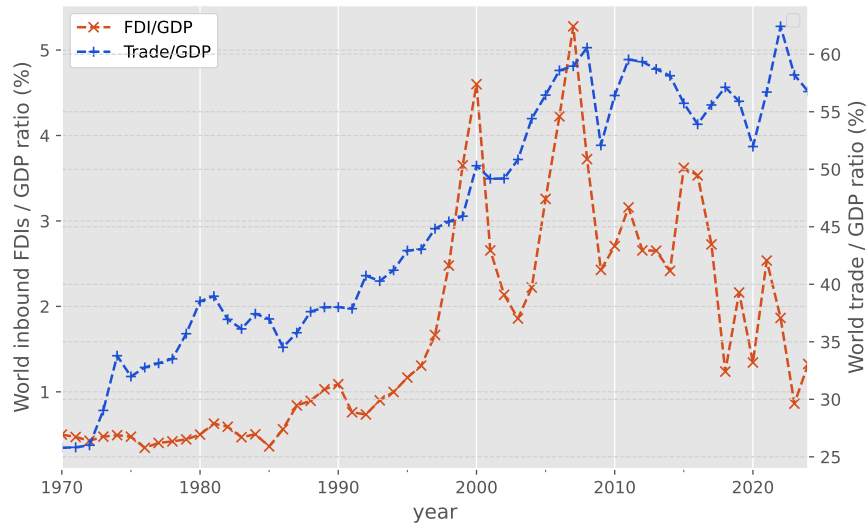


Figure 3.2: Total inbound FDI to GDP ratios and trade to GDP for the entire world over time. Data from the World Bank (2026b), plot by the author.

distinguish investments which correspond to a relatively high degree of control over the foreign firm from those which do not. FDI's are thus investments that have a double motive: bringing profits to the parent firm as well as controlling the company which receives the investment, for instance to integrate it in transnational production (Ietto-Gillies 2019, pp.23–24). The emergence of international production in its contemporary form has been concomitant with an increase in the volume of FDI's relative to GDP (Figure 3.2 until the 2010s). One of the telltale signs of the current stalling of the globalisation process, which we introduced in the previous chapter, is the reduction in volume of new FDI's (Figure 3.2 from the 2010s onwards) and the plateau of the volume of trade over time (both figures are taken relatively to world GDP). This is explained by the fact that FDI's serve to co-ordinate global production, which itself increases the volume of trade relative to GDP.

The attention to control in foreign investment can be traced back to Steven Hymer (1970)'s seminal work on TNCs and relates to the interpretation that FDI's are *strategic* investments that are usually associated with the expansion of the parent company's supply network, the acquisition of strategic assets or the removal of a competitor (these are the motives identified by Hymer). While the arbitrary threshold of 10% for identifying FDI's is useful to characterise which part of foreign investment may be deemed *strategic* at the

aggregate level, more granular characterisations of foreign equity control are also in use. It is usually said that a foreign firm owned with a 10% to 50% stake can be called an *associate* of the *parent firm* while a firm owned with a 50% or more stake is called a *subsidiary*. The distinction, of course, being that if the stake is 50% or higher, the parent firm has complete guaranteed control over its foreign investment, whereas control may be more limited otherwise. In principle, FDIs can be made by the corporate sector as well as individuals and state actors, but in practice, the largest volume of FDIs is attributable to the corporate sector, and thus to TNCs since a single FDI is enough to make a firm a TNC.

Historically, equity control was not always sufficient to allow TNCs to co-ordinate the activities of their subsidiaries or associates; ownership did not automatically imply *managerial control*. This changed with the emergence of modern information and communication technologies (ICT), which are therefore understood to have been instrumental in the birth of the modern international production network (Ietto-Gillies 2019, Ch.1–2). Here it is useful to introduce a theoretical distinction, first proposed by Charles Bettelheim (1975) between *possession*: the capacity to control the productive process; *economic ownership*: the capacity to allocate resources (crucially, means of production) to production and to capture profits; and *legal ownership*: holding the formal property title.

Equity control in the contemporary world implies some degree of *economic* (some profits are recovered by the company owning the equity) and *legal ownership* (the company legally holds part of the property title to the subsidiary) as well as *possession*. The owner company can direct the productive process of the foreign firm it owns, hence we say it possesses or partially possesses the subsidiary or associate. This is not the case if we consider the form of equity which existed before efficient long-distance communication, or if we consider portfolio investment. In both of these cases the firm which has invested abroad can be said to have *legal* and *economic ownership* without *possession* of the foreign (partially) held entity.

3.1.2 Outsourcing and Global Value Chains

While FDIs have been understood to be, and arguably are still, the main defining modality of the transnational operations of TNCs, there are others. UNCTAD (2011) refers to such modalities of TNC operations as *non-equity modes* (NEMs). The list includes but is not limited to: *contractual relationships*, where TNCs integrate contractors which are not *associates or subsidiaries* in their production networks, licensing, franchising, and strategic alliances (for instance joint ventures, which is an aspect we will come back to in Chapter 5) (Ietto-Gillies 2019, Ch.2). Here we will focus on the broad practice of *outsourcing* parts of a TNC's activity, in other words, of a TNC deciding to *buy* an intermediate product instead of *making* it. This modality of production has become more and more prevalent since the late 1990s (Miroudot and Ragoussis 2009). Seemingly paradoxically, a process of *vertical disintegration*, where TNCs abandon parts of the production process to subcontractors, has happened simultaneously as *the level of international economic integration has increased* (Milberg and Winkler 2013, Ch.2). In other words, TNCs have become, at the turn of the 2000s, more and more reliant on foreign subcontractors for carrying out (partially or completely) their production activities, while at the same time production has become more and more spread out geographically and more and more interdependent across borders.

This phenomenon, which was termed '*trade in tasks*' by Grossman and Rossi-Hansberg (2008) and '*the great unbundling*' by Baldwin (2006), is perhaps best explained by illustration. In a seminal work, Linden, Kraemer, and Dedrick (2009) dissect the production chain of one of the early twenty-first century's most archetypal commodities: the iPod. The device is made up of about 400 individual components, which are purchased and assembled without going through a single Apple-owned manufacturing plant. The entire production is outsourced until the point where the finished products are shipped to Apple distributors. What Apple handles is R&D, design, branding, marketing and, to some extent, distribution. This makes Apple a so-called *factoryless producer*: a manufacturing firm which has completely forgone ownership of any manufacturing facility. Even if it is completely dependent on subcontractors, Apple is extremely profitable. In the entire production network of the iPod, it is the company which extracts the highest margin in the entire operation: \$80 over the \$299 price of the iPod model of interest, with the rest of

the value being captured by various subcontractors in East Asia, for instance, by Toshiba in Japan for the manufacturing of the hard drive. While Apple’s production network is a bit of an extreme example of the disintegration of production² it is a good illustration of the two ‘*seemingly incongruous tendencies*’ (Milberg and Winkler 2013, p.123) which characterise modern globalised industry: increasingly disaggregated production networks with apparently less market concentration simultaneously with extremely high margins for a small set of firms at the top of the food chain. This is an example where the *lead firm* does not own legally the firms which do the actual production: they are neither *legal owners* nor formally *economic owners* but in which they *possess* the subcontractors (in the sense of Bettelheim (1975)).

3.1.3 ‘Unequal Capture of Value’ in Global Value Chains

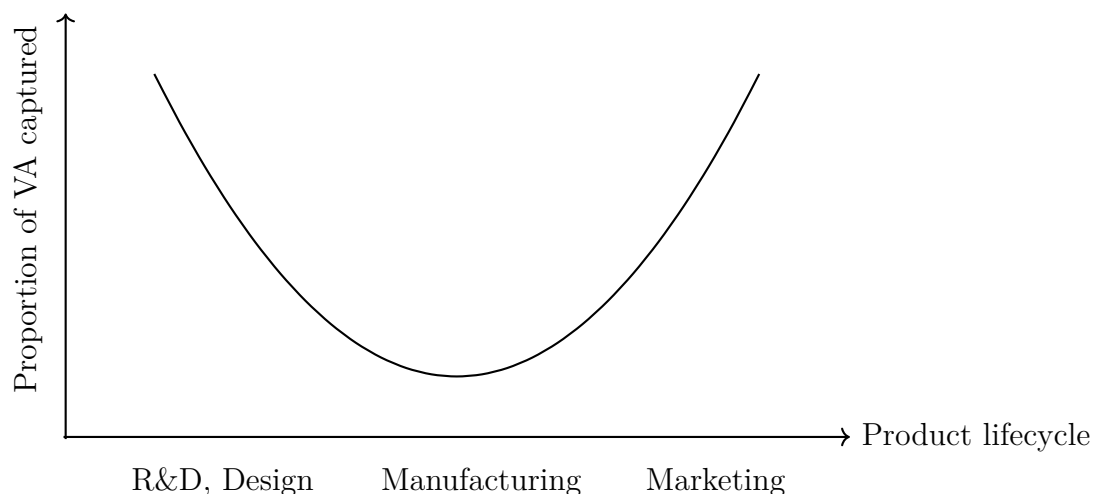


Figure 3.3: The so-called ‘smiling curve’ which schematically describes the notion that often in international production, more value added is captured at both ends of the product lifecycle (relative to the full, finished commodity price).

This pattern, whereby a disproportionate share of the value added is captured ‘at both ends’ of the product lifecycle: either in the early stages, such as R&D and design, or in the latest stages, such as distribution and marketing, at the expense of the ‘middle stages’ of production and distribution, has been referred to as a *smiling curve*³ of value

²As Milberg and Winkler (2013) note, in many cases internationalised production still largely goes through wholly owned subsidiaries abroad.

³The idea of the *smiling curve* is attributed to Acer Inc.’s founder Stan Shih. Acer is a company

added capture (Figure 3.3). It is not an isolated pattern; it has been identified in many company-level case studies (Linden, Kraemer, and Dedrick 2009; Shin, Kraemer, and Dedrick 2012), using large datasets of firm financial data (Rungi and Del Prete 2018), and at the macroeconomic level, using input-output methods (Meng, Ye, and Wei 2020).

In the context of this work, we are interested in how the structuration of international production along GVCs relates simultaneously to imbalances and to the formation of the profit rate. The phenomenon of asymmetric capture of value-added, which is what the *smiling curve* illustrates, is central to that question, and we must ask ourselves what conditions make the *smiling curve* possible. Elements of an answer are provided in Milberg and Winkler (2013), Ch.4, who discuss the structuration of GVCs in relation with the strategy of *lead firms*: the firms which co-ordinate production in the production network (for instance, Apple, in the iPod example). Their argument is one of oligopolistic competition. They note that GVCs are structured in such a way that at the level of *lead firms*, the markets are highly concentrated and that at the level of subcontractors they are not. TNCs are thus oligopolistic sellers of final goods and oligopsonistic buyers of intermediate products (Milberg and Winkler 2013, pp.111–123).

At the level of subcontractors, the degree of competition is understood to be very high, which has the effect of cutting the prices of intermediate inputs down, reducing the profit rate of the subcontracting firms, which reduces their ability to challenge the *lead firms* (Milberg and Winkler 2013, pp.116–122), and of splintering the labour force across different companies, lowering its bargaining power and, consequently, real wages (Nathan and Sarkar 2011). This stylised fact is supported by the observation that the intensity of competition seems to be higher in the *periphery* than in *core* countries (Glen, Lee, and Singh 2002). At the level of the lead firms, on the other hand, it is understood that product diversification, branding and accumulation of intangibles play a role in raising the barriers to entry to the final goods market, increasing the oligopolistic character of competition (Milberg and Winkler 2013; Durand and Milberg 2020). This leads to a situation where markups for *lead firms* are consistently high, whereas they are not for their subcontractors. The unequal share of value added can be reduced to unequal ability to raise markups and corresponds to an unequal ability to raise profits. This is a point

which, like Apple, has come to completely rely on a network of arm's length relation subcontractors to handle the production of the commodities it sells.

which we will come back to in Chapter 4.

3.2 Profits, Profit-Shares, Profit-Rates

Profits can either be defined at the level of the firm or at a macroeconomic level, where the profits of all firms in a given geography are aggregated. At the firm level, measuring profits is an accounting question and corporate accounting admits multiple measures for profits. One very common measure is *earnings before interest and taxes* (EBIT), which, as its name suggests, measures profits before taxes and before interests are paid (Berk and DeMarzo 2024, Ch.2). It is computed, for a firm k at time t , as:

$$\Pi_{k,t} = s_{k,t} - w_{k,t} \cdot l_{k,t} - \chi_{k,t} - \delta K_{k,t}, \quad (3.1)$$

where $\Pi_{k,t}$ denotes the EBIT, $s_{k,t}$ the sales, $w_{k,t}, l_{k,t}$ the labour hourly cost and labour hourly expense (their product gives the labour cost), $\chi_{k,t}$ the cost of inputs, and $\delta K_{k,t}$ the capital depreciation allowances.

At the aggregate, national, level, for a given time-step t the following identity relates aggregate profits (Π_t), aggregate labour compensation (W_t), the total value added produced in an economy (Y_t) and the current account balance in a country's economy (Mitchell, Wray, and Watts 2019, p.412):

$$\Pi_t = Y_t - W_t + CAB_t. \quad (3.2)$$

This measure of aggregate profits corresponds to the aggregate of the EBIT of firms in the country of interest. Unlike EBIT at the company level, Π_t is simply computed by deducting employee compensation from the economy's total value added Y_t . No input costs, overhead costs or capital depreciation are accounted for, except those which are part of the current account. This is because in a closed economy, some company's input or overhead costs are another's earnings (deducting intermediates and overhead costs would constitute a fallacy of composition).

The main reason we care about profits when characterising economic behaviour is because

profits are the main driver for decision-making in a capitalist economy. Capitalists (or *entrepreneurs* in the language of Keynes) who are ‘*at the helm*’ of the capitalist economy conduct their activity seeking to extract profits. They invest with the specific intent on turning a profit on the capital they invested. This idea is perhaps best understood when presented following Marx’s famous scheme of capital accumulation:

$$M - C - M', \quad M' \gg M,$$

according to which a capitalist mobilises money-capital, which they invest (acquiring input equipment, commodities and paying wages) in order to produce commodities (goods or services), to be sold for profit (Marx [1867], 1887, Ch.4). Upon that sale, and if all goes well, the capitalists retrieve a higher quantity of money-capital, turning a $M' - M$ profit on the operation.

Here what is central is that the capitalist aims to ensure a return on its investment, which is, in principle, as high as possible. This notion, of how much profit is returned on a given investment, is measured through the idea of a profit rate. Suppose a capitalist invests a quantity I_p in a project which, over the course of its lifetime, returns a profit Π_p . Then the return on investment of the capitalist over that specific project p , the profit rate of that project, is given by $r_p = \Pi_p/I_p$. The investment which yields the highest profit rate is simply the one which yields the highest profits, assuming finite access to funding. This is why it is understood that the fundamental logic of capitalist investment is to strive to maximise the profit rate, not the volume of profits or the profit share (although they are related).

The return on investment Π_p/I_p is a sensible way to understand ‘*how worth it it is*’ to invest in a one-off project, but, roughly speaking since the eighteenth century, firms have an indefinite lifetime and maintain permanent capital. This complicates the computation of the profit rate, since, as firms receive a flow of investment over the long run, it becomes impossible to measure return on a specific investment.

The way this difficulty is overcome is through the notion that investment in a firm becomes ‘*immobilised*’ for a certain purpose. In modern accounting we refer to this investment

(the monetary value of the immobilised things) as a *capital stock*⁴, which, at the level of the firm, but also at the aggregate level is accounted for through a method of ‘*perpetual inventory*’. The idea is that at any given time, the *capital stock* which had previously been invested loses value, due to wear and tear, the need for replacement etc. (this is called the depreciation of capital) and new capital gets added to the stock (in the form of investments). The stock of capital (K_t) is thus simultaneously depleted and increased (depending on the investment and depreciation flows). At the aggregate level, the perpetual inventory method for measuring the capital stock is expressed as follows:

$$K_t = K_{t-1} + I_t - Z_t, \quad (3.3)$$

where K_t denotes the capital stock, I_t investment and Z_t the depreciation, all at the same time t , K_{t-1} denotes the capital stock in the previous time period. One common special case is the case where it is assumed that the capital stock depreciates at a fixed rate $0 < \delta < 1$, in this case depreciation can be rewritten as $\delta \cdot K_{t-1}$ and the perpetual inventory method can be reformulated as:

$$K_t = I_t + (1 - \delta) \cdot K_{t-1}, \quad (3.4)$$

which, in the absence of new investments describes a geometric degrowth of the capital stock. Regardless of the flavour of the perpetual inventory method which is used, it is required that investment and depreciation be expressed in real terms (properly adjusted for inflation) for the capital stock to be meaningfully interpretable. Otherwise, we would be adding together capital investments in ‘different units’. Once we have constructed this measure for the stock of capital we can use it to compute our profit rate, which is given by:

$$r_t = \frac{\Pi_t}{K_{t-1}}. \quad (3.5)$$

By convention in that kind of computation the capital stock is lagged by one time period (usually a year); this reflects the notion that the profits earned on year t are the results of investments made before that year (some of the investments contributing to K_t have likely

⁴It should be noted that this accounting definition of capital (stock) does not match the Marxian conception of capital.

been made after some of the profits earned during that same year). The definition of the profit rate given by (3.5) roughly corresponds to the definition of return on assets (ROA) used in finance (and measured over portfolios of financial assets), the only difference being that in the finance formulation of ROA, would account dividends as revenue as well as profits from changes in valuation of securities (Berk and DeMarzo 2024, Ch.2). Here it is relevant to introduce a useful decomposition of the profit rate, sometimes known as Weisskopf's identity (Lavoie 2022, p.378):

$$r_t = \pi_t \cdot u_t \cdot \kappa_t, \quad (3.6)$$

where the profit rate r_t is decomposed by the product of the profit share $\pi_t = \Pi_t/Y_t$, the rate of capacity utilisation $u_t = Y_t/Y_t^{\max}$, and the capital intensity of production (which is defined at maximum capacity) $\kappa_t = Y_t^{\max}/K_t$. In other words, the parameters influencing the average profit rate are: 1) the markup of the representative firm (which determines the profit-share), 2) the rate of capacity utilisation, 3) the output-to-capital ratio (at engineered capacity). This expression clarifies in which way there can be such a thing as *overinvestment*. Indeed, if more capital is invested than is required to meet demand, the rate of capacity utilisation u_t ratio will fall (as some productive capacity becomes unused). Furthermore, if investment in new vintages is done under *Marx biased technological change* (substituting labour costs with capital costs) the output-to-capital ratio κ_t is expected to fall as well.

At this point it should become clear why the *aggregate*, inflation adjusted, profit rate holds the importance it does in economics; it is simply a measure of average, economy-wide return on investment. If it increases, it indicates that businesses are more viable (at the aggregate level); if it decreases, the opposite is true. If it goes to 0, it means that every investor's profits happen at the expense of another.

3.3 Trade Imbalances and Firm Profitability

Now that we have clarified what we mean by firm profitability, let us turn our attention back to the relationship between transnationality, trade and the profit rate. Our aim in

this section is to clarify why the effect of a trade deficit on domestic firms' profitability is an inherently ambiguous one.

3.3.1 Supply, Demand and Transnationality

To properly conceptualise how a firm's profitability relates to trade imbalances, let us write out the equation of a firm's profit and decompose terms which relate to international trade. Let $\Pi_{k,t}$ denote the firm's profit, $s_{k,t}$ its total sales, $w_{k,t}$ hourly wages and $l_{k,t}$ the quantity of domestic labour expended in production, and $\chi_{k,t}, \chi_{k,t}^*$ its input costs corresponding to domestic and foreign suppliers⁵ and $f_{k,t}^*$ the revenue from the eventual foreign investments of the firm:

$$\Pi_{k,t} = s_{k,t} - w_{k,t} \cdot l_{k,t} - \chi_{k,t} - \chi_{k,t}^* + f_{k,t}^* \quad (3.7)$$

$$= (1 + \mu_{k,t}) \cdot \overbrace{(w_{k,t} \cdot l_{k,t} - \chi_{k,t} - \chi_{k,t}^*)}^{UNC_{k,t}} + f_{k,t}^* \quad (3.8)$$

When things are laid out in such a straightforward way, it becomes clear that there are three channels simultaneously affecting profitability $\pi_{k,t}$ and the trade balance.

E.1. Demand, an increase in the trade deficit can correspond to a reduction of demand and thus to the sales component $s_{k,t}$ of our equation. For instance because the domestic firm is outcompeted by foreign competitors.

E.2. Supply, an increase in the trade deficit can correspond to an increase in the volume of intermediary products which are imported ($\chi_{k,t}^*$) relative to exports.

E.3. Foreign investments, in the case where a firm invests abroad, it may earn revenue $f_{k,t}^*$ from the activities of its foreign contractors leading to imports in the domestic economy. This corresponds to a trade deficit either measured in the sales of intermediate or of final-demand goods.

At this point, it is clear how two drastically opposed conclusions may be drawn. To a trade deficit may correspond a pure fall in demand (s_t) and hence an unambiguous fall in profitability, or an effect on supply which alters the cost structure of production, possibly

⁵For simplicity we count fixed capital related costs as input costs, so here there are no separate capital depreciation allowances in this section.

increasing the profitability of domestic firms. The fact that firms may invest abroad and repatriate profits generated abroad, possibly from importing into the national economy, opens another pathway for what we may call *profitable trade deficits*. In what follows, we are going to discuss both possibilities as they are approached in the post-Keynesian and Marxian literature.

3.3.2 Competitive Imports

Let us start by considering the simplified case where trade is performed in final goods only, which means that unit normal costs stay mostly unaffected by trade deficits. This is the assumption on which early (post-)Keynesian models of the relation between trade imbalances and profitability were constructed. We will build on them to start our discussion of the issue and show that, under this assumption, there is an unambiguous effect of profitable trade surpluses and costly trade deficits.

First let us note that once we assume all trade is performed in final goods, any trade deficit (surplus) must at the micro-level correspond to foreign (domestic) firms outcompeting domestic (foreign) firms in their own product markets, leading to a ‘demand-leakage’ in favour of foreign (domestic) firms. In other words, part of the domestic demand, which could be profitable for domestic firms if they managed to meet it, is met by foreign actors.

This can be shown at the macro-level with a very simple model, constructed directly from the multiplier equation (Blecker 2002). Let the worker’s savings be 0 (equivalently, assume that workers consume their entire wages) and the capitalist’s consumption be given by a linear function of the form: $C_\pi = \beta_\pi \Pi$. We find that aggregate profits relate linearly to investments, government deficit and trade surpluses as follows:

$$\Pi = \frac{I + (G - T) + (X - M)}{\beta_\pi}. \quad (3.9)$$

In other words an increase $\Delta(X - M)$ in the trade surplus linearly maps to a corresponding increase in the volume of profits $\Delta\Pi$:

$$\Delta\Pi = \frac{\Delta(X - M)}{\beta_\pi}. \quad (3.10)$$

Of course, this model alone does not explain much, if only because it takes the trade balance as an exogenously given parameter, but it does capture the fundamental idea we just outlined that a trade deficit corresponds to a ‘demand-leakage’ which is reflected in profits. This relationship also points to a very significant conclusion from early Keynesian discussions of trade. As a simple consequence of (3.9) and stock-flow consistency, there exists a fundamentally conflictual dimension to trade imbalances. In the words of Kalecki: *‘[t]he capitalists of a country which manage to capture foreign markets from other countries are able to increase their profits at the expense of the capitalists of the other countries.’* (Kalecki 1989, p.85).

One key aspect which is not made obvious by such a simple model is the notion that foreign competition need not lead to a massive trade deficit to significantly reduce profits. This is because, in the case where foreign competitors are in a position to drive prices down and engage in price wars, it should be expected that domestic producers will be forced to drive down prices as well. Therefore, the reduction in profit earnings may be coming from a reduction of the markup as well as from decreased demand. This is a phenomenon which has been well documented in the 1970s and 1980s as competition from upcoming industrial powers (mostly Japan and Germany) was affecting U.S. manufacturing. A classic study in that regard is the study of pricing behaviour of the U.S.-American steel industry by Blecker (1989), which identifies a structural break in pricing patterns in response to foreign competition around the turn of the 1970s. A detailed historical account of foreign price competition over the same time-period can be found in Brenner (2006), Ch.8–9. This is a question which we will come back to in Chapter 4.

There are two other aspects which the simple multiplier model, which I have introduced, does not clearly outline and which are worth mentioning. The first, and perhaps most obvious remark, is that we should not expect *competitiveness* to be uniform across companies and sectors of the economy. Therefore, sustaining a deficit in one sector does not mean that an open economy is suffering from a negative effect on profits overall. This is the central mechanism mobilised by comparative advantage arguments. The second aspect relates to the profit rate. So far, we have characterised how trade affects aggregate demand which in turn affects aggregate profits. Yet, from the perspective of investors, what matters most is the profit rate. The key difference in the dynamics of aggregate

profits and of the profit rate is that the profit rate depends on past capital outlays. This means that an unexpected fall of aggregate demand (as a result of foreign competition and of a trade deficit) risks leading to a larger fall in profit rate than aggregate profits, by virtue of its unexpected character, and of the sunk costs of already-expended capital outlays. Indeed, if companies have invested in productive capacity and there is a negative demand shock from foreign competition, not only does the aggregate profit term Π_t go down, but the capital stock K_t does not, leading to a more pronounced fall of the profit rate $r_t = \Pi_t/K_t$ than the aggregate profits Π_t . This is a point to which we will come back in Chapter 4. It is central in the argument of Brenner (2006), who argues that a foreign competition shock in the 1970s drove profit rates down significantly and led to a durable slowdown in growth (what he calls ‘*The Long Downturn*’).

Together these observations lead to an understanding of trade as fundamentally being a zero-sum game, a view which is very much opposed to the comparative advantage story. This perspective is quite common amongst the founders of post-Keynesianism, for instance Joan Robinson writes that:

‘[...], everyone is keen to sell and wary of buying. Every nation wants to have a surplus in its balance of trade. This is a game where the total scores add up to zero. Some can win only if others lose. The beautiful harmony of the free-trade model is far indeed to seek.’ (Robinson 1979, p.205).

More recent post-Keynesian models of trade, such as those presented by Blecker (2002), showcase this central property: one country’s surplus and corresponding above-normal profits from trade correspond to another country’s deficit and correspondingly below-normal profits. At the aggregate, macroeconomic level, there is of course a third variable to be taken into account, that of government deficit, which can compensate the effect of a trade deficit by increasing government debt. Similar arguments have been made by Marxist scholars (Brenner 2006; Shaikh 2016).

Early post-Keynesian models such as the one we just presented are consistent with an understanding of trade being first and foremost a question of *competitive advantages*. They predict that if the firms in a given country manage to outcompete the ones in another, this simultaneously results in trade imbalances and pressure on profits. But

this argument is constructed on the hypothesis of no trade in intermediate goods and no direct investment abroad by domestic firms. Thus, it reaches its limits in a world such as ours where these assumptions break down. This is an omission which we need to address, and it is the topic of the following section.

3.3.3 Importation of Intermediates and Firm Profitability

As soon as the possibility for importing intermediate goods and for collecting revenue from foreign production opens – in other words, as soon as we accept that transnational production is a possibility – the picture becomes more nuanced.

In what follows, we first discuss the implications of introducing cheap intermediate goods from foreign-owned companies (which corresponds to outsourcing-offshoring) in a model with a single representative firm. We will see that this scenario opens the door for the same firm-level decisions to prop up profitability while deepening a trade deficit.

We build our model⁶ starting with the following price equation:

$$p = (1 + \mu)(wa + p_m a_m), \quad (3.11)$$

where p is the price level, μ the markup, w the domestic real wage, a the domestic unit labour requirement, p_m a price index for the imports of intermediate goods and a_m the unit import requirement. This is a simplified model with no explicit cost of fixed capital investment and no domestic intermediate goods, where we consider a representative firm for a national economy which is reliant on importation of intermediate products. Since the firm-level equation of (3.11) applies for a representative firm, it is rather straightforward to derive the domestic profit-share in a Kaleckian fashion:

$$\pi = \frac{\Pi}{W + \Pi} = \frac{\mu wa(1 + z)}{\mu wa(1 + z) + wa} = \frac{1}{1 + (\mu(1 + z))^{-1}}, \quad (3.12)$$

where $z = (p_m a_m)/(wa)$ is a ratio of the unit material costs of intermediate imports to the

⁶This presentation draws on the presentation of the outsourcing offshoring model of Schröder (2020), but diverges in a few key ways; it also takes hints from Woodgate (2023). Equations here have been derived by the author; detailed derivations are left to Appendix C.

unit labour costs. It is assumed that $p_m a_m < w a$, which is just the necessary condition for the decision to offshore to be guaranteed to reduce costs. It should also be mentioned that this inequality in pricing is essentially a very simplistic model of the asymmetric capture of value-added, which we previously discussed. Let us define the *offshoring ratio* $\zeta = a_m/a$, which is trivially positively related to the ratio z . Now for the level of aggregate demand, let us write down the three components of aggregate demand:

$$Y = (C + I) + (X - M) + (G - T), \quad (3.13)$$

where C is aggregate consumption, I aggregate investment, X exports and M imports. Following Schröder (2020), let us assume w.l.o.g. the government runs a balanced budget and write out $A = C + I$. Further assuming that the three remaining components of the left-hand side of (3.13) ($A + X - M$) react linearly to an increase in the offshoring ratio ζ , we get the following relation between aggregate demand and the *offshoring ratio*:

$$Y = Y_0 + \zeta \cdot (\beta_X - \beta_M - \beta_A) \quad \beta_X > 0, \beta_M > 0, \beta_A > 0. \quad (3.14)$$

Where Y_0 is the level of aggregate expenses assuming no offshoring ($\zeta = 0$). Depending relatively on how strongly exports, imports and domestic demand react to an increase in the *offshoring ratio*, the level of domestic expenditures may increase or decrease (specifically it will increase if $\beta_X - \beta_M - \beta_A > 0$).

Formulated in such a manner, offshoring can be understood as a ‘*labour-saving import-using technical change*’ (Schröder 2020). Substituting domestic production with foreign imports (equivalently increasing the offshoring ratio ζ) has two effects: first, on the profit share (and correspondingly the profit rate) as per (3.12) and second, on the level of aggregate demand, as per (3.14).

It is rather straightforward to derive the condition under which an increase in the *offshoring ratio* leads to an increase in aggregate profits *and* to an increase in the trade deficit. To see how, note that the condition for aggregate profits, given by:

$$\Pi = Y \cdot \pi, \quad (3.15)$$

to increase with the *offshoring ratio* is met when (detailed derivation for this inequality in Appendix C):

$$\frac{\partial}{\partial \zeta} (Y \cdot \pi) = Y \cdot \frac{\mu p_m / w}{(1 + \mu (1 + \zeta \frac{p_m}{w}))^2} + \pi(\beta_X - \beta_M - \beta_A) > 0. \quad (3.16)$$

This can be affected by an exogenous increase in the level of aggregate demand, in the form of government deficit. The way this model behaves is as follows (see Figure 3.4). There are three configurations which depend on the choice of parameters $\beta_X, \beta_M, \beta_A$. In one case increases in offshoring decrease the aggregate profits and aggregate demand. In one case, the effect on aggregate profits is ambiguous while aggregate demand unambiguously decreases with increases in the offshoring ratio. In the last case aggregate profits and demand both increase. The effect on the profit share is not ambiguous: it always increases with the *offshoring ratio*.

The main merit of such a model is that it provides a synthesis of the dynamics which are playing out. Let us outline a few of its conclusions. Replacing domestically produced intermediate goods by cheaper imported alternatives has an unambiguously positive effect on the profit-share, and an ambiguous effect on aggregate profitability. If offshoring depresses domestic demand too much, it may reduce aggregate profitability while increasing the profit share (and possibly the profit rate) (Schröder 2020).

There are two possible channels through which these intermediates may affect profits. Either the reduction in unit normal cost allows for cutting down prices, which may lead to an increased market share by domestic firms. Or the reduction in unit normal cost allows for increasing the markup at level prices. These are termed a *scale effect* (if prices go down, enabling higher domestic and foreign sales) and a *markup effect* (if prices stay level and markups go up) by Milberg and Winkler (2013), pp.152–156. It should be mentioned that the ‘markup effect’ may be reinforced by an effect on wages. Indeed, offshoring reduces the demand for labour, and introduces the possibility of threatening to offshore, which should be expected to lead to wage stagnation (Harrison 2002).

These two possibilities open up the door for multiple outcomes regarding the relationship between trade imbalances and the *offshoring ratio*. If importing cheap foreign intermediaries allows for price reductions, it may be that it ends up propping up the competitiveness

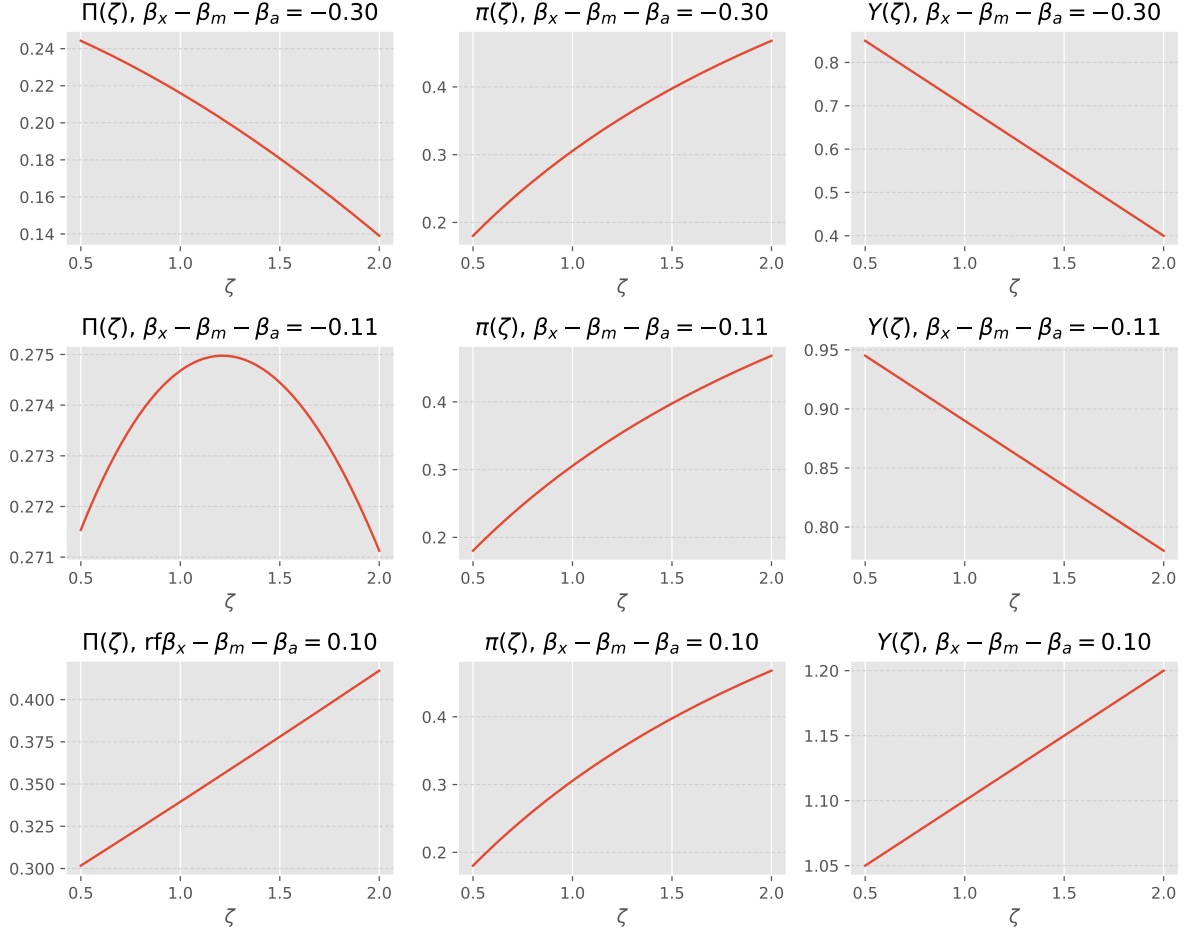


Figure 3.4: Plots of the relation of the aggregate profits Π , the profit-share π and the aggregate demand Y to *offshoring ratio* ζ (in columns from left to right) under different parameters $\beta_X, \beta_M, \beta_A$. The parameters are from the top to the bottom row chosen such that $\beta_X - \beta_M - \beta_A = -0.30$, $\beta_X - \beta_M - \beta_A = -0.11$ and $\beta_X - \beta_M - \beta_A = -0.10$. The remaining parameters are set across all examples to $Y_0 = 1$, $\mu = 0.2$, $w^*/w = 0.2$.

of national firms, which may lead to an increased trade surplus. This is ‘the German scenario’ where part of the domestic industry relies on offshoring to deliver high-value added goods that then get exported, leading to a net trade surplus. On the other hand, if it does not, it may lead to an increased trade deficit (or reduced surplus), in a way which may or may not threaten aggregate profits.

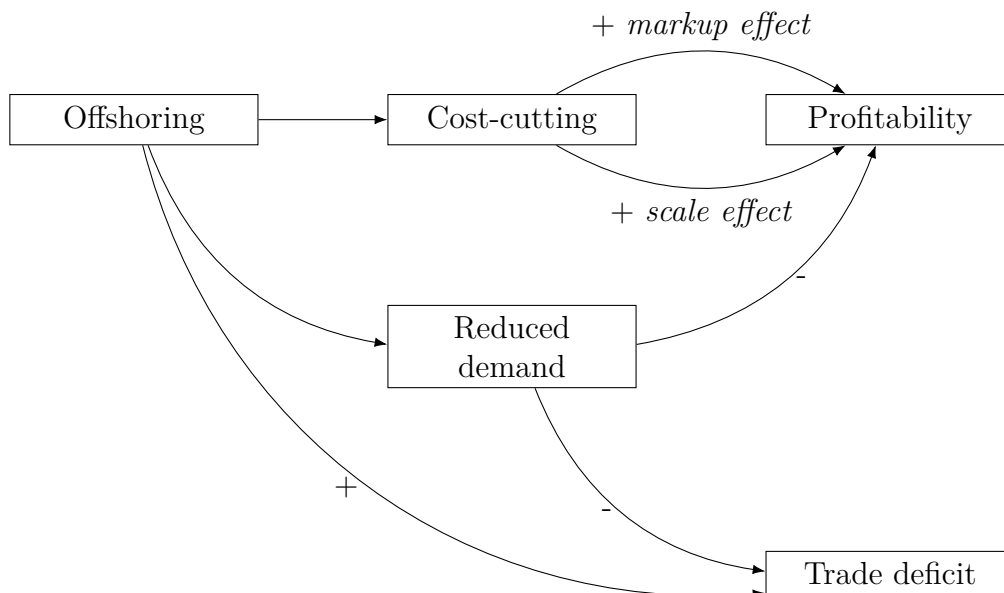


Figure 3.5: Summary schematic of the relation between offshoring, profitability and the trade balance. The effects are the following. Offshoring has two direct effects: it cuts costs for TNCs but affects domestic demand through its effect on wages. The cost-cutting alone unambiguously affects profitability positively through *scale* and *markup* effects. But the reduced demand depresses profits, which has the contradictory effect. On the other hand, the reduced demand may have a moderating effect on the trade deficit if it translates into reduced consumption. Altogether this leads to the multiplicity of regimes which we describe in this chapter.

This picture is not heavily complicated by the introduction of in-house offshoring. The main difference between in-house offshoring and outsourcing offshoring at the macroeconomic level is that while outsourcing offshoring involves a ‘profit-leakage’ (subcontractors take a share of the profits that would be captured by the importing firms), in-house offshoring does not (if imports are cheaper because of cheaper inputs abroad, the increased profits are all appropriated by the importing TNC). Conversely, some of the effects on wages through workforce fragmentation may not apply and sunk costs are obviously higher for the in-house offshoring firm. The contradictory tendencies of increased markups and demand leakage persist (Woodgate 2023).

The implications of in-house offshoring at the macro-scale provide a resolution to the often discussed *profit without accumulation paradox* (Durand and Gueuder 2018), which designates the apparent loss of correlation between profit level or rates and investment in fixed capital in core economies⁷. While most approaches to the paradox focus on the *problématique* of financialisation, Woodgate (2023) justly remarks that in-house offshoring alone is in principle sufficient to provide a resolution to the paradox.

One other aspect related to offshoring (in-house, or outsourcing) is a change in the capital intensity of production. Sufficiently low wages abroad may allow for transnational corporations to substitute more labour-intensive techniques with capital-intensive ones, propping up the profit rate. This corresponds, from the point of view of the firm, to the inverse of *Marx biased technological change* (Blecker and Setterfield 2019, pp.80–83). We may term this effect a *capital substitution effect*.

Lastly, let us remark that there clearly exists in (3.16) the possibility for a situation where an increase in the *offshoring ratio* simultaneously creates a greater trade deficit and increases aggregate profits. This is the case when $\beta_X - \beta_M < 0$ and $\frac{\partial}{\partial \zeta} (Y \cdot \pi) > 0$ are both true. This plausibility of a *profitable trade deficit* scenario is sometimes mentioned in the literature on international flows of profits and on transnational corporations (Ali 2015). It clearly corresponds to the kind of patterns of internationalization of transnational firms which we discussed in Section 3.1. The conditions under which this happens relate to the degree to which increasing the import dependency of the national economy increases the leakage of demand as well as the degree to which profitability increases with the *offshoring ratio*. Furthermore, let us remark that firms are in competition with each other and act to maximise their firm-level profitability, not the aggregate profitability. Hence, even if outsourcing reduces aggregate profits, it may be that individual firms still offshore as it is guaranteed to increase their own profitability (their profit rate and share) *ceteris paribus* even if it lowers aggregate demand and thus aggregate profitability.

The possibility of both a *profitable trade deficit* regime and of a foreign competition compressing profits helps to understand the ambiguous relationship between trade, trade-policy and business profitability. In the context we are interested in for this thesis, that of the China-U.S. rivalry, it is particularly enlightening. One key contribution we

⁷This stylised fact is arguably less descriptive of reality in the era of massive AI-driven investments.

must outline is that of Milberg and Johnston (2025) who propose to distinguish between *capital-supporting imports* (imports which raise the profitability of domestic firms) and *capital-competing imports* (imports which lower the profitability of domestic firms). In their contribution Milberg and Johnston (2025) investigate whether imports into the US have been profitable or not for U.S.-American companies by studying the relationship between the profit share across industrial sectors and the degree of import penetration of Chinese manufactured products in corresponding sectors. What they find is that while higher import penetration from Chinese goods was, in most sectors, associated with a higher profit share during the 2002–2010 period, the tendency is much less clear, and appears to have reversed in many sectors for the 2011–2023 period. They also find that while increased offshoring intensity (computed at the sectoral level, from input-output tables) raised the profit-share in the 1998–2006 window, there is no clear effect anymore since 2006. This apparent exhaustion of the *profitable trade deficit* regime suggests that the capacity for firms which are offshoring to make abnormally high profits is waning and warrants further investigation.

3.4 Conclusion

	Foreign competition	Offshoring
Effect on $X_t - M_t$	Decreases	Ambiguous
Effect on π_t	Decreases	Increases
Effect on Π_t	Decreases	Ambiguous
Effect on r_t	Decreases	Ambiguous

Table 3.1: Summary table of the expected effects of foreign competition and the two types of offshoring which we discussed in this chapter on our variable of interest: the trade balance $X_t - M_t$, the profit-share π_t , aggregate profits Π_t and the profit-rate r_t .

Summing up (see Table 3.1), relating firm profitability to trade imbalances reveals a complex relationship between the two. There are two main causes which may be expected to create a trade deficit: competition from abroad reducing the demand of domestic firms or the decision to offshore (whether in-house or not) taken by these same domestic firms. In the former scenario, the effect of increased competitive pressure on the trade deficit and profitability is unambiguous, and we should expect that a larger trade deficit indicates

pressure on domestic firms' profits. In the latter scenario, the effect on the profit share for domestic firms is unambiguous, but this cannot be said for aggregate profits and the profit rate. This is because substituting domestic with foreign labour simultaneously reduces unit normal costs, domestic demand and possibly changes which production techniques are viable, affecting the capital requirements of production.

At this point, the limitations of the trade deficit as an economic indicator are clear, and we have seen that a similar degree of trade deficit can translate into vastly different realities from the perspective of a country's firms. The model which we introduced in Subsection 3.3.3 showcases that a trade deficit can be profitable under two central assumptions. First, that importing intermediates from abroad is cheaper than producing them domestically: because of a difference in wages, or of a difference in the capacity to raise the markup. Second, that foreign competition does not threaten domestic (and possibly foreign) demand for the production of the domestic firms. In other words, that the domestic *lead firms* manage to preserve the barriers to entry which protect them from competition. The work of Milberg and Johnston (2025) suggests that these barriers may not be as high as they used to be, which motivates the next step in our discussion of the relationship between transnationality, trade and profitability: a discussion of the process of competition itself.

We have reached a clearer hypothesis concerning what may have happened as the Chinese-U.S. relationship has progressed. The position of U.S.-American lead firms, *vis à vis* their Chinese competitors, has gradually worsened, either in terms of their ability to maintain a high markup or to import cheap intermediates. We also have a clearer picture of the central importance of inter-capitalist competition in the issue at hand. Exploring the question of the different possible modalities of inter-capitalist competition will be the subject of the next chapter.

Chapter 4

Competitive Regimes and International Competition

We have thus far arrived at two broad conclusions. First, there exists no automatic mechanism ensuring that current account and trade balances are self-correcting; imbalances are therefore to be expected as a structural feature of the global economy. Second, in a world where transnational production and trade in intermediaries are preeminent, trade deficits need not hurt the profitability of domestic capital. Rather, trade deficits may result from business strategies aimed at raising the profit rate – such as offshoring for cost-cutting purposes – or from domestic firms being outcompeted by foreign actors. These are two situations with drastically different outcomes for domestic firm profitability. Whether trade imbalances are associated, for domestic firms, with a profitable organisation of production thus hinges on a single element: their capacity to sustain their markups without being displaced by foreign competitors. Because that capacity – the market power of domestic firms – is itself an outcome of the competitive process, which is inseparable from the dynamics of accumulation and technical change, understanding the implications of international competition requires a formal investigation of competition itself, which is the object of this chapter.

Our central argument will be that the capacity of *lead firms* to raise markups is best understood through the lens of *competitive regimes*. Such regimes can change in space and time and a shift from one regime to another provides a powerful explanation for the

changing nature of international commercial relations.

To build this argument, we first introduce the decision problem of the firm, which is driven by the imperatives of survival and growth under conditions of fundamental uncertainty. Building on the seminal work of Crotty (1993) on a Marxian model of investment and competition we delineate the concept of *core industries* (Crotty 2017b), sectors characterised by high sunk costs which set structural conditions that protect incumbents and incentivise the moderation of competition. Some of the largest *lead firms* which operate in *core-industries* are (so-called) tech giants. We therefore refine the analysis by investigating the specificity of the tech sector, engaging with debates on the importance of intangibles in the tech sector (Foley 2013; Durand and Milberg 2020; Starosta, Caligaris, and A. Fitzsimons 2024; Durand [2020], 2024). We introduce a formal distinction between *relatively scarce* and *absolutely scarce* intangible assets and show that many – but not all – tech firms whether through heavy investment in fixed capital or through the ownership of relatively scarce intangible assets, can be characterised as belonging to *core industries*.

This sets the stage for the main analytical proposition of this chapter: the differentiation between two ideal-typical competitive regimes. First, a ‘*corespective*’ regime characterised by restrained competition allowing for higher markups and for firms to adopt a long-run strategic horizon. We argue this is the competitive regime *lead firms* come to be accustomed to. Second, a ‘*coercive*’ regime of intense competition, rapid constrained investment and aggressive price-warfare. We outline how a transition from one regime to another may unfold. We argue that this framework is essential for understanding the asymmetrical power dynamics within GVCs and, crucially, for identifying the effects of international competition. Indeed, foreign competition is a primary pathway that can trigger a destabilising transition from one competitive regime to another.

4.1 The Firm's Decision Problem and the Conditions of Competition

Following what is the dominant view in the post-Keynesian (Galbraith [1967], 1972; Lavoie 2022) and Marxian (Crotty 1993; Shaikh 2016) traditions, we will assume here that the firm's management is attached to the *survival* (avoiding bankruptcy) and the *growth* (increasing sales) of the firm itself, rather than the sole objective of maximising the profit rate, that neoclassical theory posits. Profits do matter but they are either conceptualised as a means (towards growth and avoiding bankruptcy) rather than an end, or as a constraint, imposed by investors, under which managers operate, while striving to avoid being put out of control of the firm.

At the very core of inter-capitalist competition lies the decision problem to which the management of firms is confronted. In the context of this discussion of competition, we can roughly organise firm decisions into two broad categories.

1. **Investment** (as well as localisation and integration) **decisions**, which set the conditions for the firm's production (and therefore cost) structure for the medium- to long-term future.
2. **Pricing decisions**, which, given the organisation of production that the firm has adopted as well as the corresponding costs of production, determine the profitability of the firm, given the previously set cost-structure.

These two decision classes, which relate to two different time frames, are deeply interrelated and are taken in accordance with a firm's broader strategy.

4.1.1 Fundamental Uncertainty

The crux of the problem of the firm's strategy lies in so-called *fundamental* (or Keynesian) uncertainty to which the firm is necessarily confronted when investing. *Fundamental uncertainty* arises as a problem because the act for firm managers to invest (equivalently

to offshore and/or outsource) with the aim of maximising return on new investments requires them to forecast demand¹ (Shaikh 2016, Ch.13). Once the investment has been made, the firm incurs the risk that whatever commodities the new investment has enabled the production of cannot be sold at a high enough price or in high enough quantity so that the investment is made profitable. What is more: it is a sunk cost. If the fixed capital the firm has invested is illiquid, it cannot be sold back if demand is not met (except perhaps at a loss). Hence, it is unlikely that the firm can recoup the value of its investment if it turns out it was made according to a wrong prediction (Crotty 1993). While the notion of *fundamental uncertainty* has become strongly associated with Post-Keynesianism, a similar idea is to be found in the Marxian notion of the *salto mortale* of the commodity – which would perhaps more aptly be called the *salto mortale of the producer* (Cartelier 2023) – Marx’s colourful description of the risk faced by the seller who is ignorant of exchange values as they produce their commodities (Marx [1867], 1887, Ch.3, Sec.2). To Marx the *salto mortale* emerges as a result of the independent (uncoordinated) decision to produce taken by the private producer (here the capitalist) in a world where production is an inherently social fact. Indeed, the realisation of surplus value is conditioned to some consumer finding use value in the resulting good, something that cannot be forecasted on an anarchic (unplanned) market (Benetti 2019).

The intrinsic difficulty in forecasting demand arises for multiple reasons. First, because of the sheer difficulty of accessing, aggregating and interpreting information on such a complex system as the entire economy. Second, because demand is conditioned by the aggregate investment decision (of all firms), but investment decisions at the level of the firms are (usually) not taken in a coordinated fashion. Hence, firm managers are ‘taking a bet on the collective investment’ when they decide to invest in fixed capital. Since demand is higher when aggregate investment is higher, *it is always best to invest when others are investing* and not to invest when others are not investing. Aggregate investment and aggregate demand therefore show patterns of self-reinforcing instability,

¹Here what the investor is to forecast is the level of demand for the specific commodity produced using the fixed capital they are investing in, which is something radically different from the level of aggregate demand. Other factors such as technical progress, as well as the intensity of competition at the time the produced commodities hit the market are central. Even in a setting of high aggregate demand, a highly competitive environment, or a fast technological improvement which renders commodities produced by an old vintage of capital obsolete may strongly depress the demand for products of a given firm, and prevent some investments from being made profitable.

which makes forecasting them fundamentally difficult (Shaikh 2010). This is an example of what Davidson (1996) refers to as a theory of *ontological uncertainty*. The economic actor's ability to forecast the future is limited not only by the information they have access to and by their limited cognition², but also by the fact that every agent's interpretation of the world, and subsequent decision is modifying the conditions in which the economic process takes place.

4.2 Sunk Costs and Core Industries

Let us narrow down our discussion of capitalist competition to discuss the conditions under which it operates for the lead firms which we discussed in Chapter 2. The view under which we operate is that most (but not all) *lead firms* are concentrated in so-called *core industries* (Crotty 2017b), such as *vehicles, electronics, semiconductors, aerospace, consumer durables, shipbuilding, petrochemicals, pharmaceuticals*. Crucially, and as will soon be clear to the reader, *core industries* are not defined according to *what they produce* or *how they produce it* but rather according to *what kind of investments are required* to be competitive. This in turn delineates very specific conditions of competition. One of our key working hypotheses in this work is that *core industries* provide an excellent description of the characteristics of sectors exposed to the Chinese competition (more on which sectors are of specific interest to us in Chapter 5).

The characteristics of so-called core industries are such that they are often sectors in which securing a technological advantage is key in the competitive process. Yet, as we will develop below (see Subsection 4.2.2) it is not technology *per se* which makes a sector *core*, but rather specific characteristics which pertain to risk, uncertainty and the competitive process.

Core industries display the following peculiarities³:

1. They benefit **from large economies of scale**, and their factors of production do not substitute well. Once an investment has been made, the production

²In which case Davidson (1996) would refer to that uncertainty as *epistemological*.

³Which are identified in (Crotty 2017b).

function is essentially locked down to a Leontief production function.

2. They are subject to **a large barrier to entry** as a result. The amount of capital required to benefit from economies of scale, and attain low enough unit normal costs to become competitive is so large that entry is very difficult. This can be understood as the result of the necessity to construct either expensive productive facilities (for instance in shipbuilding) and/or because of the need to go through a long and costly R&D process (for instance semiconductors manufacturing or aerospace).
3. **Required investment in these sectors entails large sunk costs:** it is irreversible. This is in great part as a result of *asset specificity*. Take the example of a semiconductor production plant: once it has been built, there are not many possible buyers, especially if there is not enough demand for semiconductors relative to supply. Therefore, overinvesting is extremely costly.

These are sectors which are fundamentally hard to contest. Indeed, entering a *core sector* likely requires adopting a price-cutting strategy to create demand, but this will lower the overall profit rate, making the outcome less profitable, by an unknown amount (because of fundamental uncertainty). Because of the large volume of assets required to enter a core sector, and because these are very specific assets which are extremely illiquid, the sunk costs make the eventual losses of a failed entry prohibitively large. Conversely, for incumbents, the opportunity cost of preserving the usage of existing assets is much lower than that of investing in new ones (again, because of asset specificity, the assets can hardly be resold). We thus have a situation where incumbent firms are willing to sell at a much lower price than entrants would be able to accept, as a strategy to increase their market share. Thus, entry into a *core industry* is extremely rare.

Furthermore, from the perspective of incumbents, rapid technological change has an inherently threatening character, because, by enabling more efficient production, it risks devaluing already invested fixed capital. This is a classical Marxian argument whereby the capital invested at the current technological level is under permanent threat to be devalued (or destroyed) by the emergence of the next technological wave (Marx [1867], 1887, Ch.15, Sec.3). This phenomenon, which is referred to as *the vulnerability of fixed capital* by Brenner (2006), p.32, gives incumbent firms an incentive to slow down the pace

of technological progress to ensure that their investments have time to yield returns.

4.2.1 Core Industries and Natural Monopolies

This makes core industries examples of *natural oligopolies* (Crotty 2017a). But the two categories are not mutually interchangeable. While usual characterisations of *natural oligopolies* often list two central characteristics which are shared by core industries: large economies of scale and high sunk costs, some other classical characteristics of *natural oligopolies* are not. For instance, non-fixed capital related aspects such as network externalities and benefits from standardisation are traditionally associated with the notion of *natural oligopolies* (Mosca 2008) but have little to do with our aforementioned characterisations of *core industries*. Here it's worth highlighting that when considering examples from the first or second Industrial Revolution (train and telegraph networks, large public infrastructures) network externalities and benefits from standardisation are usually associated with large sunk costs as well. This is not the case for some contemporary examples, especially in the tech sector, where network externalities and benefits from standardisation can be attained without large fixed capital investment. This is especially relevant for so-called two-faced (or multi-faced) markets in which platform companies operate (Armstrong 2006).

4.2.2 Core Industries and the Tech Sector

Many *lead firms* (in particular the largest, see Table 4.1) have been broadly characterised as 'tech' companies. Because of the importance of these companies in the discussion surrounding competition with China, as well as the clear overlap between the categories of *lead-firms*, firms in *core industries* and *tech firms*, we dedicate a short section to clarifying how our notion of *core industries* relates to the tech sector. Let us preface this by stating what is perhaps the obvious: the 'tech' sector is not one which is easily pinned down or defined. This lack of a clear agreed-upon characterisation is rather problematic for the analysis. Nevertheless we can identify some particularly important features attributed to tech companies:

2000			2026		
firm	val	sector	firm	val	sector
Exxon	\$362	oil	<i>Nvidia</i>	\$5,724	semiconductors
GE	\$348	industrial	<i>Apple</i>	\$5,045	consumer electronics
<i>Microsoft</i>	\$279	software	<i>Alphabet</i>	\$4,230	advertising/software
Citigroup	\$231	finance	<i>Microsoft</i>	\$4,190	software
BP	\$226	oil	<i>Amazon</i>	\$3,055	retail/advertising/software
Shell	\$204	oil	Aramco	\$2,520	oil
P&G	\$197	consumer staples	<i>Meta</i>	\$2,180	advertising/software
HSBC	\$193	finance	<i>SpaceX</i>	\$2,020	aerospace
<i>Pfizer</i>	\$192	pharma	<i>Broadcom</i>	\$2,001	telecom
Walmart	\$189	retail	<i>TSMC</i>	\$1,821	semiconductors

Table 4.1: Market capitalisation of the ten largest firms in the world in 2000 and in August 2026. The emergence of tech sector giants (italicized) is striking. These figures are reproduced and updated from Durand (2018). Capitalisation data for the year 2000 is reproduced directly from Durand (2018); the August 2026 data is from the Financial Times.

1. they are firms which are heavily invested in intangible assets,
2. they are companies benefiting from specific network externalities, arising from their use of information and communication technologies (ICTs).

We go through these features one by one, discussing how they relate to the other categories of *lead firms* and of core industry firms.

Intangible Assets One very specific aspect of the activities of tech firms (but this goes beyond the sole tech sector) is that they hold and sell large amounts of *intangibles*. For the purpose of clarity we define intangibles as follows: they are ‘*nonfinancial assets that lack a physical substance, are nonrival in consumption and are at least partially appropriable*’ (Durand and Milberg 2020). These include *intangible assets*: digitally stored data, technological know-how, artistic or technological intellectual property, brands, employer-provided training and organisational structures which enable the production of certain commodities or give the firm *specific advantages* in its economic activities. They also sometimes sell *intangible commodities*⁴ which are sold (Foley 2013; Durand and Milberg 2020). In Chapter 3 we introduced the notion that contemporary manufacturing firms are often *factoryless*. Such firms are necessarily heavily reliant on intangibles, but they

⁴We find reference to these as *cognitive commodities*, *knowledge commodities* as well in the literature (Foley 2013; Durand and Milberg 2020; Starosta, Caligaris, and A. Fitzsimons 2024).

are not the only such examples; firms which manufacture in-house may also be heavily reliant on *intangibles* in the form of software, of specific know-how required for the manufacturing of their products, etc. Here a close relationship between transnationality and holding *intangible assets* appears.

The discussion around *intangibles* is the locus of rich and quite heated debates, where conceptual propositions are many and there is a need for clarification (Foley 2013; Durand and Milberg 2020; Durand [2020], 2024; Vasudevan 2022; Starosta, Caligaris, and A. Fitzsimons 2024; van den Ecker, Reitz, and Seignani 2025). Our aim here is to clarify what we mean by *intangibles*, and how the ownership of large amounts of *intangible assets* or the sale of *intangible commodities* relates to the position of a given firm in competition.

As Starosta, Caligaris, and A. Fitzsimons (2024)⁵ helpfully remarks, in Marxian terms the specificity of *intangible commodities* is that ‘*their use-value can be appropriated as a means of production of further identical use-values*’, p.159. In the language of Coasian transaction cost theory, this means that intangible assets in general suffer from an *appropriability hazard* (Oxley 1997). This makes the ‘*salto mortale*’ especially dangerous as well as especially rewarding for a company selling an intangible commodity. Indeed, a rival firm could directly copy the commodity and sell it for itself (this is the appropriation of the use-value as a means of production), but it also implies that the price of their reproduction is 0 (or rather near 0), which means that they return increasing returns to scale. The importance of the *appropriability hazard* makes the protection of intellectual property particularly important for firms which sell *intangible commodities*, as without such legislation, it becomes impossible for them to earn a return on their investment.

The observation of the increasing return to scale in intangible-heavy sectors is not sufficient to properly understand the dynamics at play. To better conceptualise them, let us turn our attention to their larger production process. Usually, the production of the ‘*prototype*’ (the first copy, for instance the source code of a particular piece of software) of an intangible usually entails the expansion of a large amount of qualified labour. In that sense, there is a striking similarity between the production of a new intangible asset and the production of new fixed capital. The firm which invests in R&D to produce such an asset faces a large fixed cost which it may hope to recover through sales.

⁵Who refers to intangibles as cognitive commodities.

At this point we must point to what is perhaps obvious: intangible assets do not depreciate in the traditional *wear and tear* understanding of depreciation. But they do suffer from what Marx referred to as ‘*moral depreciation*’ (Marx [1867], 1887, Ch.15, Sec.3), which is given by the changes in ‘*social needs for the finished marketable product and whether those products have been displaced by similar ones*’ (Starosta, Caligaris, and A. Fitzsimons 2024, p.156). This notion of ‘*moral depreciation*’ was originally conceptualised by Marx when considering tangible means of production⁶ which is fitting because of the unique character of the intangible commodities as means of production themselves. Software, for instance, depreciates as the hardware on which it is expected to run changes, as competitors add features which make previous vintages of software less appealing, etc. This gives a limit to the increasing returns *in time*, that of the replacement of the current vintage of intangible assets. On the other hand there is essentially no limit to the increasing returns *to scale*, except of course, as Sraffa used to remark, the limited character of demand itself (Sraffa 1926). The production of these commodities is thus reminiscent of the production of capital-intensive commodities (Figure 4.1), albeit with an even smaller unit normal cost as the production scale increases.

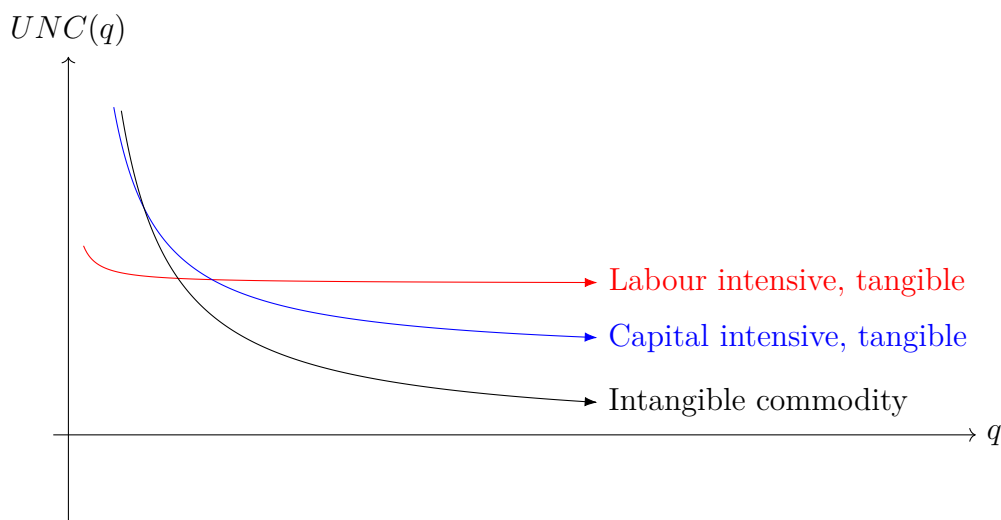


Figure 4.1: Schematic plot of the evolution of unit normal costs (costs including R&D, fixed capital expenditures and per-unit direct labour and input costs) for 1) a *labour-intensive tangible commodity* 2) a *capital-intensive tangible commodity* and 3) an *intangible commodity*.

When intangible assets are conceptualised in this way they appear to share the same

⁶Marx [1867] (1887), Ch.15, Sec.3 is a section about the social transformations brought about by the emergence of complex machinery in manufacturing. This is usually cited in Marxian arguments about the vulnerability of fixed capital, such as the one we make in this text in the last paragraph of Section 4.2.

characteristics as fixed assets in the description we previously made of *core industries*. They are possible, but costly to reproduce, requiring very large initial sunk cost to be created (they have a large barrier to entry), benefiting from large economies of scale. Such large investments into the creation of intangibles are arguably quite illiquid as well. To see why, take the example of software and suppose a company has invested in developing a full alternative office-work suit to compete with Microsoft's, if that company has failed to turn a profit by doing so, it is unlikely that it will find a buyer for the intellectual property of the newly made software to recoup its original R&D investment. Our key argument is that companies which own large amounts of intangibles which behave in the way we have described, and perhaps sell them as well are companies facing the same kind of competitive environment as Crotty (2017b)'s characterisation of *core industries*.

Absolutely Scarce Intangible Assets, Network Externalities So far we have discussed the situation where intangible assets are costly, but possible to reproduce. In doing so we have overlooked the other possibility, that of an intangible asset being *absolutely scarce*, that is, truly impossible to reproduce. This is a point made by Durand [2020] (2024):

Strategies for conquering cyberspace rely on the control of data flows: they need access to our telephones and computers, sensors in machine tools and vehicles, and even in our homes (...) There are not infinite points for capturing the data generated by individuals' and organisations' activities, any more than oil wells are infinite in number. The same is true when it comes to capturing our attention. So, *there is a kind of absolute scarcity of original data* (Durand [2020], 2024, p.175)

And indeed it appears that there is a specific category of intangible which has a property (akin to land, or to mineral resources) to be absolutely scarce. The chief example of such intangibles is *large quantities of original data*, of the kind which are crucial for running a data-driven advertising business such as that of Alphabet, Meta or Amazon (Durand [2020], 2024, pp.170–189). The existence of such absolutely scarce intangibles delineates a qualitatively different kind of logic of competition. Where barriers to entry are not set

by sunk costs, there is no clear path of entry for competitors.

The second point is that owning access to this kind of data essentially amounts to benefiting from positive network externalities, specifically for firms which operate on two-faced (or multi-faced) markets. These are markets where an intermediary platform facilitates interactions between two or more distinct groups of agents (users of the platform), and where the use-value that an agent of either group derives from joining the platform depends on the number of participants on that same platform (Armstrong 2006). The network externalities here act on two levels. First there is the aforementioned effect which drives users to flock towards the most used platform, second there is an effect where the data which users generate enable the platform to improve both the quality of its service (the use value for users) and the profitability of its operation using the large amount of data so-generated. Firms such as Alphabet, Meta or Amazon exploit their data with the purpose of optimising advertising – which is one of the largest sources of revenue of these firms (Stan 2025) – as well as sales and the design of contracts with their consumers (Varian 2014).

Here two key remarks are in order about this specific kind of intangible asset. First, we should note that such *large quantities of original data* are not cheap to collect and hold. The kind of data centres required to hold the petabytes of data required to run Google or Amazon’s advertising services requires large fixed capital investment. We would argue for this reason that there is no appropriability hazard which affects these large datasets in the same way as it affects *intangible commodities*. In this they are qualitatively different from relatively scarce intangibles as well.

In conclusion, the large quantities of original data associated with the platform economy have qualitatively different implications regarding the character of competition over intangibles obtained from research and development, as well as over the sale of intangible commodities.

Conclusion on the Tech Sector and Intangibles Concluding this discussion of the tech sector and its characteristics, we have isolated the key role of intangibles, and highlighted that not all intangible assets have the same implications for the competitive

process. We have isolated two ideal-typical categories of intangibles.

1. **Relatively scarce intangible:** which can be reproduced with qualified labour (for instance, software, industrial and organisational processes, technological advances) and consequently which requires investment akin to investment in fixed capital.
2. **Absolutely scarce intangibles:** which cannot be reproduced with qualified labour, usually because they arise as a result of a firm ‘*conquering cyberspace*’ (Durand [2020], 2024, p.175) and which are thus subject to dynamics very much distinct from fixed capital.

Why introduce these categories? Because they locate where, within the tech sector, Chinese competition actually operates. We argue that Chinese competition in the tech sector is concentrated on firms whose revenues rest on relatively scarce intangibles, rather than on firms which depend on absolutely scarce ones. Since the former – whose competitive edge requires investment akin to investment in fixed capital – can be understood as belonging to *core industries* in the sense developed above, this justifies our choice to focus on this specific category of companies in our investigation of U.S.-China competition. Firms which derive most of their revenue from absolutely scarce intangibles, on the other hand, operate under fundamentally different conditions of inter-capitalist competition and some of the conclusions we are about to draw on core industries should not be generalised to them.

4.3 Corespective Competition

We have seen in Section 4.2 that specifically in *core industries*, there exist structural conditions which protect the incumbents from the entry of new competitors into the market. Furthermore, there is – amongst incumbents – a strong incentive to limit the intensity of competition, especially by refraining from engaging in price wars, in overinvestment or in too fast technological improvement⁷.

⁷One can think of oligopolistic competition as an iterated prisoner’s dilemma where firms collectively benefit from collectively refraining from overinvesting and price-cutting, but would be penalised if they refrained from that behaviour individually. In the iterated (infinite, unknown iterations) prisoner’s

In the words of Crotty:

‘Because the rapid pace of technical progress threatens the viability of the accumulation process, capitalists have an ever increasing material incentive to attempt to control the character and pace of technical change through collusive action. Thus, competition creates forces seeking to negate it.’ (Crotty 1993, p.8)

This occurs through the development of what Crotty (1993) calls, after Schumpeter, ‘*corespective*’ relations, which roughly correspond to the notion that firms enter a ‘tacit contract’ in which they agree to refrain from price cutting and overinvestment (Crotty 1993, p.8). This agreement is aimed at preventing the advent of ruinous competition between core industry firms. Crucially, once such an agreement has been struck, competitors face a greatly reduced uncertainty and become able to plan with greater ease. The ‘*salto mortale*’ is now much less of a risky endeavour. This is a point which was already noted in the work of Galbraith [1967] (1972), who posited that the emergence of the firm itself was directly related to the necessity to tame fundamental uncertainty (Dunn 2011, Ch.7).

Technology and the companion commitments of capital and time have forced the firm to emancipate itself from the uncertainties of the market [...] the firm, to the best of its ability, has made the market subordinate to the goals of its planning. (Galbraith [1967], 1972, p.122, cited in Dunn (2011) p.183)

What we are describing here is a ‘*regime of competition*’ which we will call the *corespective regime* (after Crotty (2002)). It is a regime in which *most competition is non-price competition*. Here, price wars are avoided by firms because they are understood to be ‘*likely to squeeze business profits, become ruinous, and lead to bankruptcies*’ and to risk inducing ‘*lower product quality, as the consequence of desperate attempts to cut into unit costs*’ (Lavoie 2022, p.132). There is therefore good reason for businesses to avoid

dilemma cooperation emerges as a viable strategy. This can be shown formally (Aumann 1959), experimentally with human players (Cooper et al. 1996) as well as experimentally with learning algorithms (Kasberger et al. 2026). It is thus quite unsurprising that it emerges in economic systems.

them, even more so than in other sectors. Indeed, the properties of *core industries* make oligopolies easy to stabilise which makes it likely that attempts by firms to moderate the intensity of competition will be successful. Prices are therefore understood to be set not by supply and demand, but according to the direct or normal costs, plus a margin⁸ (*cost-plus pricing*) which is the result of corespective agreement more than that of competition.

The *regime of competition* which we are describing is therefore a regime in which the main instruments of competition are innovation (but at a paced rate) and product differentiation. Cost-cutting is prevalent but it is of limited intensity, in the sense that firms are not willing to accept too large a reduction in their profit rate in the name of cost cutting.

4.3.1 The Growth-Safety Trade-off Model of Investment

Under the conditions of the *corespective regime*, the well-known Keynesian assertion that firms' investments are autonomously set applies. In other words, firms do not face immediate consequences for refraining from investing. Assuming that first, management is primarily concerned with the growth and survival of its own firm (as we introduced in Section 4.1) and second, that fixed capital investment is illiquid (as we introduced in Section 4.2), we now reach the following conclusions: a trade-off between growth and safety becomes central to the investment decision.

Firms decide to invest in fixed capital expenditures (which are illiquid) because it is a required condition for the growth of their sales. As we previously introduced, we assume that the firm's management is concerned with the firm's survival, as well as the preservation of its own position; hence, this is something we should expect them to avoid. By doing so, they take on the risk that no demand meets the newly constructed productive capacity. Here, demand, and correspondingly revenue, is an inherently uncertain quantity. On the other hand, increasing investment requires forgoing liquidity or taking on debt, which the firm is sure to be required to pay. If it cannot pay its interest, then management's position is threatened and/or the firm will exit the market. These expenses

⁸Alternatively they can be set so as to ensure a certain profit rate given the amount of capital invested, which is known as *target-return pricing*.

are a certain quantity: for instance, assuming investment is financed through fixed-rate bonds, the cost which the firm will incur is exactly known in advance.

The firm is therefore, when it invests, certain to face new costs and uncertain to face new revenue. But it has no other choice than investing if it is to grow at all; in other words, it is certain that whatever level of investment it chooses ‘upper-bounds’ its future profits. In summary, each new investment has the effect of increasing the growth potential of the firm, as well as uncertainty and risk, from the management’s perspective. This is the logic behind the growth-safety trade-off model of investment (Crotty 1993).

This model can be synthesised in the form of the following relation describing investment behaviour of a firm applying the growth-safety trade-off:

$$\frac{I_{k,t}}{K_{k,t}} = f(\alpha_k^+, r_{k,t}^+, \varphi_{k,t}^+, i_t^-, \sigma_t^-), \quad (4.1)$$

where $I_{k,t}/K_{k,t}$ is the rate of accumulation for firm k , α denotes preference for growth relative to safety (in other words, risk appetite), r the current profit rate, φ a measure of financial solidity of the firm (this could for instance be proxied by an equity-to-liability ratio) i the interest rate and σ is a measure of (perceived) uncertainty. The variables $r_{k,t}$ and $\varphi_{k,t}$ are indicators of the firm’s capacity to fund future ventures either with its internal funds ($r_{k,t}$) or through debt ($\varphi_{k,t}$). Increases in the interest rate i_t are associated with costlier borrowing, and therefore with a higher risk associated with the same level of debt. Finally, the variable σ_t measures perceived uncertainty, which is associated with a higher perceived risk taken when investing. This relation will be central to our empirical investigation of competitive behaviour.

4.4 Coercive Competition

So far, we have shown that the specificity of investment in core industries makes price competition especially destructive. We have discussed why this incentivises firms in core industries to enter a regime of *corespective competition* instead, where they avoid price wars and overaccumulation, maintain high markups and pace the rhythm of technological

improvements to guarantee that fixed assets yield a return. But here we must introduce a key insight from Crotty (1993)'s interpretation of the Marxian theory of investment: *corespective competition* is not the only regime of competition which can rule competition in *core industries*. What is more, the *corespective competition* is a fragile compromise that can easily be jeopardised.

Recall that *corespective* competition prevails mostly because of the asymmetric nature of entry: it is much riskier for a new entrant to attempt to compete with incumbents than it is for an incumbent to defend its position. This all hinges on asset specificity, large economies of scale and the fundamental uncertainty of demand.

But in certain situations, these conditions break down, for instance:

1. In a situation where *excess demand is very large and growing* (since it is unlikely that incumbents can meet all future demand). In which case entry appears viable for outsiders.
2. In a situation where *the entrant has a drastic advantage* (either very efficient cost-cutting technology which makes starting a price war worth it, or a revolutionary new product which depreciates the capital of the incumbents). In which case entry appears viable for outsiders with that advantage.
3. In a situation where *the incumbents have overreached their market power* (markups are so high that entering while starting a price war appears to be a viable option). In which case entry appears viable for outsiders.
4. In a situation where fixed capital assets are already accumulated (for some exogenous reason) and cannot be made viable *in the face of reduced demand* unless competitors are pushed out of the market. In which case engaging in aggressive competition appears viable for incumbents.

When this is the case, large investments and price cutting become viable options for firms. Once the 'tacit contract' of maintaining low intensity of competitive pressure breaks, the basic premise of the growth-safety trade-off according to which investing less reduces the risk stops being true⁹.

⁹Circling back to game theory, a well-known near-optimal strategy for the iterated prisoner's dilemma is 'tit for tat' (Axelrod 2006, Ch.1). This is a strategy of picking the individual choice if and only if the other player refuses to collaborate, and suggests that it is a sensible decision for a firm to engage in

Once price-cutting competition starts becoming the norm, old vintages of fixed capital stop being able to yield sufficient earnings. This is simply because, as market prices go down, driven by the firms which enjoy a cost advantage and use it in price wars, the unit normal costs associated with producing on old vintages become too high for their products to be sold with any profit. At this point, it becomes clear to all competing firms that the condition not to be priced out of the market is to invest heavily into the latest vintages of fixed capital. In such conditions, firms are faced with two possible choices: either accept an exit or increase their investment in the face of falling profit rates. In the latter case, firms play an invest-or-die game where investment in fixed assets is their main weapon. We call this regime of competition the *coercive regime*, because firms are coerced to invest if they are to survive.

In that regime, the priority put on cost cutting is not limited to capital accumulation; it also has consequences for labour, as attacking the labour force is central to reducing production costs for the firm. The *coercive competition* regime is thus also associated with ‘*vicious attacks on workers*’ as well as with undertaking ‘*major new investment projects designed to lower costs of production through labour substitution and enhanced labour productivity*’ (Crotty 1993, p.17). Because this all occurs under conditions where firms are likely facing low, or even negative profit rates, new investments must be debt-financed. Here we must highlight that the relationships which applied in the *corespective regime*, in particular the growth-safety trade-off, do not apply any more. This means that the investment function (4.1) should stop describing firm behaviour accurately, or at least lose explanatory power. Identifying a structural break in the investment behaviour (modelled by an investment function) thus constitutes strong evidence for a shift in competitive regime (Crotty and Goldstein 1992).

One striking pattern here is that we should see (amongst firms which do not exit) simultaneously an increase in investment in fixed assets, an increase in the financial fragility indicators (such as the debt-to-equity ratio) and a fall in the profit rate.

Before concluding this section on *coercive competition* we must make a detour through the notion that the regime of competition as a whole is nothing but the aggregate result of the strategies adopted by the distinct firms which are competing Crotty (2017b). Since aggressive competition once it has broken out.

for a firm, changing strategy takes time and is risky in its own right, we should expect that there is inertia in changes of the regime of competition, as well as some degree of hysteresis (a ‘ratchet effect’).

Furthermore, it should be noted that an additional layer of complexity arises from the fact that investment strategies for cost cutting are themselves the result of a certain worldview held by firm management about what will effectively function to give firms a competitive advantage in the future. For instance, as of 2026 we are seeing massive (arguably coercive) investments in computing power for artificial intelligence while there is so far little evidence that AI yields the productivity improvements that motivate these investments (Challapally et al. 2025). In other words, what constitutes a competitive edge is itself subject to fundamental uncertainty.

4.5 Heterogeneity in Competitive Regimes

Formulating our description of competition in terms of competitive regimes becomes especially fruitful when we recognise the possibility of the coexistence of multiple competitive regimes separated either by space, which provides us with a way of describing GVCs, or in time, which opens a discussion about structural breaks in the nature of competition itself.

4.5.1 Global Value Chains and Competitive Regimes

At this point it is worth coming back to our previous discussion of competition and markups in GVCs. Recall that in Chapter 3 we introduced the idea that GVCs are characterised by ‘unequal capture of value added’ (schematically represented by the smiling curve), which comes down to unequal market power, or an unequal capacity to raise markups. Studies of the intensity of competition such as the one performed by Glen, Lee, and Singh (2002) show that competition is drastically more intense in countries of the periphery than it is in those of the core. In the words of Milberg and Winkler (2013):

On one side, there continues to be a high degree of markup pricing power and concentration of industry for global lead firms. On the other side, there is evidence of persistently high levels of dispersion as more developing countries entered lower- and medium-tech industries in manufacturing and services through the 1990s and continued (although at a much slower pace) through the mid-2000s. The result is an asymmetry of market structures within GVCs, with oligopolistic *lead firms* at the top, and competitive markets among the lower-tier suppliers. (Milberg and Winkler 2013, p.123)

What's being described in the aforementioned quote is what makes trade in intermediaries, as presented in the outsourcing-offshoring model which we introduced in Subsection 3.3.3, viable. Indeed, it is the heterogeneity in the conditions of the competitive process between lead firms and their subcontractors which makes the condition $p_m a_m < w a$ (see Subsection 3.3.3) hold. Offshoring is worth undertaking because the price abroad is sufficiently lower than what it is domestically.

It thus seems fitting to assume that the *asymmetry of market structures* which Milberg and Winkler describe translates to differences in competitive regime. This hypothesis is very clearly illustrated by A. L. Fitzsimons and Starosta (2026) in the context of the automotive sector. They outline that different capitals with various degrees of concentration of capital yield distinct capacities to valorise said capital, which leads to a phenomenon of stratification of value chains, where profit rates are proportional to the amount of accumulated capital, including in the form of intangibles. Under that lens GVCs can be understood as a two (or more)-sector organization of the economy where producers subject to two distinct regimes of competition interact. This is the geographically fragmented version of a relation between an '*oligopolist*' and a *competitive* sector, which Mandel [1972] (1999), pp.540–548 described as characteristic of the coexistence of both sectors under *late capitalism*.

4.5.2 The Foreign Competition Path into Coercive Competition

Crucially for our discussion here, Crotty (2017b) hypothesises that one of the paths from corespective to coercive competition lies in the emergence of efficient foreign competitors. Writing about the entry of Japan, Korea and Germany as competitors to U.S.-American heavy industries in the 1970s, Crotty and Goldstein (1992) argues that the development of new industries within these catching-up countries, coupled with lower wages for their workers (relative to the United States), triggered a switch from *corespective* to *coercive* competition in the United States.

By doing so, he highlights a pathway which ties international competition to competitive regimes. This argument is heavily reminiscent of Brenner (2006)’s argument about the onset of what he calls ‘*The Long Downturn*’, at the turn of the 1970s, in which he argued that ‘*the new prey on the old*’ (Brenner 2006, pp.33–34). Specifically, Brenner argued that the investment in fixed capital tended to happen in waves and that these waves were usually synchronised in a given ‘block’ (a country or a group of countries). He remarked that:

‘one can envision the dangers which arise with the advent of new blocs of interrelated productions at higher levels of productivity, or simply lower costs. Large masses of capital can quickly become vulnerable, as firms of the older bloc are obliged to face reduced prices for their higher-cost output and to confront the necessity of premature scrapping and lowered profit rates on their fixed capital.’ (Brenner 2006, p.33)

Concerning pricing, studies such as that of Blecker (1989) highlight that the pricing patterns of steel producers within the United States changed during the 1960s as competition from Japan, Korea and Germany intensified. Specifically, Blecker (1989)’s study shows a structural break in 1962 before which steel producers appeared to practice target-return pricing, and pricing patterns did not respond to supply and demand; and after which pricing clearly responded to supply and demand, and markups were reduced.

This alternative pathway ties in with our previous discussion of international competition, and we can hypothesise that it relates to the entry of Chinese firms as competitors to

European, advanced Asian and U.S.-based *lead firms*.

4.6 Conclusion

Regime of Competition	Corespective	Coercive
Markup pattern	High markups, set through cost-plus or target return pricing.	Low markups, as a result of intense price competition. Prices respond fast to demand shocks.
Mode of accumulation	Investment decision follows the growth-safety trade-off, paced investment.	Firms are coerced to invest rapidly, the growth-safety trade-off does not hold.
Rate of exit	Low	High (firms have a choice between exiting and investing)
Labour relations	High-road (attempt to retain skilled labour)	Attacks on labour (cost-cutting)
Role of innovation	Central to competition, but adoption is relatively slow	Constrained rapid adoption of new cost cutting techniques

Table 4.2: Summary table of the two ideal-typical regimes of competition.

Summing up, we have outlined two ideal-typical regimes of competition (see Table 4.2), which we have constructed following Crotty (2017b). These regimes of competition concern *core industries* which we defined according to the conditions of investment in the sector. We argue that these conditions describe well the conditions of firms engaged in intense competition with Chinese actors.

These two regimes are the following. First, a *corespective regime*, characterised by competition of low intensity, with firms refraining from engaging in price wars and with a low exit and entry rate as well as paced adoption of new technological vintages. In this regime, firms planning over the long term and ‘high-road’ labour relations may prevail, and the markup of firms should be expected to be stable and high. In the second regime, the *coercive regime*, competition is much more intense, with price wars common, the strategic horizon is therefore much shorter, the exit rate is high, investment is also high but constrained (by cost-cutting competition).

We then argued that the coexistence of these two modes of competition provides both a way of explaining the divergence of markups which enable unequal capture of value added in GVCs and, more crucially to our argument, a way of understanding how the entry of foreign competitors could trigger the transition from one regime to another. Our point is that this ‘road to the coercive regime’ through the emergence of strong foreign competitors is relevant to understanding the emergence of Chinese competitors into U.S.-American firms. Circling back to the changing relation between U.S. and Chinese firms and its political ramifications, the lens of shifting and heterogeneous competitive regimes helps to clarify how U.S.-American attitudes towards China could have been so ambiguous and conflicted. In the following chapter, we will see how the two countries’ relation changed over time, which, in the light of our theoretical framework, will help us better understand the character of the two countries’ relationship.

Chapter 5

Towards an understanding of ‘The China Shock 2.0’

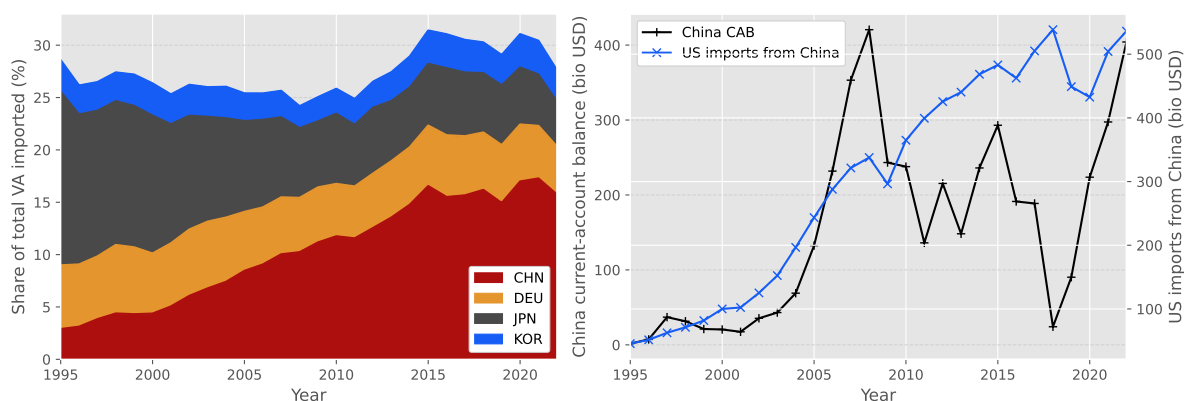


Figure 5.1: Share of gross imports into the U.S. by origin of value added for China, Japan, Korea and Germany. Data from the OECD’s TiVA database (OECD 2025).

At this point we are ready to return to our research object: the so-called ‘*China Shock 2.0*’. The phrase was coined by Autor and Hanson (2025) in a New York Times op-ed, to describe the following development. Having become the workshop of the world by hosting the offshore production of U.S. and European transnationals, China is now moving out of the low-value-added sectors it had occupied since its opening – assembly, textile manufacturing – and into high-value-added industries such as machine tools, transportation equipment, electronics and renewables, where it threatens the dominant positions, and therefore the profitability, of U.S., European and advanced Asian firms.

The trade data bear out this transition, but with an important asymmetry. China's exports have continued to grow overall, while their growth towards the United States has slowed markedly (Figure 5.1). At the same time, the share of Chinese domestic value added embodied in Chinese exports has been rising fast, so that the amount of value added of Chinese origin absorbed by the United States kept increasing even as bilateral gross trade waned. As Figure 5.1 shows, however, the total value added originating in China and imported into the United States rose steadily until the mid-2010s and has been stalling since – a point to which we will return. What we may be witnessing, in other words, is the exhaustion of the '*profitable trade deficit*' configuration whose possibility we outlined in Chapter 3: Chinese firms are now becoming competitive in precisely those sectors whose sheltering from foreign competition had made the deficit profitable for U.S.-American *lead firms*. This, as we have discussed in Chapter 4, implies that Chinese firms are able to rival (at least to some degree) lead firms from the core. In other words: China's ascension to a leadership position in critical sectors may be threatening the sustainability of the very model which it had itself previously enabled.

In the remainder of this chapter we first turn our attention to characterising precisely what is meant by the '*China Shock 2.0*', before discussing what factors explain China's fast catching up with the United States (and other advanced economies). Finally we mobilise the conceptual apparatus which we previously produced to hypothesise how Chinese competitive pressure may be affecting U.S.-American firms. In conclusion we formalise testable hypotheses, to be investigated in the following chapter.

5.1 Characterising the Shock

To properly characterise what we refer to as the '*China Shock 2.0*', we highlight three main points:

1. First, **China sustains a massive current account surplus** (Figure 5.1). It is the world's biggest net exporter of goods by a significant margin.
2. Second, **Chinese exports are becoming dominant exporter in high-value-added sectors**, sectors which, until recently, were only accessible to firms from the

United States, Europe and advanced Asian economies.

3. Third, Chinese *lead firms* are becoming dominant in these sectors¹, which is **threatening the position of firms from the United States and Europe** in markets they previously exclusively dominated, threatening their profitability and perhaps survival.

We have already explored the first of these points in Chapter 2. Before turning our attention to the third point, which is at the core of this investigation, let us make a quick detour through the second. We already mentioned that the domestic value-added content of Chinese exports was rising (Figure 5.1). This implies that China is successfully producing high-value-added goods, in other words goods which are ‘hard to produce’ in the sense that they are protected by large barriers to entry (often as the result of large sunk costs). Most of these industries are well characterised as *core industries*, as we introduced them in Chapter 4.

The increasing export dominance of China in sectors it previously could not compete in is perhaps best illustrated by exploring product-level trade statistics. Defining product-level dominant positions as a share of more than 50 per cent of worldwide exports held by a single country, Jean et al. (2023) show that China is the country which is dominant in the highest number of products, and that its dominance is increasing over time. Building on this idea, we measure how dominant a position China has in its trade with the world and with the U.S. across the products manufactured by different industrial sectors (Figure 5.2) delimited using two-digit (the so-called ‘major-groups’) SIC (Standard Industrial Classification) codes.

Three main stylised facts emerge: first, growth in low value-added sectors, such as furniture and apparel (respectively SIC major groups 23 and 25), has slowed down or even reverted around 2010. Second: China’s dominance on the world market in exportation of some key high-value-added sectors has accelerated since 2010. For instance, this is very clearly the case for industrial machinery and computer equipment, as well as electronic equipment (respectively SIC major groups 35 and 36). This is consistent with the hypoth-

¹Note that this is different from China becoming a dominant exporter in these same sectors as it implies that the firms in question are headquartered in China, and are not subsidiaries of U.S.-American or European firms.

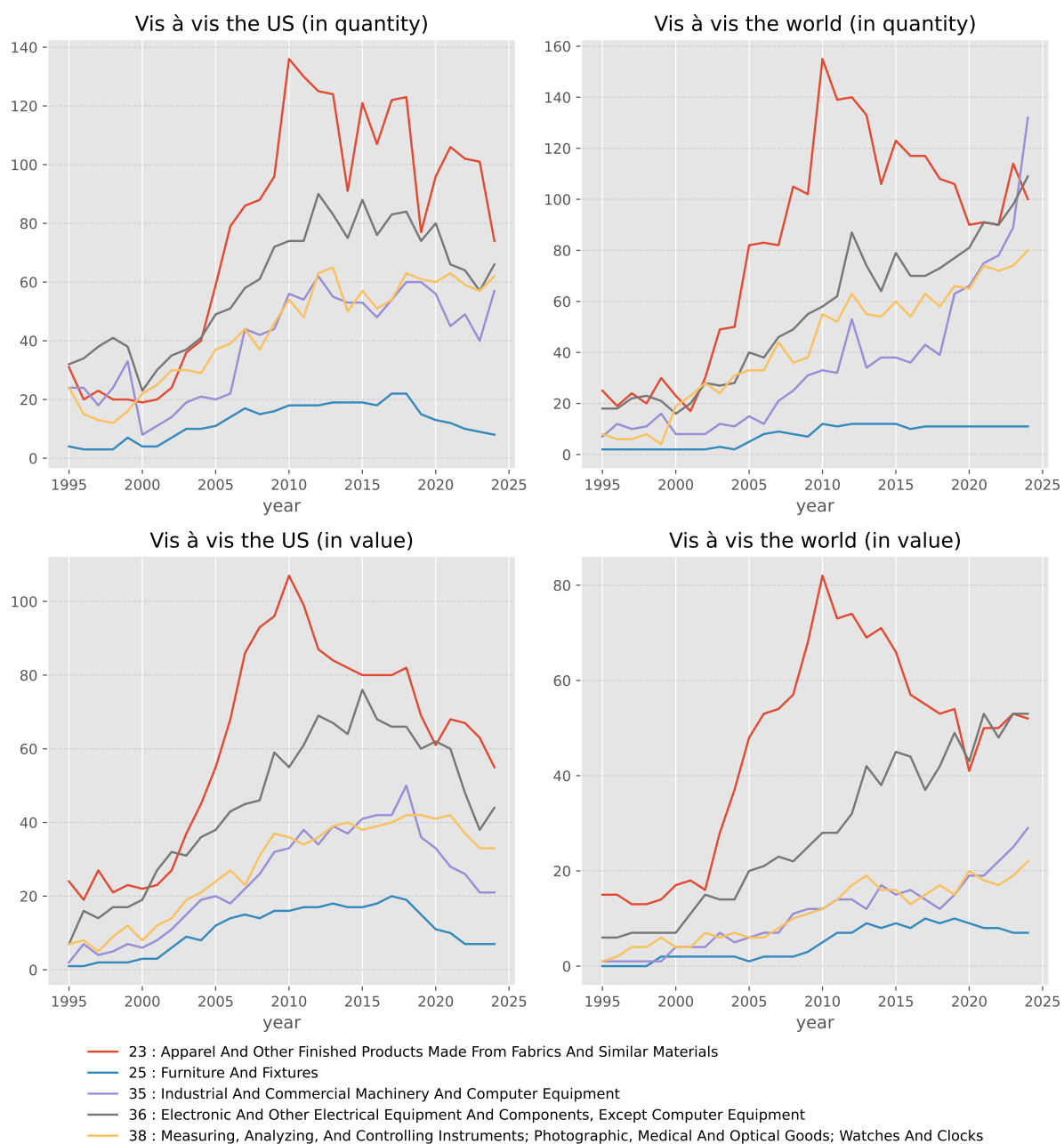


Figure 5.2: Number of products in which exports from China represent at least 30% of trade with the U.S. (left) or with the entire world (right), in five key industrial sectors (SIC major groups: 23, 25, 35, 36 and 38) measured in quantity units (tons of goods traded, top) and in monetary units (dollars, bottom). Plot constructed by the author from the BACI database (Gaulier and Zignago 2010) using the 2024 version of Pierce and Schott (2009)'s concordance map to assign HS product codes to SIC sectors.

esis that China is pivoting towards high-value-added sectors and successfully securing a large market share as it does so. Third: China’s dominance on the U.S. market has been waning since around 2010. This latter observation suggests that to some extent, there has been decoupling between China and the U.S. But if there has been decoupling, it is only a partial one.

The volume of Chinese exports to the U.S. is still very high (as Figure 5.1) attests, but the growth in volume of trade has been reducing. As García-Herrero and Tan (2020) attest, the picture is clearer in terms of FDI: new investments across both economies have slowed down². What is more, quite a large volume of trade happening between the U.S. and China appears to have been ‘rerouted’ through third countries such as Vietnam and Mexico, adding confusion to the trade statistics (Ning, Qin, and Ma 2025).

While the second point (China is becoming a dominant exporter in high-value-added sectors) is best documented, the third, which is most significant in this discussion, and currently also the most difficult to investigate empirically, is not. Autor and Hanson (2025)’s paper is an op-ed and mostly relies on anecdotal evidence to argue their claim that Chinese firms are becoming more and more able to compete with their U.S.-American, European and advanced Asian counterparts in high-value-added sectors. A more rigorous approach is adopted by Milberg and Johnston (2025) who show that Chinese import penetration correlates with decreased profit-share for U.S.-American firms, suggesting that the ‘*China Shock 2.0*’ may already have started. But at the time of the writing of this thesis, the literature relating the profitability of U.S. or European firms to the emergence of Chinese *lead firms* is still relatively sparse.

5.2 China’s Catching Up

The main explanations which can be found in the literature to explain how China has become capable of challenging the lead firms of the most advanced economies are the following.

²With the notable exception of Chinese investments into U.S. treasury bonds, but these can be understood as a consequence of the trade imbalances more than as a strengthening of integration.

1. A technological upgrading narrative.
2. The idea that China simply benefits from the economies of scale yielded by its huge market size and its status as the world's factory.
3. The notion that price-competition in China is much more aggressive than elsewhere in the world (in the Chinese context, this is sometimes referred to as *involution*).
4. The notion that China is benefiting from a well-tuned industrial policy which gives its firms a competitive edge.

As we discuss these explanations, we will see that they are intrinsically tied together in the historical process of the development of China and can hardly be analytically separated. Nevertheless, for the sake of readability we will go through these different explanations one by one.

5.2.1 Technological Upgrading

The first, and most obvious point explaining the shock is that Chinese firms have managed to enter sectors in which they previously could not possibly compete because they were lacking the relevant '*human capital*' (qualified engineers, researchers and technicians). For many U.S.-American and European firms, there was thus a technological barrier to entry which made parts of their activity immune to Chinese competition. This is no longer the case, which means that Chinese firms are now able to challenge the protected position of *lead firms*. This is essentially the point which Autor and Hanson (2025) make. Similarly, Milberg and Johnston (2025) write:

There is no great mystery about what is driving [increased competition from China]. China has been able to upgrade its technological capabilities and to innovate in the high-scale production of new technologies that are vastly more cost competitive than similar goods produced in the U.S. (Milberg and Johnston 2025, p.26)

One aspect of this story is that China has managed to technologically upgrade so rapidly

by imitating the processes of U.S.-American and European firms. This was enabled by the fact that some of them had moved their production lines to the country and were either relying heavily on Chinese subcontractors, or involved in Joint Ventures (JVs) with Chinese companies. The importance of the JV model in the process of technological upgrading should not be understated. Indeed, the policy requiring foreign companies investing in China to form joint ventures with Chinese firms (usually large state-owned enterprises) was implemented with the explicit goal of ‘*Trading Market For Technology*’ (市场换技术) (Y. Li and Lazonick 2022, pp.16–17). This strategy was implemented as a response to the failure of the import substitution policy focused on importing capital equipment to develop domestic industries, which had previously yielded disappointing results in improving the level of technology and the productivity of Chinese production lines (Feng 2019, Ch.4).

Examples of Chinese ‘copycat’ firms are not hard to come by. For instance, AMSC, a U.S.-American manufacturer of electronics required for the production of wind turbines, nearly went bankrupt after the Chinese wind-turbine manufacturer Sinovel (which used to be AMSC’s biggest consumer) bribed an AMSC employee to download its product source codes and started producing AMSC’s main product in-house. Sinovel was found guilty of theft of trade secrets and ordered to pay around \$59 million by a U.S. court (Hung 2022, p.36). Another instance is the Chinese heavy-agriculture and industrial manufacturing firm XCMG, which allegedly manufactured copycat products from the U.S. firms Vermeer and Caterpillar. Unlike in the example of Sinovel, in this case no legal procedure was attempted (Hung 2022, pp.37–40). Many other such examples exist, and indicators of the volume of intellectual property legal actions from U.S. and European firms against their Chinese competitors suggest that this has been a prevalent practice. However, the importance of this controversial channel of industrial espionage is often overstated relative to the significant role of legitimate technology transfer, which has coexisted with these practices (Y. Li and Lazonick 2022, p.24).

Crucially, upgrading technologically – be it through imitation – can only happen if a sufficiently large base of educated workers is available to perform the imitation. Thus, the basis of any such upgrading lies in the education and research systems. The Chinese education system has relied on a consistently high level of investment from the Revolution

onwards, albeit with a short period of divestment at the time of the Cultural Revolution³. Mass education has raised the level of education of the average worker higher than in other countries with comparable development paths. For instance, while India and China have a comparable share of higher education graduates, India has a much higher share of its population which has failed to receive any education at all, while China has nearly eradicated illiteracy (Y. Li and Lazonick 2022, pp.8–9). In the realm of higher education and of research, which have been closely linked since China put universities at the centre of its research effort, in the 1990s, the country has succeeded in creating a competent highly qualified R&D workforce. The number of university graduates has increased rapidly and the country’s focus on engineering programmes has ensured that this education effort immediately transfers into production capacities (Y. Li and Lazonick 2022, pp.10–15).

Like other emerging economies, China has faced the issue of the ‘*brain-drain*’ phenomenon where its most qualified college graduates seek graduate education and employment opportunities abroad, in particular in the United States through H1-B and L-1 work visas. After first approaching the issue by penalising students which did not return to China, the country has changed its approach and has focused instead on poaching back its best overseas scientists and engineers, offering them competitive opportunities at home (Zweig 2006). This has led to a pattern where returnee scientists – known in Chinese as 海归 literally ‘sea turtles’ – would come back to China, founding companies in crucial sectors such as ICT (Yu Zhou 2008), renewables (Hopkins and Y. Li 2016). Returnees have also been recruited by the country’s universities and research institutes, contributing to the construction of a very strong fundamental and applied research sector.

Indeed, China has acquired a leading position in scientific research over a number of fields, which of course translates to an increased capacity to introduce technological advances into its production. One metric which is often mobilised to highlight that reality is the increased output of Chinese universities in research on so-called strategic technologies (Burbaumer 2024; Autor and Hanson 2025). For instance, the Australian Strategic Policy Institute’s ‘*Critical Technology*’ tracker (Gaida et al. 2023) establishes, using metrics

³Although the Cultural Revolution saw a reduction of budgets allocated to education (Y. Li and Lazonick 2022, p.7) and is thus often interpreted as an event causing loss of human capital by economists, there is evidence that the send-down movement had the effect of accelerating alphabetisation in the rural countryside (Chen et al. 2020), painting a more nuanced picture of this period in that regard.

based on the impact of published scientific research, that China is leading technological development in 37 out of the 44 *critical technologies* it evaluates⁴, with every single one out of the seven critical technologies where China does not dominate being dominated by the United States. Along similar lines, García-Herrero, Krystyanczuk, and Schindowski (2025) investigate the capacity of the Chinese economy to lead technologically by looking at newly introduced patents. They define ‘*radical novelty*’ patents as patents replicated at least five times and compare the number of ‘*radically novel*’ patents in China, the U.S. and the EU. Using this metric they find that China appears to be slightly lagging behind the U.S. in most ‘*critical technologies*’ while still being a more than credible competitor.

5.2.2 Economies of Scale

When China opened up its economy to the world and offshoring from U.S.-American and European firms towards China first picked up, most of the export-oriented production within China was dedicated to relatively labour-intensive manufacturing (in sectors such as the textile and apparel industries) as well as assembly tasks where factories in China would assemble products for U.S.-American or European TNCs. This is easily seen in the evolution of the capital-output ratio of the country, which, in the 1990s, was much lower than that of developed economies (see Figure 5.3, left). At this point, Chinese firms were sourcing most of their high-end (high-value-added) components through importation, which is the reason why China was a much larger exporter in trade statistics than it was in value-added. In that regard, Y. Xing and Detert (2010)’s classic study decomposing the value-added components of an iPhone is quite telling. In 2007–2009, Apple was selling iPhone 3Gs assembled in China, yet no key component originated in China and the country was capturing a very low fraction of value added. China was either exporting commodities which it assembled but did not manufacture, or exporting commodities it did manufacture, but which yielded low value added (as was the case in the textile industry).

What is singular about China is how large its population, territory and therefore ability to receive foreign investment are. Given the conditions created by the market reforms,

⁴The report evaluates 44 research fields which it deems to correspond to ‘*critical technologies*’ in the broad categories of ‘*Advanced materials and manufacturing*’, ‘*Artificial intelligence, computing and communications*’, ‘*Energy and environment*’, ‘*Quantum*’, ‘*Biotechnology, gene technology and vaccines*’, ‘*Sensing, timing and navigation*’ and ‘*Defence, space, robotics and transportation*’.

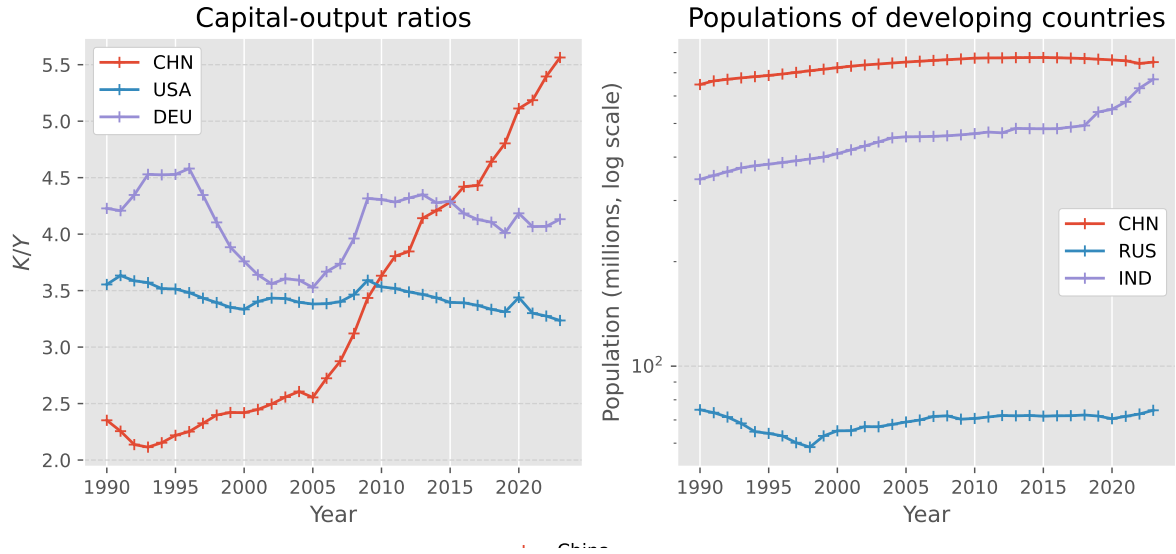


Figure 5.3: **Left:** evolution of the capital-output ratios of China, the United States and Germany over time. Notice how China went from being much more labour-intensive than its counterparts to being much more capital-intensive. **Right:** population of China, Russia and India, three countries which opened their markets at the same time. Data for both plots is from the Penn World Tables 11 (Feenstra, Inklaar, and Timmer 2015).

the quantity of available labour in China was considerable compared to other developing countries (Figure 5.3, right). As production became increasingly fragmented along GVCs, China became central in the middle segment of the *smiling curve*, collecting comparatively low amounts of value added relative to the amount of labour performed in China (Y. Xing 2022). This is one of the reasons why the country became central to many supply chains, providing TNCs from around the world with compelling reasons to relocate more of their production to China, in order to benefit from accessible cheap labour and a developing network of competent subcontractors.

This positive feedback loop of fast growth, high demand and high investment (both domestic and foreign) coupled with improvements in productivity is in itself an explanation for the high competitiveness of Chinese firms through dynamic increasing returns to scale; this is known as a Kaldor-Verdoorn effect (Guo, Dall’erba, and Gallo 2013; Nabar-Bhaduri and Vernengo 2025). Once this positive feedback loop was started, fixed capital investments picked up and the country entered a self-sustaining loop of increasing productivity and capital-intensity of outputs. The latter stylised fact is easily readable from Figure 5.3, left. These dynamic effects on the productivity of Chinese manufacturing concretely translate to lower unit normal costs and hence increased competitiveness of

manufacturing in China.

5.2.3 ‘Involution’

What makes Chinese competition so threatening for U.S.-American, European and advanced Asian firms is the capacity of Chinese firms to sustain extremely low price levels. A central point in the literature concerned with Chinese competition is that the exports of Chinese-headquartered firms are much cheaper than those of U.S.-American and European ones (García-Herrero, Krystyanczuk, and Schindowski 2025; Bayoumi and Gagnon 2026). This is reflected in the fact that China’s producer price index (PPI) is consistently below that of the United States and Europe (Figure 5.4, left). Over time, manufacturing costs in China have begun to decline relative to those in Europe and the United States, and the gap is not closing but widening as China enters high-value-added sectors (Figure 5.4, right).

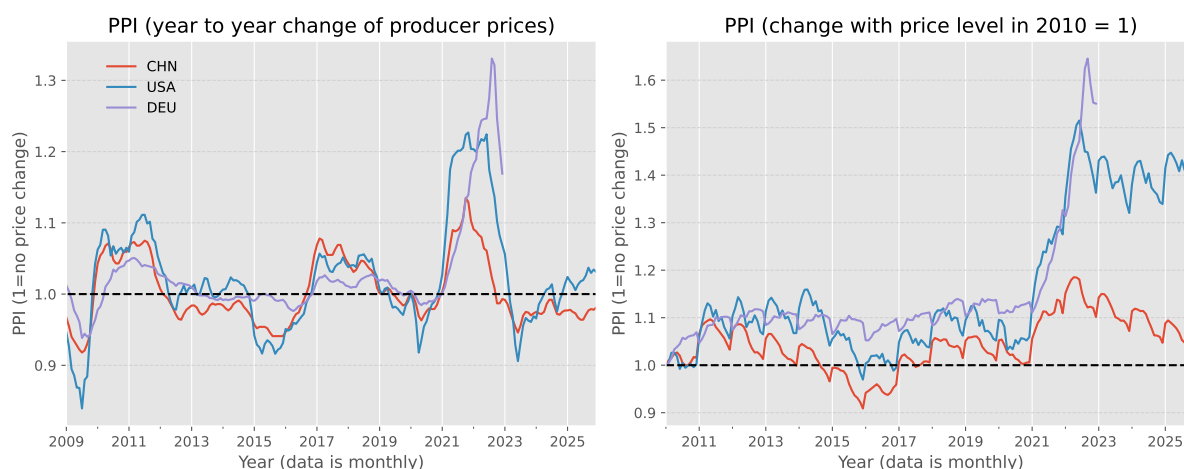


Figure 5.4: Producer price index (PPI) in China, the USA and Germany across all industrial goods. **Left:** year-to-year change. **Right:** PPI in levels, rebased in 2010. Sources: U.S. Bureau of Labor Statistics (2026); Organization for Economic Co-operation and Development (2023); National Bureau of Statistics of China (2026).

This is not only explained by lower labour costs or higher productivity in China, but also by lower markups as a result of intense price competition coupled with persistent overinvestment in the Chinese market. Chinese business owners report a climate of constant price wars which fuel fast improvement of manufacturing techniques, enabled by a high sustained level of cost-cutting investment. A Chinese businessman answering the

Financial Times declares that ‘[his company] would update [their] processes two or three times a year [...] the five-year product cycles with annual price negotiations that the auto industry once ran on have disappeared’ (McMorrow et al. 2026). Ten years ago, this intense cut-throat competition/overinvestment nexus was already identified by Robert Boyer (2011) as characteristic of the Chinese mode of regulation. It appears that the intensity of competition has only been increasing since then.

What we qualitatively describe here closely matches Crotty (2017b)’s description of *coercive competition*, where firms are constrained by intense competition to simultaneously cut their prices and to invest into fixed capital with the purpose of cost-cutting. Quantitative research on the matter is scarce, with García-Herrero and J. Xu (2026) being a notable exception. The authors investigate the markup of Chinese firms⁵ and find that, first, the median markup of Chinese firms is falling down, suggesting that firms are cutting prices faster than they manage to cut costs and that this occurs regardless of the productivity of firms. They also note that the number of firms paying more interest than they earn in revenue (so-called *zombie-firms*) is increasing fast in China. They estimate that about 12% of Chinese firms are in that situation, which is especially prevalent in the renewable sector, where almost 30% of firms are *zombie-firms*. The authors suggest that this is due to ‘profit-less’ investment fuelled by debt and facilitated by the access these firms have to low interest rate credit. This tracks closely with Crotty (2017b)’s description of debt-fuelled investment and brings a different light on the findings of the OECD’s MAGIC database report (OECD 2026).

Since 2024, the Chinese Communist Party (CCP) has adopted the term ‘involution’ (the English translation of the Chinese word *nèijuǎn* (内卷)) to describe the phenomenon. In a 2025 article published in *Qiushi* (求是), the theoretical journal of the CCP, ‘involution’ is defined as: *a vicious competitive phenomenon where economic entities continuously invest large amounts of energy and resources to maintain market position or compete for limited markets, yet fail to bring about overall revenue growth* (J. Li 2025). The reason the CCP makes an effort to analyse this phenomenon in its theoretical journal is that combating it has become one of its main stated political priorities. Likely because, past a certain level, this degree of intensity of competition clearly has a destabilising effect on

⁵Using the *De Loecker* method for estimating markups.

the Chinese economy and the government’s doctrine of *macro-control* (see the following subsection) gives quite a lot of importance to avoiding market instability.

5.2.4 Macro-Control and Industrial Policy

At the fulcrum of China’s development and technological upgrading lies the Chinese state’s approach to economic policy. Unlike economies adhering to the Washington consensus’ doctrine of ‘*not picking winners*’, the Chinese state asserts that its economy is ‘*market-driven, and government guided*’ (市场主导政府引导) (Naughton 2021, pp.101–102). This entails an understanding according to which market forces are the primary driver of the allocation of resources in the economy, but that this allocation can be directed through government intervention (Naughton 2021, p.100). The Chinese state thus performs what it refers to as ‘*macro-control*’ (宏观调控): a coordinated use of monetary, fiscal, credit and industrial-policy instruments to ‘*maintain the fundamental balance of economic aggregates, accelerate the optimization of the economic structure, guide the national economy’s sustained, rapid and sound development, and push forward comprehensive social progress*’ (CCP 1993, cited by Sperber 2023). The government is thus in this understanding in a position to accelerate the country’s development by steering market forces in the right direction, for instance in the direction of fast technological upgrading.

This understanding of the role taken up by government in China’s industrial upgrading translates to a set of policy-mechanisms and institutions built with the intent of enabling this government steerage of market forces. We can roughly summarise the construction of this set of institutions by considering three main levels at which the country’s industrial policy operates.

At the top level, lies the planning process which, starting at the national level, outlines ‘*vision statements*’ communicating the main goals of policy makers as well as a few binding targets to be achieved⁶ (Naughton 2017, pp.11–12). These general guidelines then drive the production of sectoral and local plans which tackle the implementation of the broad,

⁶Well-known examples of such targets are a 20% energy conservation target in the XIth 5-year plan (2006–2010) or the goal of bringing 55 million Chinese above the poverty line in the XIIIth 5-year plan (2016–2020).

overarching goals set by the national-level five-year plans.

Below the planning levels lie the different instruments used by the state to achieve the goals of the plan. Practices such as price control or the setting of production quotas have largely been abandoned (Aglietta and Bai 2012), instead, the government intervenes in two main ways. Through direct intervention aimed at propping up the country's '*basic capabilities and infrastructures*' and through control of the investment flow across the economy.

What is meant by direct investment in *basic capabilities and infrastructure* is: on the one hand the construction of a strong education and research system (which we have already touched upon), and on the other hand investment in physical infrastructure such as the energy sector and the transportation sector (Y. Li and Lazonick 2022, pp.6–15). This investment in basic infrastructure reaches significantly further than what may be envisaged in a Washington consensus adhering state, as since the end of the centrally planned economy, the Chinese state has maintained partial control over some key upstream industrial sectors (for instance the steel industry) (Y. Li and Lazonick 2022, p.15). Strategic investment in upstream sectors ensures that fundamental inputs of the Chinese industrial apparatus are kept accessible and secure and participate in a strategy to ensure the conditions in which private businesses operate are kept favourable (Jabbour and De Paula 2021).

The other arm of Chinese industrial policy is the state's management and steering of the flows of investment, which largely comes from credit (Sperber 2023). The state's exercise of control over the flow of investment manifests through a variety of institutional forms and practices. Public banks, such as the Chinese development bank (Y. Li and Lazonick 2022, p.13), as well as private banks and non-bank actors pool capital in various types of investment vehicles which are piloted in such a way as to adhere to the plan (Sperber 2022; Zhang and T. Lan 2023). Some controlled by the local governments, some by the national (X. Lan 2024). Notable such funds include land-collateralised debt for infrastructure (X. Lan 2024, pp.156–170) and guidance funds for technological ventures (Zhang and T. Lan 2023). These vehicles act as a source of patient capital which expects low return on investment (at least at first) and allow for the development of large infrastructure projects and of risky technological ventures (Y. Li and Lazonick 2022). Crucially they

come with requirements in terms of localization and adherence to the plan (Y. Li and Lazonick 2022, pp.16–18). Finally the state’s monetary policy is guided by objectives of the plan, interest rates are administered along the whole yield curve so as to ensure a cheap abundant flow of capital (Aglietta and Bai 2012, pp.171–172).

Explaining China’s breakthrough in high-value-added sectors requires discussing the country’s relevant plans. Of particular relevance is the ‘*Made In China 2025 plan*’ (hereafter MIC 2025) which was a sectoral plan, implemented as part of the XIIth 5-year plan (PRC State Council 2015). Approved in 2015 in an executive meeting of the State Council of the PRC⁷, the MIC 2025 plan outlined a path for the modernisation of China’s manufacturing capacity in key sectors: the IT industry, high-end manufacturing CNC and robots, aerospace, offshore engineering and ships, advanced rail and transportation equipment, ‘new energy vehicles’, agricultural machines and equipment, new materials and the biotech/pharma sector (Malkin 2022). The MIC 2025 plan explicitly lays out as an objective for industrial policy that Chinese firms in the aforementioned sectors move away from being assemblers to becoming owners of IP assets and technological leaders. According to the plan, one central aspect of this transition lies in increasing the capacity of China to develop the ‘*basic core components*’ of these technologies, in other words that Chinese firms become technologically less dependent on imports from U.S.-American, Asian and European counterparts.

To fulfil this objective, Chinese authorities have deployed a wide arsenal of policy instruments. Upstream, the Chinese state organises government-funded research initiatives which follow the priorities of the plan. These are meant to lead to commercial adoption by the private sector (Malkin 2022). Further down the ‘innovation pipeline’, the Chinese state explicitly supports research and development efforts for private manufacturing firms, and facilitates the investment required for bringing innovations to market. ‘*Government guidance funds*’ (at the national level) and ‘*local government financing vehicles*’ (at the provincial level) play a role akin to venture capital in the U.S.-American economy, bringing together private and public sources of capital and investing in early and mid-stage high-tech enterprises directing investment according to the objectives of the

⁷This is a government organ akin to a council of ministers for the PRC (although drawing direct comparisons between U.S.-American and European liberal democracies and the structure of the PRC’s government quickly reaches its limits).

plan (Zhang and T. Lan 2023).

This approach to economic policy is one which leads to what a recent OECD study (OECD 2026) refers to as large market distortions. The Chinese state heavily subsidises its national champions in sectors which it deems strategic (many of which are the core industries which we described in Chapter 4), for instance by giving them access to loans at a much lower interest rate than the national benchmark. This of course gives Chinese firms which benefit from such state support a clear competitive edge investing large amounts in fixed capital and R&D expenditures. García-Herrero and J. Xu (2026) interpret that the extension of these loans favour the continuation of this regime of *coercive competition*⁸. This is an argument we agree with: indeed, reducing the cost of capital for firms in strategic sectors effectively lowers the entry barriers in *core sectors* which should, by the theoretical framework we laid out in Chapter 4 have a destabilising effect on any *corespective competition* regime.

As should be clear to the reader by now, the Chinese approach to industrial policy (or rather to economic policy at large) is central to the analysis of Chinese development and the catching up of its companies with the United States' *lead firms*. It has been designed with the explicit intent of facilitating the technological upgrading of Chinese private firms both by funding fundamental research and by supporting its introduction to the market.

5.3 Relating Chinese Competitive Pressure to Firm Behaviour in the U.S.

Having discussed how the current environment in China provides context for the shock, let us now consider the shock itself from the perspective of our theoretical framework.

To this end, we bring together the different building blocks we have assembled in the preceding chapters. Chapter 2 crucially clarifies that because it is ruled by *absolute*, not *comparative* costs, competition across national borders is not qualitatively different from competition within national borders. Chapter 3 adds that trade deficits have an ambigu-

⁸García-Herrero and J. Xu (2026) talk about *involution* not coercive competition.

ous relation with firm profitability. It outlines how a situation of trade deficit can prove profitable for firms operating in the country that runs the deficit, but only under specific conditions. These conditions boil down to the ability of U.S.-American firms to maintain a high markup relative to their Chinese counterparts. This is a question which, to be apprehended properly, needs to be discussed at the level of inter-capitalist competition. Chapter 4 thus discusses in detail how the cohabitation of two competitive regimes under the backdrop of specific structural conditions can explain the aforementioned differences in profitability, and discusses how transitions can occur from one competitive regime to another, for instance through the effect of foreign competition.

At this point, and with the context we have provided on the ‘*China Shock 2.0*’, putting these three components together into an explanation of the shock is rather straightforward.

During the ‘Chinamerica’ period, U.S.-American (as well as European and advanced Asian) *lead firms* (which are largely firms operating in *core industries* in the sense of Chapter 4) enjoyed protection from competitors in their core domain of activity. This was secured by the prohibitive costs of entry and sheer difficulty of catching up with them technologically. This enabled low-intensity competition between *lead firms* operating in *core industries*, and we could describe the competitive regime between *lead firms* as *corespective*. At the same time, their subcontractors, which were operating in sectors with much lower barriers to entry and which were pitted against each other by *lead firms*, were engaged in much more intense competition. The coexistence of these two regimes – on the one hand, intense cost-cutting competition amongst subcontractors, and on the other, mild competition between *lead firms* – coupled with low wages in the countries which hosted the subcontractors, simultaneously enabled the maintenance of high markups for *lead firms* as well as low inflation in the United States.

In China, where many of these subcontractors were located, competition was fierce, but demand was also extremely high and growing, because of China’s centrality in GVCs. This, coupled with a well-tuned developmental policy, resulted in fast accumulation in China and facilitated the rapid technological upgrading of China. The arrangement which we just outlined ensured the conditions of *profitable trade deficit* were preserved in the United States. It was sustainable as long as the technological gap between China and the

U.S. was sufficiently large so that the barriers to entry in *core industries* were maintained.

But the fast development of China makes it so that this is not the case anymore. What is happening now is that the intense regime of cut-throat competition, which was prevalent in sectors that were not *core industries* – in part because *lead firms* made sure it was the case (Milberg and Winkler 2013, Ch.4) – is becoming prevalent for *core industries* as well within China’s market. The persistence of a complex set of conditions (high demand stemming from centrality in GVCs, access to subsidies and very low-interest rate loans, high availability of qualified personnel) makes the *coercive regime* persistent in China, which leads to prices rapidly falling and productivity increasing fast. Because international competition is not ruled by comparative but by absolute advantage, the competition from Chinese firms is exerted on their non-Chinese competitors as well, which threatens the model that prevailed in the U.S. (as well as European and advanced Asian economies) until now.

The hypothesis that we formulate is that for firms in the U.S. (as well as in the EU and in advanced Asian economies, but this is beyond this work’s scope), competition with Chinese firms is likely to trigger a switch from *corespective* to *coercive competition*. If this transition is indeed occurring, we should expect that it does in a manner similar (up to a certain extent) to the one which was described by Crotty (2017b) and Brenner (2006) in the context of foreign competition to U.S. firms in the 1970s.

If the hypothesis which we are formulating above is correct, we should see corresponding indicators regarding:

1. Pricing: prices in core industries outside of China should be pushed down by the entry of Chinese competitors.
2. Profitability, measured through profit rates, should be falling as a result of Chinese competition and coerced investment.
3. Investment behaviour should move away from a strict growth-safety trade-off; we should see a structural break in investment patterns.

We are now in a position to state our testable hypotheses more precisely, laying down the

foundations for later empirical analysis. The narrative of a transition from a *corespective* to a *coercive* competitive regime, driven by the entry of Chinese *lead firms* into *core industries*, yields the following testable implications for the population of U.S.-American firms operating in *core industries* and exposed to Chinese imports:

- (H.1) **Reversal of the exposure-profitability relationship.** During the ‘*Chinamerica*’ period, a high exposure to imports from China was associated with *higher* profitability, as it typically signalled a successful offshoring strategy on the part of *lead firms*. If the model of *profitable trade deficit* outlined in Chapter 3 is indeed being exhausted, this association should weaken over time. If Chinese competition is indeed becoming more and more intense, it should eventually turn *negative*: firms operating in product markets where Chinese exports are dominant should tendentially see their profit rates compressed relative to less exposed firms.
- (H.2) **Intensification of competition within the U.S.** If competition is intensifying as in the scenario we laid out, in the form of price-cutting competition and coerced investments, we should observe that the profit-rates of firms exhibit lower persistence and faster mean-reversion dynamics in recent periods than in earlier ones, reflecting the erosion of the sheltered positions which *lead firms* previously enjoyed in *core industries*.
- (H.3) **Weakening of the growth-safety trade-off.** Under *corespective* competition, investment is ruled by the growth-safety trade-off: it responds positively to profitability and negatively to financial fragility. If more and more investment becomes coerced, this relation should appear much less clearly. We expect some kind of structural break in investment behaviour.
- (H.4) **Investment is associated with Chinese imports.** If the investment pattern which we are measuring is indeed driven by competition from China, we should see a positive association between a firm operating in a sector exposed to Chinese imports and its investment rate.

Two remarks are in order. First, we do not expect these hypotheses to hold sharply: the framework we laid out in Chapter 4 suggests that if such a transition is occurring, it is likely to be partial and uneven across firms and sectors, so that traces of both regimes

may coexist in the data. Second, while pricing is the most direct channel through which coercive competition operates, we do not test it at the firm level. The aggregate PPI evidence presented above (Figure 5.4) suggests that price competition is happening, but we leave a firm-level price study for further research.

5.4 Conclusion

In this chapter we have characterised the ‘*China Shock 2.0*’ as the conjunction of a massive current account surplus, a growing Chinese dominance in exports of high-value-added goods and – most significantly for our purposes – the emergence of Chinese firms capable of rivalling their U.S.-American, European and advanced Asian counterparts in *core industries*. We have discussed the main explanations offered in the literature for China’s fast catching up – technological upgrading, economies of scale, a regime of intense *coercive competition* known in China as *involution* and a very successful developmental industrial policy – arguing that these are best understood as jointly constitutive moments of a single historical process. Mobilising the conceptual apparatus developed in Chapters 2, 3 and 4, we have argued that this process threatens the arrangement which underpinned the profitability of U.S.-American *lead firms* during the globalisation era, and hypothesised that it may be triggering a transition from *corespective* to *coercive* competition among U.S. firms.

The research question to be answered in the next chapter can now be stated explicitly: *has the rise of Chinese competition in high-value-added sectors altered the competitive regime in which large U.S. firms operate?* More precisely: *is the relationship between exposure to Chinese competition, on the one hand, and the profitability and investment behaviour of U.S. firms, on the other, evolving over time in the direction predicted by our four hypotheses?*

To answer this question, the next chapter constructs a firm-level panel of large, publicly listed U.S.-American firms over the 2000–2024 period.

Chapter 6

Empirical Investigation of the Regime Shift Hypothesis

This chapter puts the regime shift hypotheses formulated in Chapters 4 and 5 to the test. Our aim is to investigate four testable implications of the narrative which we present in Section 5.3: a reversal of the association between exposition to Chinese imports and profitability (H.1), an intensification of competitive pressure amongst U.S. firms (H.2), a structural break in investment behaviour (H.3) and coerced investment in the sector most exposed to Chinese import penetration (H.4).

The approach we devise in this chapter is the following. We construct a firm-level panel of large listed U.S.-American firms over the 2000–2024 period, whose originality lies in an indicator of exposition to competition from Chinese imports obtained by mapping product-level trade data onto individual firms. We then split the observation period into three successive windows – 2000–2008, 2009–2016 and 2017–2024 – so as to let a regime shift appear, if there is one, as a change across windows.

The chapter is organised as follows: Section 6.1 discusses some methodological considerations which pertain to the testing of our theoretical framework using a panel regression; Section 6.2 documents the construction of the dataset and of the exposition indicator; Section 6.3 covers the econometrics required for estimating profit-rate dynamics and investment functions across the three windows; finally, Section 6.4 draws the conclusions

of the empirical work.

6.1 Methodological considerations

Testing our hypotheses requires making a series of decisions regarding which data we deem relevant to investigate and which indicators we use to operationalise the concepts discussed in the previous chapters. Because our theoretical framework insists heavily on the importance of sunk costs and of illiquidity in *core-industries*, we choose to concentrate on ‘*large*’ firms, quoted publicly in the United States. The choice to focus on publicly quoted firms is simply made to ensure data-availability. We define as ‘*large*’ any firm which, in the timespan of our panel, always holds more than \$1 billion in total assets.

Furthermore we narrow down the scope of the firms which we investigate by removing certain sectors, which we identify by SIC classification of firms in our panel. Because profitability in the Finance, Insurance and Real Estate (FIRE) sectors¹ follows a strictly distinct logic from productive activity (Shaikh 2008), we remove companies operating in these sectors from our sample. Similarly, competition, profitability and pricing in the primary sector are ruled by a logic of ground rent and associated geopolitics; we consequently remove firms operating in the primary sector² from our dataset. Following a similar logic, we exclude transportation and communications, wholesale and retail as their exposition to foreign competition if there is any must occur through foreign investment, rather than through trade in goods and services, making it irrelevant to our hypothesised mechanism and our focus on trade imbalances³.

Our theory supposes that if foreign competition is to hurt a firm’s profitability, it will do so as foreign competitors import final goods equivalent to the ones that firm sells on the domestic market. Trade statistics, however, also record rising imports when that very firm has offshored its production and imports final goods made by its own subcontractors or subsidiaries. Since there does not exist readily available data which distinguishes between these two types of imports, we measure both possibilities with a single variable,

¹Which we identify through their SIC code as $SIC \geq 6000$.

²Which we identify through their SIC code as $SIC \leq 1500$.

³Which we identify through their SIC code as $SIC \leq 4000$.

which we call *exposure*. We say that a firm is *exposed* to Chinese imports in proportion to the volume of imports from China of products equivalent to those it sells. We describe below how we compute this indicator.

Operationalising our research question requires identifying precisely which variables stand in for the quantities of our theory. This matters most when computing economically meaningful quantities – such as profit rates – from financial data. We measure profits with EBIT, and the stock of capital with total assets (book value). The more restrictive alternative, net property, plant and equipment, would capture only the tangible fraction of the capital advanced. This is inadequate in our framework, since our discussion of core industries and of the tech sector has insisted on the importance of intangible capital, which is by definition absent from property, plant and equipment. For all their limitations – historical-cost valuation and the inclusion of financial assets – book-value total assets offer the closest firm-level counterpart to the capital advanced by the firm, wherever that capital is located and whatever form it takes.

We measure the intensity of competition through the persistence of profit rates, following the empirical tradition initiated by Mueller (1986) and Geroski and Mueller (1990). This approach is consistent with the classical conception of competition as the turbulent equalisation of profit rates through the mobility of capital (Shaikh 2016): persistent profit-rate differentials are the empirical trace of barriers to entry and of restrained, corespective competition, while rapid mean-reversion of profit differentials is the signature of intense competition. The autoregressive structure of the firm-level profit rate is thus the direct empirical counterpart of our conception of competitive regimes.

Implementing this measurement strategy requires a firm-level panel dataset, which we construct at an annual frequency (the variables are summarised in Table 6.1). All financial variables in the panel are directly downloaded from the *Compustat* database (S&P Global 2026), which provides standardised, good-quality company-level financial data; all variables used in this work, except those relating to exposure to Chinese imports, are sourced from this database. Measuring whether a specific firm operates in a market where Chinese firms are dominant, or whether it relies on offshoring for its production, is a nontrivial task; the following section documents how we address it.

6.2 Data collection and preparation

Our approach starts by constructing a firm-level panel dataset (the variables are summarised in Table 6.1), sampled at an annual frequency. All of the financial variables in the panel are directly downloaded from the *Compustat* database (S&P Global 2026) which provides us with a standardised, good-quality database of company level financial data. All variables used in this work, except for the ones relating to exposition to Chinese imports, are sourced from this database.

Measuring whether a specific firm operates in a market where Chinese firms are dominant, or whether it is reliant on offshoring for its production is a nontrivial task. In the following section we document how we address it.

6.2.1 Metrics for Exposition to Chinese Imports

We construct our indicator relying on the CEPII-BACI database (Gaulier and Zignago 2010), a highly detailed, product-level database which measures bilateral trade flows across 200 countries at the 6-digit HS code⁴ level. For our purposes the 6-digit HS code level⁵ is too specific, therefore we construct a less granular version of the BACI database, by aggregating it down to 4-digit HS codes.

We define indicators for product-level importance, on the U.S. market as follows:

$$\text{IP}_{h,t}^{\text{US}} = \frac{X_{h,t}^{\text{China,US}}}{X_{h,t}^{\text{World,US}}}, \quad (6.1)$$

Where X denotes exports. The h, t pair denoted in subscripts indicates a specific product, by 4-digit HS code (h) at the time t . The superscripts A, B indicate the exporter(s) (A) and the importer(s) (B). For instance, the notation $X_{h,t}^{\text{China,World}}$ thus denotes all products

⁴*Harmonized System* (HS) codes are standardised 6-digit hierarchical numerical codes used globally to classify traded products. They are used for customs, taxes and statistics. The standard for HS codes is set by the World Customs Organization (WCO).

⁵HS codes are split according to a hierarchical system into 2-digit codes (chapters), 4-digit codes (chapters + headings) and 6-digit codes (chapters + headings + subheadings). The system is constructed such that any 4-digit HS code refers to the set of products designated by all 6-digit codes starting with the same 4 digits as the 4-digit code (the same works for 2-digit codes).

exported by China anywhere in the world. Our import penetration measures thus simply measure the share of the U.S. import market held by Chinese exports. It should be mentioned that products manufactured by foreign firms in China are accounted, in the BACI dataset, as Chinese exports. This is an inherent limitation of our methodology, which to our knowledge cannot be easily overcome by changing databases⁶.

6.2.2 Mapping Firms to Products

These import penetration indicators are informative, but are not sufficient to inform us about the exposition of firms individually (which may produce one or more products). For this, we require a dataset associating firms to products. This is a relatively difficult dataset to construct or obtain, and there are three approaches which we considered for this problem. The first is to associate industrial sectors (which are attributed to firms in Compustat in the form of SIC codes) to products and then to use this correspondence to associate firms to products. This approach, while being easiest to implement, is also problematic for two main reasons. First, because there is likely misclassification of firms into industrial sectors. This case is sometimes unavoidable, as multi-product firms may be operating in two separate SIC sectors. Second, because firms which are highly specialised are bound to be wrongfully attributed the full range of products of the industrial sector they are assigned, regardless of their true output.

The second approach would be to rely on a specific database relating products to firms: the *panjiva* and *importgenius* databases would allow for this. Compared to the aforementioned products-to-industrial sector approach, using a firm-level map holds the promise of yielding much more precise results, but access to these databases is expensive and we did not secure them.

The third is to construct our own product-to-firm database using publicly available data. This is feasible as publicly listed firms in the United States are forced to report about their activities to the Securities and Exchange Commission, which means there exist publicly available data in the form of filings. These can be exploited for associating firms

⁶To the best of our knowledge, no such trade statistics are collected at the product level; it may be possible using a more aggregate database such as the AMNE database, but this would not be compatible with the kind of firm-level econometrics which we are performing here.

to products.

In this work, we explore the first and the third options. The next subsection (Subsection 6.2.3) is dedicated to explaining our approach to constructing a firm-product map from public filings, a problem which we approach by devising an appropriate data-mining method. The use of an industry-to-product database is much more straightforward and is not discussed in a separate section (we discuss it in Subsection 6.2.4).

6.2.3 NLP Pipeline for Associating Firms to Products

Our approach to constructing a firm-to-product database consists in setting up a data-mining pipeline using an ad-hoc natural-language processing (NLP) method, to construct a map from firms to products.

Our pipeline (illustrated in Figure 6.1) bears striking similarity to Daryanani (2026)’s approach, but with a few key differences. First, Daryanani (2026)’s pipeline is constructed to parse natural language descriptions of products, whereas our pipeline parses financial reports (SEC 10-K filings) from companies. Second, Daryanani (2026)’s pipeline relies on OpenAI or Anthropic APIs which are quite expensive to run, whereas ours relies on cheap, open-source models (specifically `qwen2.5:7B` and `deepseek-v4-flash`). What is shared between this pipeline and Daryanani (2026)’s is the reranking-scheme which we describe below.

Our pipeline (summarised in Figure 6.1) works as follows. First starting from a list of company IDs (SEC Ticker⁷ or CIK codes⁸) extracted from *Compustat*, we programmatically download 10-K filings from the SEC’s EDGAR database and convert the documents from XML to a markdown format. SEC 10-K filings⁹, contain a ‘business’ section describing the company’s activity which necessarily contains a description of a company’s products;

⁷SEC Tickers are short alphanumeric codes used to trade a stock on an exchange. Their standard is set by the U.S. Securities and Exchange Commission (SEC); they are usually constructed as an abbreviation of the company name.

⁸Central Index Keys (CIK) are 10-digit codes assigned by the SEC to identify corporations, funds and individuals who file public disclosures. Unlike tickers which can change with changes in the company’s name, CIK codes are stable as long as a firm stays publicly traded on the U.S. stock market.

⁹10-K filings are annual reports, detailing a U.S.-American publicly traded company’s financial performance and business operations. Their annual publication is required by the SEC which collects these reports and distributes them.

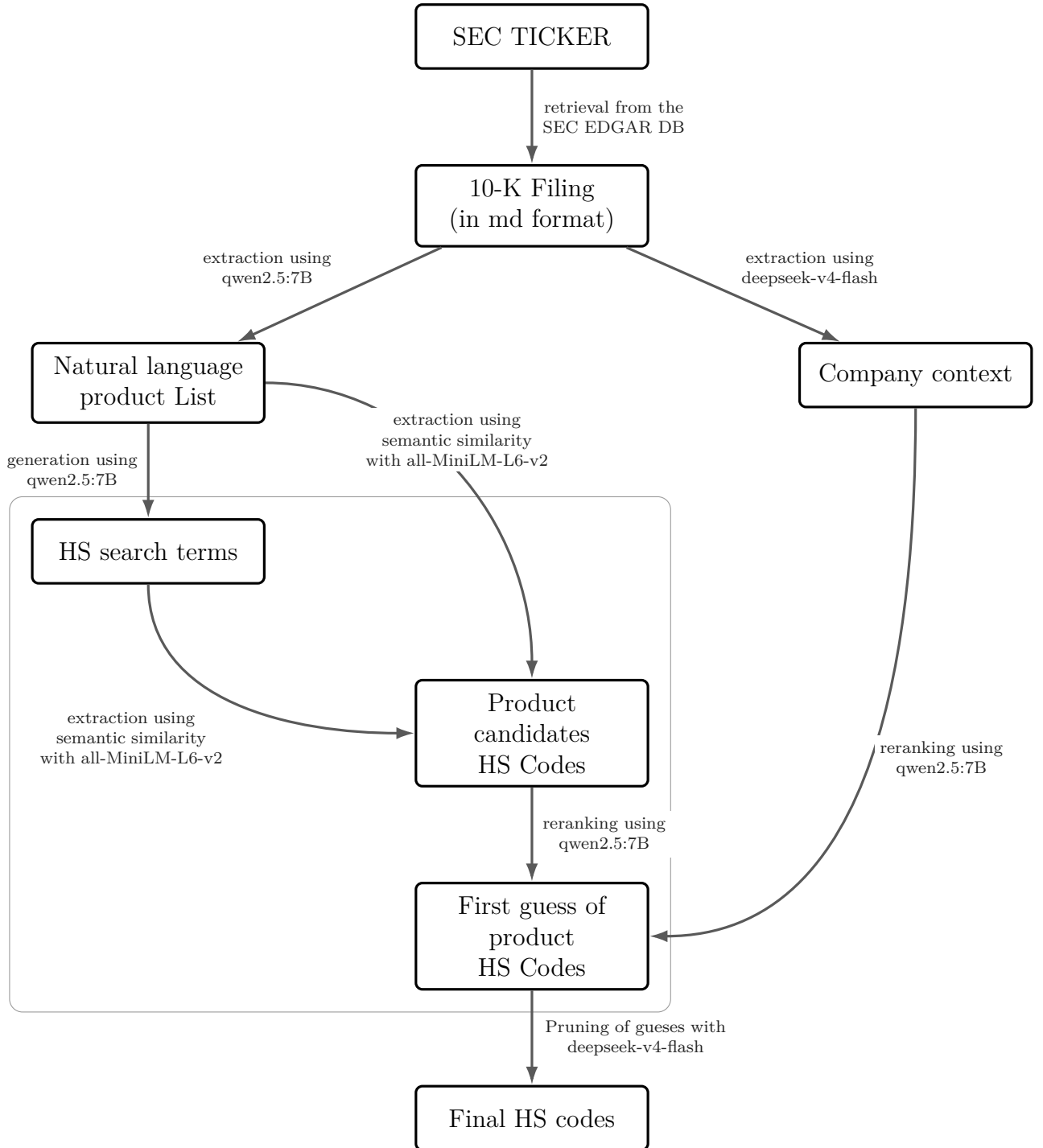


Figure 6.1: Natural language processing pipeline for extracting the company-to-product map for a single SEC-Ticker. This is run on the entire list of tickers from the *compustat* database. The steps which are enclosed in a grey box follow the same method for attributing HS-codes as Daryanani (2026).

they are the source of information from which we construct our map. Our pipeline extracts two text strings from the filing. First, we extract a natural language product list (for instance, for the ticker AAPL associated with the computer company Apple, this would be ‘*Smartphones, Personal Computers, Tablets, Wearables, Accessories*’) using an instance of `deepseek-v4-flash`, appropriately prompted. Second, we extract a *context*. A single sentence summary of the company’s activity, using an instance of `qwen2.5:7B`, appropriately prompted¹⁰. This context greatly reduces how much the models hallucinate later on, for our Apple example, the context would be: ‘*Designs, manufactures, and markets smartphones, personal computers, tablets, wearables, accessories, and related services*’. The natural language list of products is then run through a simple text search algorithm: for each product we generate a list of HS search terms using an instance of `qwen2.5:7B`.

For our Apple example, the first product is ‘*Smartphones*’, our LLM generates the search terms: ‘*Consumer goods, Telecommunications equipment, Electrical machinery, Ceramic, products, Plastics and articles thereof*’. The search terms, as well as the product name are then used to search for 20 candidates in the complete database of 4-digit HS codes. The candidates are the 20 best matches by semantic similarity across all 4-digit HS codes to the products’ natural language name. Semantic similarity is computed using the `all-MiniLM-L6-v2` model (Reimers and Sentence-Transformers Team 2026). Note that we are getting a lot of false positives. This is a result of a conscious design choice to avoid false negatives (which comes at the cost of more false positive results). This is because the next step (reranking) gracefully removes false positives, but is not suited to handle a false negative.

At this point we run one more instance of `qwen2.5:7B`, to which we feed the product names, the candidate HS code, together with their descriptions, as well as the context variable to refine the initial selection produced using our semantic similarity model. This step is known as re-ranking (Yinxin Zhou et al. 2025). For each product, the LLM outputs between 0 and 5 HS codes. The HS codes of all products are then grouped together (dupli-

¹⁰The mixed usage of `deepseek` and `qwen` instances is motivated by the fact that `qwen2.5:7B`, with 7 billion parameters, is a much smaller and cheaper to run model than `deepseek-v4-flash` which on inference uses 13 billion parameters, and is a reasoning model, which costs more computing power. Making the pipeline run fast and cheaply was a concern. In the end parsing a full dataset costs about \$5 of credits, on the LLM platform *openrouter*.

cates are eliminated) and a final pass of a well-prompted instance of `deepseek-v4-flash` is run. It is fed the output of the preceding steps (HS codes and their descriptions) as well as the context string, and it is instructed to remove incoherent HS codes. We find that this last pass by a slightly larger model greatly helps with limiting hallucinations.

For every ticker in the *Compustat* dataset, we run this pipeline to compute a firm-to-product mapping. We run it on the last available 10-K filing, which is a limitation¹¹ of our method (since a firm’s list of products may be changing over time).

6.2.4 Indicators of Firms’ exposition to Chinese Imports

We just described an approach to construct a product-to-firm correspondence map using an NLP pipeline. This approach has the clear advantage of enabling us to create large, granular firm-level datasets at very little cost. The drawback is the uncertainty brought about by the use of modern NLP techniques, especially LLMs, which are known to hallucinate. Considering the size of the dataset, methodical human checking of the pipeline’s output is not a viable approach. Instead, we run robustness tests with another, simpler method.

We use the 2024 version of Pierce and Schott (2009)’s concordance key which maps SIC (Standard Industrial Classification) to HS codes. But such an approach is, we argue, problematic in a different way. As 4-digit SIC codes are much less granular than 4-digit HS codes, a given industrial sector is bound to be associated with the production of many more products than what a single firm may actually be producing. Such an approach is thus likely to introduce quite a bit of noise in our signal.

Regardless of whether it is constructed using an NLP pipeline or from an industrial sector to product correspondence map, a mapping associating firms with products can formally

¹¹This is not so much a fundamental limitation as it is the result of budget and time-constraints. Our method could in principle be run on archived filings for every year, yielding a more granular source of data.

be written in the form of an indicator function:

$$\mathbf{1}(h, k) = \begin{cases} 1 & \text{if product } h \text{ is produced by firm } k, \\ 0 & \text{otherwise.} \end{cases} \quad (6.2)$$

We use this indicator function to compute a proxy for the exposition of each firm to Chinese competition. Specifically we define a function measuring the exposition of a given firm to Chinese competition, as:

$$\text{EXP}_{k,t}^{\text{US}} = \left(\sum_{h \in \mathcal{HS}} \mathbf{1}(h, k) \cdot \text{IP}_{h,t}^{\text{US}} \right) / \left(\sum_{h \in \mathcal{HS}} \mathbf{1}(h, k) \right). \quad (6.3)$$

Where $\text{EXP}_{k,t}^{\text{US}}$ measures the intensity of competition according to the importance of Chinese imports on the U.S. market (relative to other importing countries). The set $\mathcal{HS}$ is simply the set of all $\mathcal{HS}$ product codes included in the dataset.

A clear limitation of our method is that we do not know how important (in terms of investment and of revenue) each product is for each firm. We thus always implicitly work under the hypothesis that firms are equally competing in the markets for all the products they were making in the last 10-K filing which we have. An improvement of this metric would be to weigh the import penetration by revenue share for the firm, but this is data which we do not have access to, for the large subset of firms which we consider¹². This issue is especially problematic when using the Pierce and Schott (2009) concordance key¹³, and should be less of an issue when using the NLP-based concordance.

6.2.5 Characteristics of the Panels used in this work

Table 6.1 summarises the variables in our panel; they are obtained by merging *Compustat* data with our exposition indicators computed using the BACI dataset and either our own firm-to-product map or the one we constructed using the Pierce and Schott (2009) correspondence map. For the reasons we set out in Section 6.1, we select firms with SIC

¹²It is perhaps possible to reconstruct this information from smartly parsing 10-K filings as well, but this is beyond the scope of this work.

¹³Since it maps firms to products by their SIC codes, it likely always includes products which the firm is actually not selling in the average exposition.

Variable	Indicates	Computation	Origin
ROP Rate of profit (total assets plant and equipment)	Profitability	$r_{k,t} = \text{EBIT}_{k,t} / \text{AT}_{k,t-1}$	Compustat
AT book value of total assets	Size	$\text{AT}_{k,t}$	Compustat
CAPINT capital intensity	Capital intensity	$\text{CAPINT}_{k,t} = \text{PPENT}_{k,t} / \text{AT}_{k,t-1}$	Compustat
IR Investment rate (total assets)	Profitability	$\text{IR}_{k,t} = \text{CAPEX}_{k,t} / \text{AT}_{k,t-1}$	Compustat
LR Leverage ratio	Financial fragility	$\text{LR}_{k,t} = \text{DT}_{k,t} / \text{AT}_{k,t}$	Compustat
EXPUSV Exposure indicator	Exposure to Chinese Imports	$\text{EXP}_{k,t}^{\text{US},v}$ (see Section 6.2.4)	Compustat, BACI, SEC-EDGAR
RELCOGS Cost of goods sold (adjusted by total revenue).	Control	$\text{RELCOGS}_{k,t} = \text{COGS}_{k,t} / \text{REVT}_{k,t}$	Compustat
RELCHC Cash and cash equivalent (adjusted by total revenue).	Control	$\text{RELCHC}_{k,t} = \text{CHE}_{k,t} / \text{REVT}_{k,t}$	Compustat

Table 6.1: Variable definitions and computation

codes between 1500 and 4000, and total assets above \$1 billion. The filtered panel we thus obtain is composed of 678 U.S.-listed public firms. When computing profit rates, we apply a one-period lag on total assets, to abide by the convention that the income should be matched by the asset base that generated it.

Because we are interested in the hypothesis of a change of competitive regime, we will divide this panel into subpanels, which allow us to investigate the evolution of the relationships we care about over time. We produce three ‘*windows*’ (time-restrictions) from our original 25-year-long panel for the periods 2000–2008, 2009–2016 and 2017–2024; because firms fail and are created, we have a slight difference in the number of firms across time-windows. This periodisation is not data-driven; it is simply an even split (or nearly even, as 25 is not divisible by 3) into three subsets. Because financial datasets tend to

contain extreme outliers, they are often unduly influenced by them. To deal with this issue we winsorise observations at the upper and lower 1% (Ruppert 2014).

Time window	2000–2024	2000–2008	2009–2016	2017–2024
Number of firms k	678	531	560	522
Number of years t	25	9	8	8
Number of k,t pairs	12728	4331	3882	3717

Table 6.2: Summary table of the shape of the panels used for the regressions in this work.

Each window of our panel is a typical micro-panel with large K (number of firms, of the order of 500 units) and small T (at most 25, most often around 8); it is also heavily unbalanced as firms enter and leave the stock market at the pace of IPOs and bankruptcies.

6.3 Data Analysis: Chinese Competition and Firm-Profitability in the U.S.

We now turn to the analysis of our dataset; the following section is split up in three parts. First, we quickly go through descriptive statistics using our panel as well as aggregate data sources; second, we study the relationship between profitability and exposition to Chinese imports over time with a well-specified regression. Lastly, we study the hypothesis of a structural break in investment behaviour through the regression of an investment function.

6.3.1 Descriptive Statistics

First, let us consider the evolution of the distribution of the profit rate $r_{k,t} = \text{EBIT}_{k,t} / K_{k,t}$ in our panel (Figure 6.2). The first point to be underlined is the fact that at any time t , a large share of publicly quoted firms report a negative profit rate (between 10% and 25% until 2016, more than 25% afterwards). Furthermore, the distribution is asymmetric and fat-tailed. Firms in the lower quantiles tend to report losses of a higher magnitude than the profits reported by firms in the upper quantiles. This does not mean that the average and median profit rates are negative; they are consistently positive in our sample.

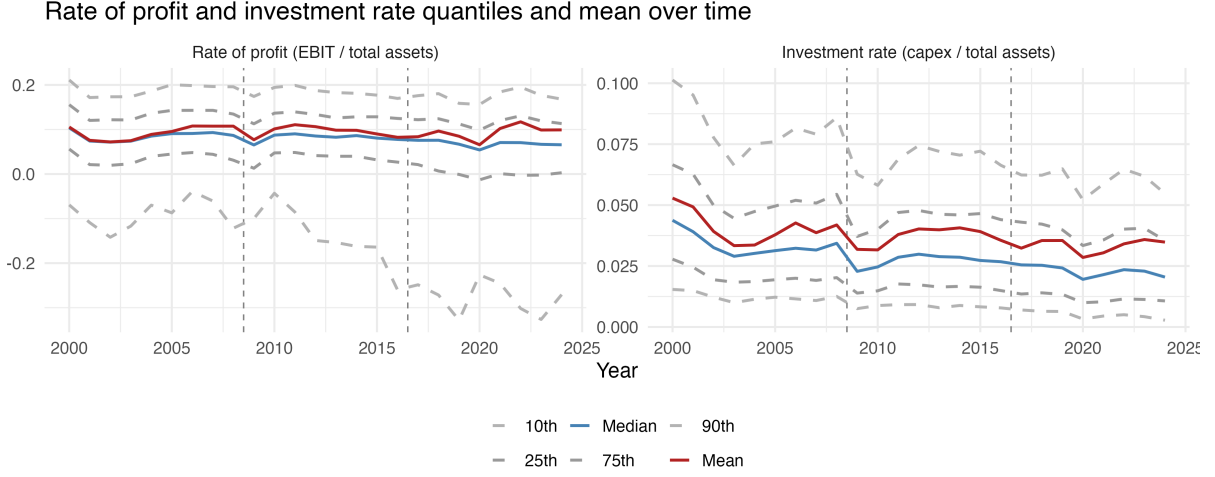


Figure 6.2: Evolution of the distributions of $r_{t,k}$ and $I_{t,k}/K_{t,k}$ over time, as a quantile time-series plot.

This asymmetry in the distribution was already observable before the 2008 crisis but has drastically picked up from 2008 until 2016; afterwards it has stayed relatively stable. The mean profit rate is also a relatively stable quantity in our sample, with a small spike after the COVID crisis and a clear dip at the time of the 2008 crisis.

Next, let us consider the evolution of the rate of investment $IR_{k,t} = I_{k,t}/K_{k,t}$. While it appears to have been on a downward trend in the 2000–2008 period, it has stayed roughly stable since then (with dips in 2008 and 2019). Since the COVID period, there appears to be a trend towards more investment in the mean but not the median, suggesting a few firms are driving investment up. Our dataset stops in 2024, largely missing the AI-driven investment surge we are currently witnessing.

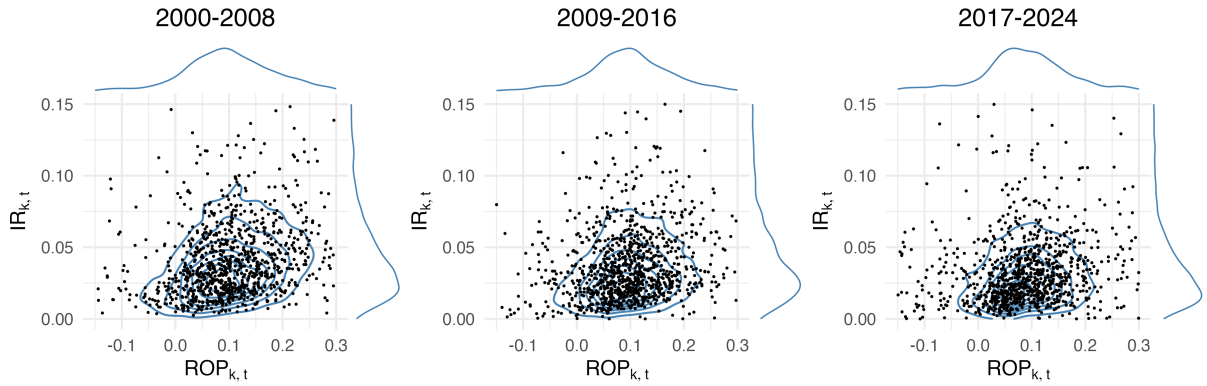


Figure 6.3: Joint density plot of the profit rates $r_{k,t}$ and investment rates $IR_{k,t}$ over all three windows of our panel. From left to right the plots are respectively for 2000–2008, 2009–2016, 2017–2024.

The joint distribution for these two variables ($r_{k,t}$ and $IR_{k,t}$) are plotted separately for each window, together with their estimated marginal distributions in Figure 6.3. Although these plots reveal what appears to be a visible positive correlation, they should not be interpreted too hastily. First, because they fail to represent any confounder variable other than the profit rate and second because they fail to cluster multiple instances of the same firm together (on each plot each firm is attributed up to 9 points for the number of years it appears in the dataset). What can be interpreted from them is the fact that the data from the 2000–2008 period appears to be more noisy (the marginal on the profit rate is especially illustrative).

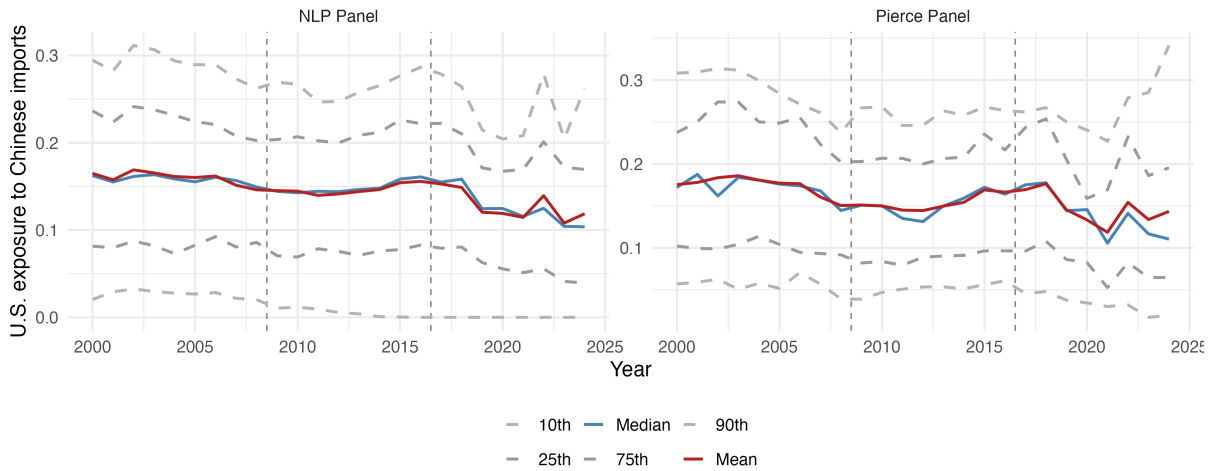


Figure 6.4: Evolution of the distribution of the exposition metric $EXP_{k,t}^{US}$ across firms using the NLP-constructed indicator (left) and the Pierce and Schott (2009) sector-based indicator (right).

As a first diagnostic of our so-produced exposition indicator, we include quantile time-series plots of both indicators on our panel in Figure 6.4. Comparing both plots highlights a few basic facts about both indicators. First, the NLP-constructed indicator gives more low values: for most of the panel time, the 0.1 quantile line plot is at 0, signifying there are always at least 10% of firms which are not exposed to Chinese imports. This is not the case using the sectoral indicator: here the 0.1 quantile is always above 0. Second, both indicators show a reduction in exposition in the 2000 to 2008 period, then exposition slowly picked up after the 2008 crisis. Perhaps most interesting is the fact that the exposition distribution widens from 2017 to 2024. In other words, some firms are becoming less exposed while some are becoming more exposed. Lastly, both indicators record a dip during the COVID crisis; this dip matches the dip in investment which is

clearly visible in Figure 6.2, showcasing that the COVID shock affected both domestic investment in the U.S. as well as international trade in general.

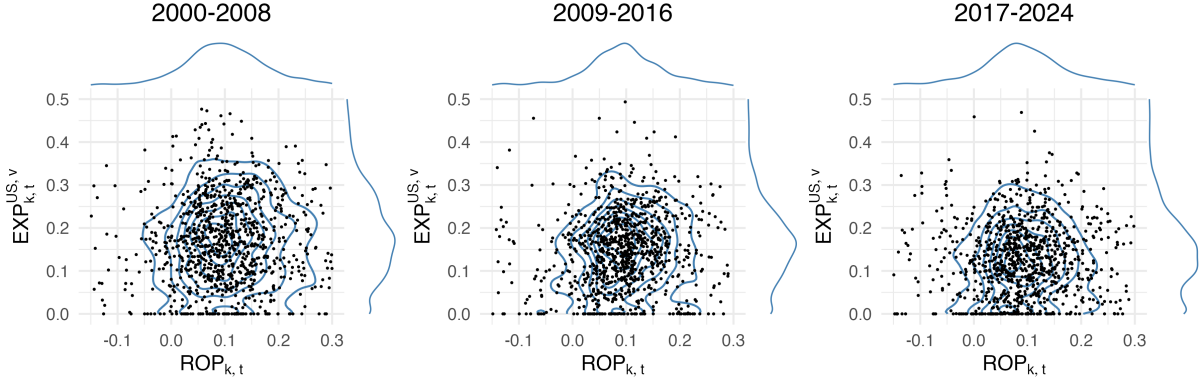


Figure 6.5: Joint density plot of the profit rates $r_{k,t}$ and the NLP exposition indicator $EXP_{k,t}^{US,v}$ over all three windows of our panel. From left to right the plots are respectively for 2000–2008, 2009–2016, 2017–2024.

We include a joint distribution plot of the profit rate against our exposition indicator (Figure 6.5). These joint distribution plots are not easily interpretable and on first appearance appear to show no clear correlation between the two variables. As for the profit-rate-investment-rate joint density plot, this is not to be interpreted too hastily.

We also include a descriptive plot (Figure 6.6) summarising the sectoral decomposition of the different companies in the dataset, measured both in number of firms and in share of total assets. What is perhaps most striking is the collapse of the importance of transportation equipment companies in total assets. Another key aspect is the increase in importance of the chemicals and allied products sector (which includes pharmaceuticals) as well as the electronics and electrical equipment sector.

6.3.2 Profitability dynamics

The first relationship we investigate is that between *exposition* to Chinese exports and profitability, both measured at the firm level. As we previously introduced, *exposition* is a measurement of whether a given firm is competing in a sector in which China is a large exporter. This is measured by the indicator variable $EXP_{k,t}^{US,v}$. Profitability is measured through the profit rate $r_{k,t} = EBIT_{k,t}/AT_{k,t}$.

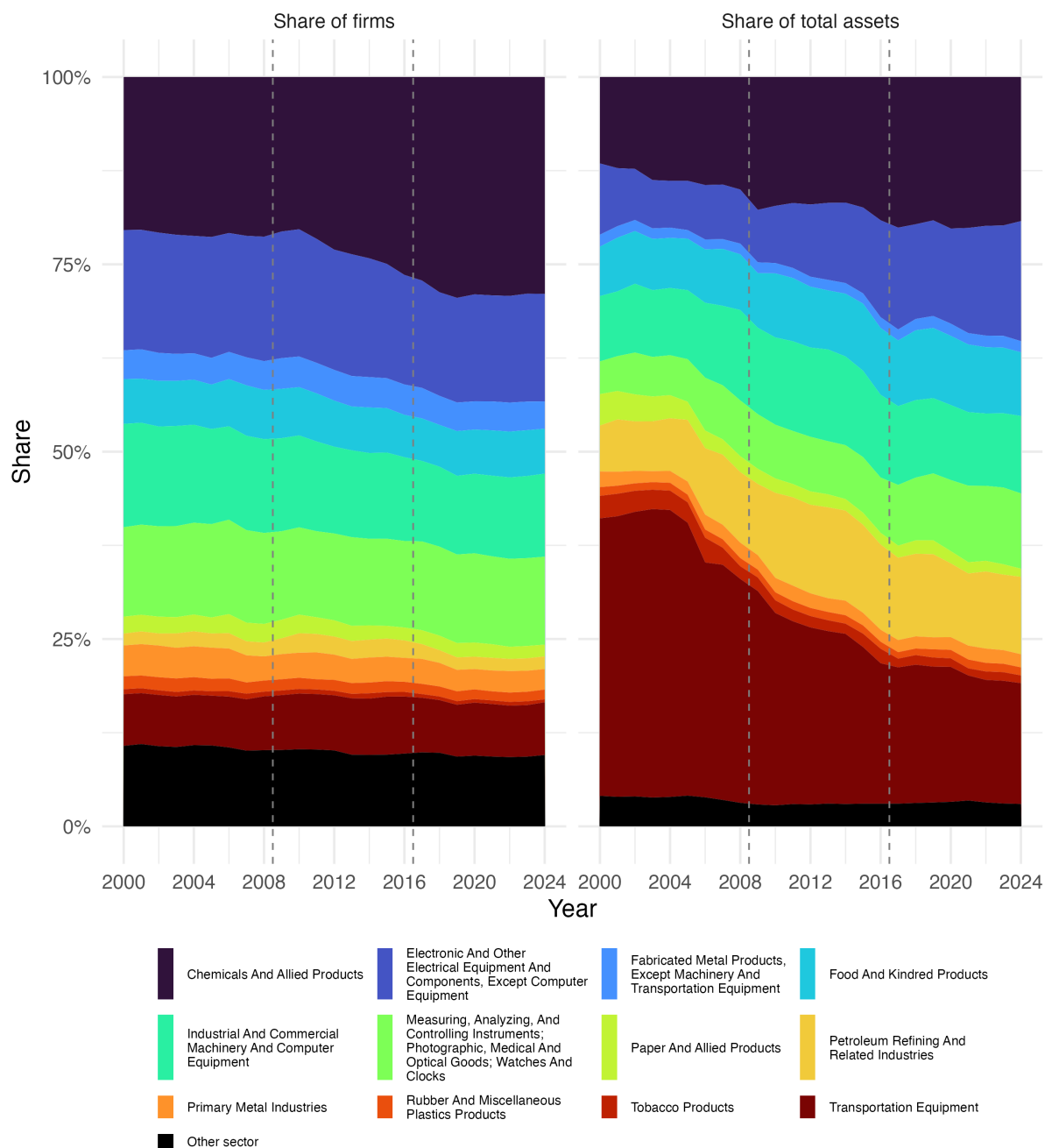


Figure 6.6: Panel composition by share of publicly traded firms (left) and by share of total assets (right) over time by SIC major group.

We are interested, specifically, in how the *exposition-profitability* relationship evolves over time. To do so we will run three identical regressions on three different time windows (2000–2008, 2009–2016, 2017–2024) drawn from our panel.

Model specification The model which we use for relating profitability to exposition to Chinese imports is the following:

$$r_{k,t} = \phi_1 r_{k,t-1} + \phi_2 r_{k,t-2} + \beta_1 \text{EXP}_{k,t}^{\text{US},v} + \beta_2 \text{EXP}_{k,t-1}^{\text{US},v} + \vec{\gamma}^\top \vec{X}_{k,t} + \lambda_t + \varepsilon_{k,t}, \quad (\text{MROP.1})$$

We include two lags of our two main explanatory variables $r_{k,t}$ and $\text{EXP}_{k,t}^{\text{US},v}$. As we have discussed at length in the preceding chapters, being active in a sector in which Chinese firms are large exporters bears ambiguity. We therefore have no clear expectation on the sign of this effect.

At this point the reader will remark that we have specified an AR(2) model which describes the evolution of profit as a dynamical process. There are two reasons why we choose to model the profit rate in such a manner. First, because this regressor significantly explains our dependent variable, consequently running our regression without the autoregressive terms would lead to omitted-variable bias which would affect our coefficient estimator, and bias our standard error estimator downwards (Arellano 1987; Baltagi 2021). But secondly and more interestingly, we include this term because competitive processes are well modelled as dynamic autoregressive processes of the profit rate variable, driven by the (unobservable) entry, threat of entry and exit of competitors (Mueller 1986; Geroski 1990). There is thus a meaningful interpretation of the AR(1) coefficient associated with the profit rate as relating to the intensity of competition. This is of particular interest to this work as we are investigating not only profit dynamics, but shifts in the regime of competition over time. Therefore, we expect a positive sign and a magnitude smaller than 1 on ϕ_1 and an unknown sign with a magnitude also smaller than 1 on ϕ_2 .

We include a vector of controls for each firm-year pair, which is defined as:

$$\vec{X}_{k,t} = \left[\log(1 + \text{LR}_{k,t-1}) \quad \log(\text{CAPINT}_{k,t-1}) \quad \log(1 + \text{RELCH}_{k,t-1}) \right]^\top. \quad (6.4)$$

First, the leverage rate $LR_{k,t} = DT_{k,t}/AT_{k,t}$, which is an indicator of financial fragility. High leverage raises fixed interest obligations, directly compressing net profitability, and makes taking on new debt more difficult. Controlling for the leverage ratio should separate the mechanical effect of financial fragility on profits from the competitive pressure channel. We expect a negative sign. Second, an indicator of capital intensity $CAPINT_{k,t} = PPENT_{k,t}/AT_{k,t}$ which is computed as the ratio of fixed capital over the total assets of the firm. As we previously discussed at length, capital-intensive firms typically have higher operating costs and more rigid cost structures (sunk costs); this may affect both their average profitability and their sensitivity to shocks. And last, an indicator of the liquidity available to the firm $RELCH_{k,t} = CHE_{k,t}/REVT_{k,t}$ which is computed as cash and cash equivalents over total revenue. By including this indicator, we control for the confounding role of pre-existing financial buffers which may enable firms to endure competitive pressure without immediately sacrificing profits. These three controls are included in logs or shifted logs because their raw distributions are heavily right-skewed and contain zero values; taking logs reduces the influence of outliers and makes the relationships with profitability more linear. For leverage and relative cash holdings, adding 1 before the log transformation avoids undefined values when the ratio equals zero, while preserving the information from those observations; this is not required for $CAPINT_{k,t}$ which is bounded away from 0. For all three controls, we also regress on the lagged variables to match the convention used for total assets. We use time dummies λ_t for each time step to capture time-specific shocks which could be acting as confounders.

Estimation methodology As our panel specification is dynamic and because our time windows are relatively short (9 or 8 years), we know that we are in the typical situation in which Nickell bias (Nickell 1981) arises in fixed-effects panel estimators. For this reason we use a Difference-GMM¹⁴ (Arellano and Bond 1991) estimator to get an unbiased estimate. Difference-GMM differentiates the model to remove individual effects and uses the lagged dependent variable as an instrument for the lagged dependent variable used in the regression. In our case we use lags 3 through 5 for the 2000–2008 time window and lags 2 and 3 for the two others. As expected for well-specified Difference-GMM regressions, for all three windows, testing for serial correlation of the error terms shows

¹⁴Sometimes also known as an Arellano-Bond regression, after its inventors.

they are exactly AR(1): we reject the null on an AR(1) serial correlation test but we fail to reject it on an AR(2) test. We use collapsed instruments, keeping our instrument count way below the $K/2$ rule of thumb for avoiding instrument proliferation (Roodman 2009). As an additional check against overidentification, we run a Sargan-Hansen test (Hansen 1982), for which we fail to reject the null. This is what we expect for a model not suffering from instrument proliferation. The p-value for these three tests, as well as instrument counts, is reported for each window in Table 6.3. We use Windmeijer standard errors for our computation of confidence intervals and significance of results (Windmeijer 2005).

Table 6.3: Profitability function: Difference-GMM across time windows. Windmeijer-corrected robust standard errors.

	2000- 2008	2009- 2016	2017- 2024
$ROP_{k,t-1}$	0.365 (0.242)	0.674*** (0.068)	0.545*** (0.066)
$ROP_{k,t-2}$	0.091 (0.099)	0.003 (0.049)	-0.307*** (0.042)
$EXP_{k,t}^{US,v}$	0.156*** (0.046)	-0.103 (0.071)	-0.111* (0.066)
$EXP_{k,t-1}^{US,v}$	-0.057 (0.045)	-0.105 (0.075)	-0.156*** (0.045)
$\log(1 + LR_{k,t-1})$	-0.048 (0.030)	-0.008 (0.045)	0.070 (0.045)
$\log(1 + RELCHE_{k,t-1})$	-0.027 (0.039)	0.082*** (0.031)	0.029 (0.020)
$\log(CAPINT_{k,t-1})$	0.048** (0.023)	0.020 (0.017)	0.034 (0.021)
K	531	560	522
T	9	8	8
Obs	4331	3882	3717
AR(1) p-value	0.040396	4.186e-07	9.7388e-14
AR(2) p-value	0.86706	0.70431	0.38844
Sargan/Hansen p-value	0.097489	0.17909	0.098923
Instrument count	29	20	20

* p < 0.1, ** p < 0.05, *** p < 0.01

Estimation results and interpretation Here the reader should be reminded that our panel is exclusively composed of large, publicly traded U.S.-American firms, and we have restricted our panel to firms that operate in the manufacturing sector. We are thus here, for a large part, considering *lead firms* playing a controlling role in GVCs, many of which operate in *core industries* and some of which may be factory-less. We expect that the results which we obtain here will likely diverge from what a researcher working with census data (such as Milberg and Johnston (2025)) would obtain, as census data covers firms exhaustively and thus contains a large number of small firms with a much lower degree of internationalisation of their activities.

Table 6.3 presents the results of the estimation; the same results are presented in the form of a coefficient-plot in Figure 6.7. The most striking finding is the evolution of the relationship between exposition to Chinese imports and profitability. In the first period (from 2000 to 2008) there is a positive significant association (at the 1% confidence level on the unlagged exposition indicator) between exposition to Chinese imports and higher firm profitability. In the second window (2009 to 2016) there is no significant association, and in the last window (2017–2024) there is an unambiguous negative, statistically significant association (at the 10% level on the unlagged exposition indicator and at the 1% level on the lagged indicator) between Chinese imports and firm profitability. In both cases the coefficient is of a low magnitude¹⁵.

This suggests the following interpretation: while in the first period (2000–2008) firms (given the composition of our panel, mostly *lead firms*) which sold products for which production was highly China-centric were benefiting from this situation. Likely because this period was one in which operating with high exposition to China was associated with being a successful offshoring firm, reliant either on in-house offshoring through FDIs or on subcontractors. For the following period (2009–2016) there is no significant association, likely because this is a transition period in which situations of increased profits from offshoring and lowered profits from competition coexist. For the last period (2017–2024) this is not the case anymore. On the contrary, we have significant evidence that firms in China-dominated sectors tendentially face lower profits, as attested by the lagged

¹⁵This is to be expected as $\text{EXP}_{k,t}^{\text{US},v} \in [0, 1]$ is a share of imports from China, across a given subset of products. For instance a β of 0.088 is to be interpreted as ‘if all the products of this firm were exclusively imported from China, then that firm would gain 8.8% in profit rate’.

exposition coefficient.

These results (sign and significance of coefficients) are not completely robust to using our alternative indicator specification: the one we constructed from the Pierce and Schott (2009)-based dataset (Table D.1 in Appendix D). When using this dataset, we are able to reproduce the positive effect of exposition for the 2000–2008 period and its disappearance starting with the 2009–2016 period, but not the negative association in the last period. This could be explained in multiple ways; perhaps our NLP measurement is biased or noisy, perhaps the lack of precision of the Pierce and Schott (2009)-based dataset attenuates the association and makes it less significant. Regardless of the explanation, we judge that this lowers our confidence in this result, and warrants further inquiry.

Our results are also coherent with the findings of Milberg and Johnston (2025) who finds using value-added by sector data from the BLS, a positive correlation between the firm’s profit-share for the 2002–2011 window and a negative correlation for the 2011–2023 window. The main difference between our results and Milberg and Johnston (2025)’s is that we use a selective sample (of publicly traded firms), largely made up of *lead firms*, while they use aggregate data which reflects the entire U.S. economy (crucially not only lead firms).

Concerning the controls, we find no significant association between (relative) cash and cash equivalents $RELCH E_{k,t}$ and profitability $r_{k,t}$ for most periods except in the 2009–2016 period; we register a positive association (at the 1% threshold). The leverage rate is not significantly associated with profitability for any time window. Capital intensity is only significantly associated with profitability for the first window. It is positively associated with profitability at a 5% significance level.

Regarding the autoregressive dynamics of the profit rate, they appear to significantly change across windows. In the first window (2000–2008) we do not find that the autoregressive coefficients are significant. This suggests that there is a large diversity of profit-rate dynamics amongst the investigated firms. The (insignificant) AR(1) coefficient is very small, suggesting profits are on average not very persistent. This is likely induced by the large number of loss-making and failing firms during the 2007–2008 finan-

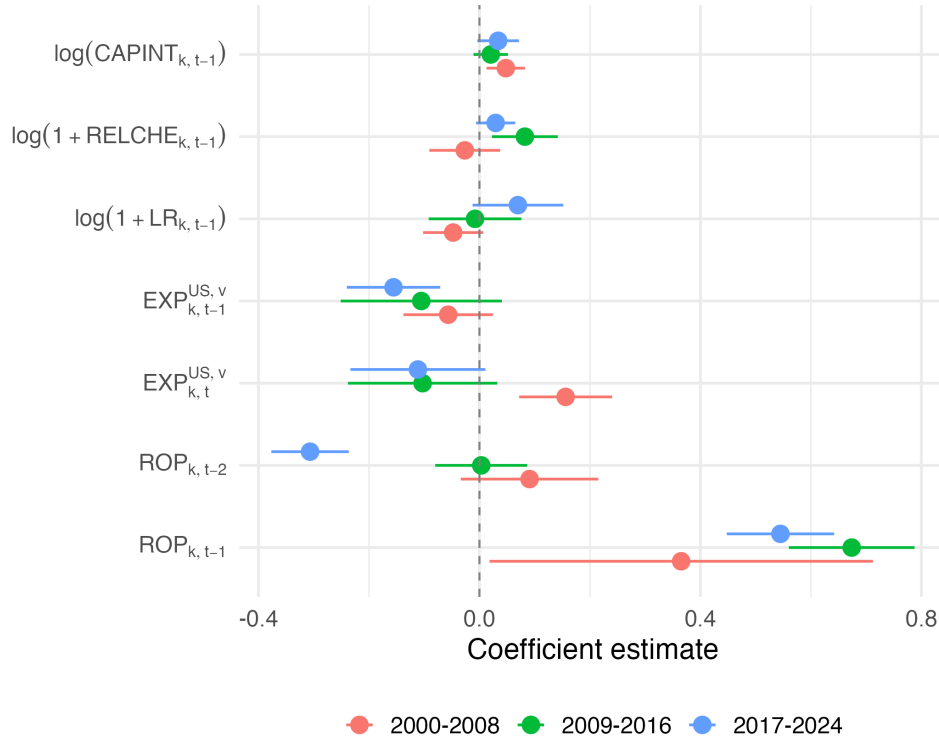


Figure 6.7: Profitability regression coefficient-plot at a 95% confidence interval for the three windows.

cial crisis, which is included in the window¹⁶. During the second window (2009–2016), we find that the AR(1) coefficient is significant and positive, while the second is non-significant (and near 0). We thus have an AR(1) process with relatively slow decay, which, as per the persistence of profits literature, is to be interpreted as an indicator of an environment of low-intensity competition (or at least lower than our other windows). During the last window (2017–2024) we find a significant positive AR(1) coefficient and a significant negative AR(2) coefficient. This describes an AR(2) process with fast mean-reversion and oscillating tendencies, indicative, as per the persistence of profit literature, of a more competitive environment. So our regression suggests that the intensity of competition increased in the last window relative to the second. The autoregressive dynamics are illustrated in Figure 6.8 which plots out impulse response functions with and without noise (where the noise has variance set according to our regression results). These plots are informative about the autoregressive component of the profit rate dynamics and ignore the other factors which influence profitability; they thus have to be interpreted with

¹⁶This can be verified by pruning the firms which have failed from the dataset and observing that the regression on this restricted dataset yields a significant, positive coefficient.

the appropriate caution.

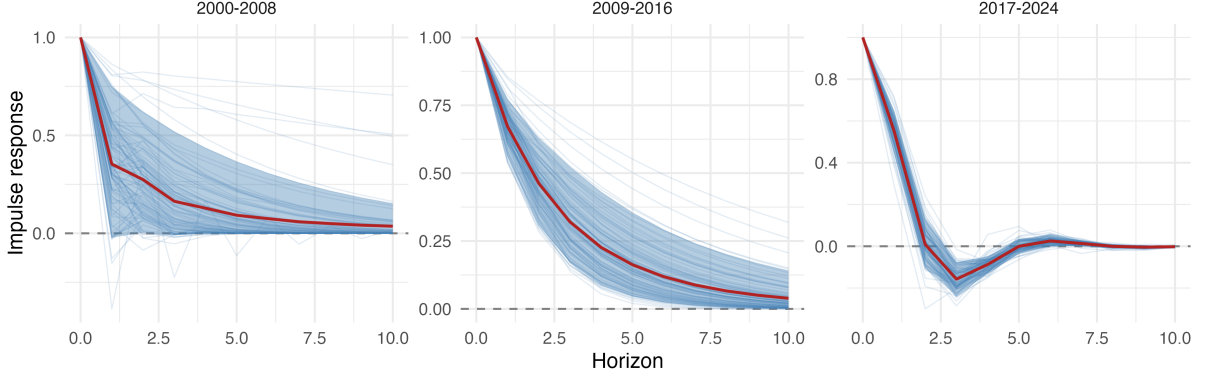


Figure 6.8: Impulse response function (in red) and simulated profit trajectories (in light blue), illustrating the mean-reversion dynamics of profit rates for the three different windows. Note the qualitative change from a highly noisy AR(1) in the first window to a slower AR(1) in the second and a fast AR(2) process in the third.

Overall these results are consistent with our first two hypotheses (H1 and H2), according to which we are seeing a reversal of the exposition-profitability relationship and an intensification of competition. This supports the thesis of an exhaustion of the profitable trade deficit model we previously described (Chapter 3). This suggests that exports from China exert a competitive pressure but it does not inform us about a possible shift in the competitive regime. In answering this question, turning to the study of investment behaviour is required.

6.3.3 Investment Functions

The second part of our investigation pertains to investment functions, as a key behavioural indicator of a shift in competitive regime. Recall the notion that the firm's investment is ruled by a growth-safety trade-off, which we introduced two chapters ago and which we expressed through the following equation:

$$\frac{I_{k,t}}{K_{k,t}} = f(\alpha_k^+, r_{k,t}^+, \varphi_{k,t}^+, i_t^-, \sigma_t^-) \quad (6.5)$$

Where $\frac{I_{k,t}}{K_{k,t}}$ denotes the investment rate, α_k a notion of risk-appetite, $\varphi_{k,t}$ a measure of the financial solidity of the firm, i_t the interest rate to which the firm is exposed, $r_{k,t}$ its current profit rate and σ_t a measure of perceived uncertainty. One of the key features of

the coercive competitive regime, according to Crotty (2017b), is that for firms engaged in coercive competition, the growth-safety trade-off stops being an appropriate model of investment behaviour. This is because what coerces them to invest – competitive pressure and prices – is independent of any given firm’s profitability.

Model specification To investigate whether the growth-safety trade-off is an accurate description of behaviour of firms in our sample, we run the following regression:

$$\begin{aligned} \text{IR}_{k,t} = & \phi_1 \text{IR}_{k,t-1} + \phi_2 \text{IR}_{k,t-2} + \beta_{\text{EXP},1} \text{EXP}_{k,t}^{\text{US},v} + \beta_{\text{EXP},2} \text{EXP}_{k,t-1}^{\text{US},v} \\ & + \beta_{r,1} r_{k,t} + \beta_{r,2} r_{k,t-1} + \beta_{\text{LR}} \log(1 + \text{LR}_{k,t}) + \vec{\gamma}^\top \vec{X}_{k,t} + \lambda_t + \varepsilon_{k,t}, \end{aligned} \quad (\text{MINV.1})$$

As with the profitability model, we model investment as an AR(2) process with two lags of the investment rate $\text{IR}_{k,t} = I_{k,t}/K_{k,t}$. Here we are not interested in explicitly interpreting the AR(2) dynamics of the investment rate, but we include it so as to avoid a dynamic omitted variable bias. To investigate the relationship between investment, transnationality and international competition, we include the exposition variable $\text{EXP}_{k,t}^{\text{US},v}$, as well as its first lag. Here, as in (MROP.1) the effect of an increase in exposition is ambiguous, and we have no expectation on the sign of this coefficient. We also include the profit rate $r_{k,t}$ together with its first lag as explanatory variables. Under the growth-safety trade-off, we expect a positive, significant coefficient. Finally we include the shifted log of the leverage ratio $\log(1 + \text{LR}_{k,t})$ as an explanatory variable, which proxies financial (in)stability. Under the growth-safety trade-off, we expect a negative coefficient. A vector of controls for each firm-year pair is also included, which is defined as:

$$\vec{X}_{k,t} = \left[\log(1 + \text{RELCH}_{k,t-1}) \quad \log(1 + \text{CAPINT}_{k,t-1}) \right]^\top. \quad (6.6)$$

Here the controls are the following $\log(1 + \text{RELCH}_{k,t-1})$ and $\log(1 + \text{CAPINT}_{k,t-1})$ which are included in shifted log form, with one lag, for the same reason as in the profitability regression. These variables are included to isolate the effect of Chinese import exposition from other firm-level determinants of investment that are correlated with it. Capital intensity $\text{CAPINT}_{k,t}$ accounts for differences in cost structure that shape the need for investment. We expect that capital-intensive sectors stay capital intensive, and thus that we record a positive, significant coefficient. Relative cash and cash equivalents likely

facilitate investment; we thus expect a positive coefficient. We use time dummies λ_t for each time step to capture time-specific shocks which could be acting as confounders.

Our goal in specifying (MINV.1) is to get as close as possible to an estimation of (6.5) while also including exposition as an explanatory factor, and avoiding misspecification (hence the dynamic specification). The interpretation of the coefficients is thus to be informed by (6.5). The parameter $r_{k,t}$ is present in the original growth-safety trade-off, and is thus to be interpreted accordingly (we expect a positive significant coefficient), and the term $\log(1 + LR_{k,t})$ is interpreted as an indicator of financial fragility, hence there should be a negative association between $\varphi_{k,t}$ and investment in the growth-safety trade-off equation. The interest rate cannot be modelled as it is shared across firms and is thus absorbed by the time-dummies which we use to rule out exogenous shocks. Risk appetite is understood to be equivalent to a firm-specific fixed effect. The fact that we use a Difference-GMM regression makes estimating firm-specific fixed effects impossible. Finally, the term for perceived uncertainty is by definition unobservable and is thus omitted from our regression.

Estimation methodology As in the previous regression, because our model is autoregressive, we know that using a naive approach would yield biased results. We thus make use of a Difference-GMM (Arellano and Bond 1991) estimator. We use three lags (2 to 4) of $IR_{k,t}$ as instruments for $IR_{k,t}$, $IR_{k,t-1}$. We use collapsed instruments, which keep our instrument count way below the $K/2$ rule of thumb. As for the previous regression, we run AR(1), AR(2) serial correlation tests and Sargan-Hansen tests of overidentifying restrictions as diagnostic tests, which validate our choice of instrument and regression technique. The p-values of these three tests are presented in Table 6.4. We use Windmeijer standard errors for our computation of confidence intervals and significance of results (Windmeijer 2005).

Estimation results and interpretation Table 6.4 presents the results of our estimation, which are also illustrated in coefficient plot form in Figure 6.9. We find that investment is persistent and follows AR(1) dynamics across windows. The AR(1) coefficient is remarkably stable across the last two windows, while it is lower in the 2000–2008

Table 6.4: Investment function: Difference-GMM across time windows.
Windmeijer-corrected robust standard errors.

	2000- 2008	2009- 2016	2017- 2024
$IR_{k,t-1}$	0.319*** (0.057)	0.617*** (0.105)	0.607*** (0.056)
$IR_{k,t-2}$	0.049* (0.028)	-0.001 (0.037)	-0.042 (0.041)
$EXP_{k,t}^{US,v}$	0.044* (0.024)	-0.024 (0.026)	0.024** (0.012)
$EXP_{k,t-1}^{US,v}$	-0.015 (0.023)	-0.019 (0.026)	0.000 (0.012)
$ROP_{k,t}$	0.078*** (0.020)	0.092*** (0.015)	0.030*** (0.010)
$ROP_{k,t-1}$	-0.004 (0.019)	-0.012 (0.012)	-0.008 (0.010)
$\log(1 + LR_{k,t-1})$	-0.053*** (0.012)	-0.049*** (0.011)	-0.033** (0.014)
$\log(1 + RELCHE_{k,t-1})$	0.007 (0.006)	-0.010 (0.009)	0.004 (0.005)
$\log(CAPINT_{k,t-1})$	-0.013*** (0.005)	-0.023*** (0.005)	-0.005 (0.003)
K	722	754	706
T	9	8	8
Obs	5849	5118	4925
AR(1) p-value	5.8345e-12	7.4015e-12	2.417e-11
AR(2) p-value	0.56622	0.36736	0.22858
Sargan/Hansen p-value	0.87292	0.094342	0.95647
Instrument count	16	15	15

* p < 0.1, ** p < 0.05, *** p < 0.01

window. Our first explanatory variable is the exposition coefficient. We find a positive, statistically significant (at the 1% level) association for the 2000–2008 window. No significant association on the 2009–2016 window. And a smaller, but significant (at the 5% level) positive association for the 2017–2024 window. This result should be interpreted cautiously as it is not robust to using the Pierce and Schott (2009) correspondence map instead of the one generated by our NLP pipeline.

The second explanatory variable is the profit rate for which we find a consistently positive and significant (at the 1% level across all windows) coefficient. Here what must be noted is that across our three windows, the coefficient on profitability is at its lowest magnitude in the last window, suggesting that the relationship between investment and profitability at the firm level is weakening during the last period relative to the other two. It should be remarked that the weakening of the profitability-investment rate association is not accompanied by a large fall in the investment rate (see Subsection 6.3.1). So firms are still investing, but current and lagged profitability is a less good predictor of whether they are still doing it or not. Lastly, for all three windows, we find a negative significant effect (at the 1% threshold) of an increased leverage ratio on investment, albeit with decreasing strength.

For two periods out of three (2000–2008 and 2017–2024) we find that an increased exposition is associated with an increased investment rate. For the 2000–2008 window, the claim holds with low statistical significance (it holds at the 10% level); for the 2017–2024 window it does with higher significance (at the 5% level) but at a lower magnitude.

Regarding controls, we find no significant relation between the investment rate and the relative cash and cash equivalent variable. Conversely, capital intensity is significantly (at the 1% level) negatively associated with investment for the 2000–2008 and 2009–2016 periods. We find no such significant association for the 2017–2024 period.

Across our three time windows, our results are coherent with investment ruled by a growth-safety trade-off in the sense that responsiveness to changes in profitability (measured through the regression coefficients on the profit rate) and in financial solidity (measured through the regression coefficients on the leverage ratio) have the expected sign and are significant. So it cannot be claimed that a significant number of U.S. firms have

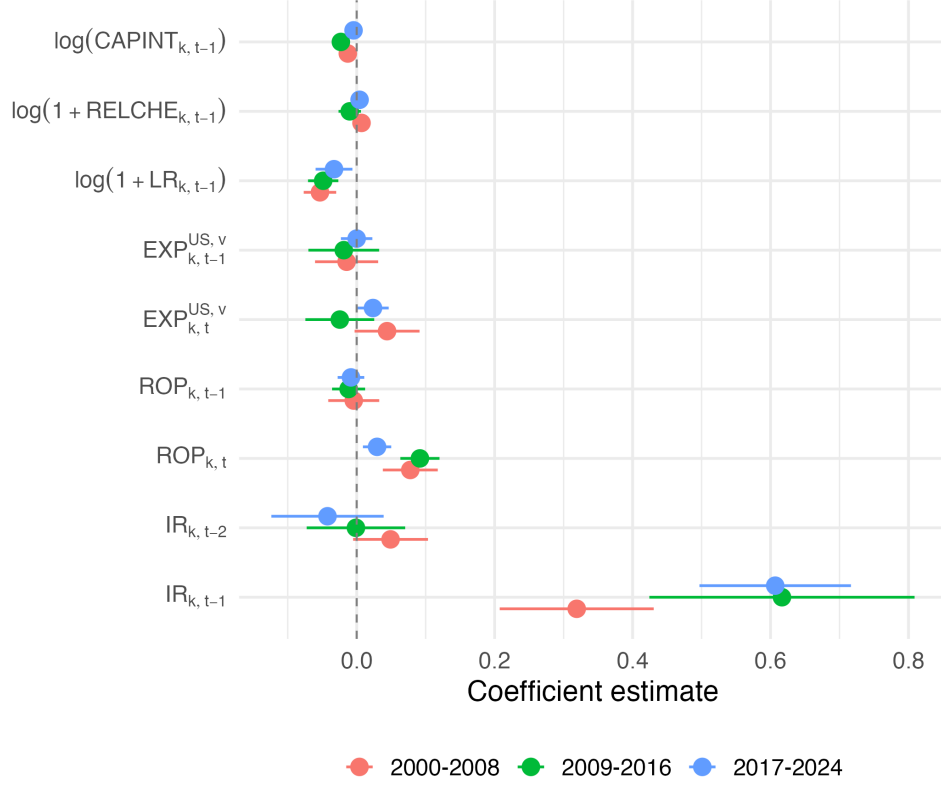


Figure 6.9: Investment function regression coefficient-plot at a 95% confidence interval for the three windows.

moved away from the corespective competition regime in the second or the last window.

But we do robustly find that the association between profitability and the investment rate has been decreasing sharply from the 2000–2016 period to the 2017–2024 window. This is clearly seen by taking a look at the coefficient plot (Figure 6.9) but can also be rigorously tested. Let $\hat{\beta}_{r,1}^{(w1)}$ and $\hat{\beta}_{r,1}^{(w2)}$ denote the estimated coefficients $\beta_{r,1}$ estimated on two windows ($w1$) and ($w2$) Computing the following Wald statistic:

$$W = \frac{\left(\hat{\beta}_{r,1}^{(w1)} - \hat{\beta}_{r,1}^{(w2)}\right)^2}{\hat{\text{Var}}\left(\hat{\beta}_{r,1}^{(w1)}\right) + \hat{\text{Var}}\left(\hat{\beta}_{r,1}^{(w2)}\right)} \sim \chi^2, \quad (6.7)$$

enables us to test if a coefficient is significantly different across windows. Running such a Wald test shows rigorously that the profitability coefficient is significantly different (lower) between the windows 2009–2016 and 2017–2024 (we reject the null with p-value 0.035), while it is not between 2000–2008 and 2009–2016 (we fail to reject with p-value 0.576). So, by this definition of a ‘structural break’ in the investment function, there

is a structural break between our last two windows. This is consistent with a general weakening of the growth-safety trade-off, or with the departure of a subset of the firms in our sample from this investment pattern. This result does not depend on our choice of product-to-firm map.

Furthermore, we find, with low confidence (this particular result is not robust to using the Pierce and Schott (2009)-based indicator), that at the same time as the profitability-investment rate association weakens, a significant positive effect between exposition to imports from China and investment arises. Again, this is evidence which is consistent with the hypothesis that firms, facing a more competitive environment driven by foreign competition, are coerced into investing more.

6.4 Synthesis of the regression results

Finding	Hyp.	DS1	DS2	Verdict
Exposition is positively associated with increased profits in the first window (2000-2008)	H1	yes, 1%	yes, 1%	Robust
Exposition is negatively associated with increased profits last window (2017-2024)	H1	yes, 1%	no (insig.)	Sensitive
Profits are less persistent in the last window (2017-2024)	H2	yes	yes	Robust
Investment association with profit is weaker in the last window (2017-2024) than the first (2000-2008)	H3	yes 5%	yes 5%	Robust
Investment is positively associated with exposition in the first window (2000-2008)	//	yes 10%	no (insig.)	Sensitive
Investment is positively associated with exposition in the last window (2017-2024)	H4	yes 5%	no (insig.)	Sensitive

Table 6.5: Synthetic table of the results, under both datasets generated from both correspondance maps. The NLP generated corresponds to DS1, and the Pierce and Schott (2009) based map corresponds to DS2. “Hyp” refers to the corresponding hypothesis, as stated in Chapter 5. Verdict synthesize how robust our results are to dataset specification (robust if they hold regardless to our choice of firm-to-product map, sensitive if the result is sensitive to that choice).

The empirical analysis, which we have carried out in this section, provides evidence consistent with the narrative of a transformation in the competitive environment facing large, publicly traded U.S. firms. In order to clarify our conclusions, we synthesise our

findings in Table 6.5.

Four main findings stand out. First, the association between exposition to Chinese imports and firm profitability has reversed sign: what was a positive, statistically significant relationship in 2000–2008 turned non-significant in 2009–2016 and became negative and statistically significant in 2017–2024. This suggests that the cost advantages that *lead firms* once derived from China-centric supply chains have eroded, and sectors dominated by Chinese imports now face mild profit compression. This result is sensitive to our specification of the dataset; if we choose the Pierce and Schott (2009)-based dataset, we only find that exposition has become insignificant. If we use our NLP-based dataset, we find it has reversed. Second, the autoregressive dynamics of the profit rate shifted from very noisy dynamics to a slowly decaying AR(1) process in the early windows to a fast mean-reverting AR(2) process with an oscillatory component in the last window. This last transition is indicative of a marked intensification of competition. This result does not depend on the correspondence map and is thus robust (in that narrow sense). Third, while investment behaviour still conforms to a growth-safety trade-off pattern, the sensitivity of investment to profitability has declined significantly in the 2017–2024 period. This is a result which holds regardless of our choice of dataset.

Lastly, we find, with low confidence, that firms which are facing higher exposition to Chinese imports were more likely to raise their investment rates in the first and the last window. But this finding does not hold when switching to the Pierce and Schott (2009) dataset.

Taken together, these results paint the following picture. Our first hypothesis (H1) holds, at least in the weak sense: exposition to Chinese firms does not support profitability anymore. It may be that it has actually reversed: exposition now hurts the profitability of U.S. firms – which was what Milberg and Johnston (2025) found – but this finding is not robust to using the Pierce and Schott (2009) dataset; it only holds for our NLP-based correspondence map. The second hypothesis (H2) clearly holds in the sense that profits are less persistent in the last window than in the post-GFC window, suggesting competition is becoming tendentially more intense for large U.S. corporations. The third hypothesis, of a breakdown of the growth-safety trade-off, is not verified in the strong sense, but it does appear that there has been a structural break in the investment function

(profitability is significantly more weakly associated with the rate of investment). Finally, our fourth hypothesis, that of *coerced* investment in the last window is verified in our NLP-based dataset but not in the Pierce and Schott (2009)-based dataset.

Together these results paint the following picture. We can confidently say that for the firms in our panel, the *profitable trade deficit* model appears to be exhausted, at least with respect to China. Firms more exposed to Chinese imports are not more profitable anymore. The picture regarding the evolution of the regime of competition is less clear but suggests a change in the competitive regime, likely attributable to Chinese competition, although results which support this thesis are sensitive to our choice of dataset. This is without a doubt a question which warrants further investigation.

Chapter 7

Conclusions

This thesis set out from the puzzle of the contemporary international political economy. The second Trump administration has broken, loudly and on several fronts at once, with the policy package that the United States had promoted for four decades: across-the-board tariffs, bilateral deals monetising access to the U.S. market and even perhaps the adoption of something akin to an industrial policy with price floors and the government taking equity stakes in strategic corporations. This is what Martin Wolf (2026b) has called ‘*the dangerous triumph of neo-mercantilism*’. Commentators have since been debating how to understand this apparent break with the previous doctrine of the U.S., and whether it was intelligible at all. The argument developed here is that if it is to become intelligible, it may only be so once one asks the question that the existing discussion – mostly preoccupied with the dollar’s hegemonic position and the tariffs – has largely left aside: *what is happening to the profitability of U.S.-American transnational corporations?* It is after all for their benefit that the U.S.-led globalised world market was constructed, if we are to believe Panitch and Gindin (2013) and Burbaumer (2024)’s accounts.

Answering that question required carefully disentangling the relationship between trade imbalances, firm profitability, offshoring and international competition. The first step was to critique the mainstream theory of trade, showing its limitations and how they can be overcome. We argued that the persistence of large imbalances is an embarrassment only for a theory that needs trade to balance: once the ‘self-righting’ mechanism of comparative cost theory is abandoned, and with it the exogenous explanations devised

to patch it up, imbalances appear for what they are. The expected outcome of the arithmetic of competition waged across borders is the same as within: the strongest competitors put the weakest out of business. On this view, a persistent deficit is financed by the foreign acquisition of debt, equity and other property titles, and cumulates into a negative international investment position; it is an expected outcome of competitive unevenness rather than an anomaly. The second step was to show that this arithmetic says nothing, by itself, about profitability. A deficit may record a leakage of demand away from domestic firms, a cheapening of their imported intermediates or of firms themselves relying on foreign production to increase their own profits. We may refer to a trade deficit as a *profitable* one under three conditions – that offshored production undercuts domestic costs, that offshoring does not affect domestic demand to the point that it reduces the profitability of domestic firms and that foreign rivals’ capacity to contest the *lead firms’* markets is limited. Furthermore, what raises the individual firm’s profit rate may still depress profits in the aggregate, as inter-capitalist competition is, of course, uncoordinated. The ambiguous welcome that U.S. business gave to China as it integrated into the world market (Hung 2022), first celebrated and then denounced, was forged by that very ambiguity. The third step of our analysis was to dissect the determinants of the third condition: what process determines whether *lead firms* can sustain their profits amidst a trade deficit or not. Here identifying what kinds of firms are most relevant to the issue of U.S./China competition is key. We argued that *core industries*, where *sunk costs* and *fundamental uncertainty* discipline incumbents into corespective restraint and deter entrants from contesting the market, make up the bulk of the *high-value-added sectors* which China is entering into. The defining feature of these industries is not their value added as such, but the kind of investment required to compete in them: large scale, high barriers to entry and substantial sunk costs. In these sectors, when the competitive regime is *corespective*, markups are relatively high and investment obeys a growth-safety trade-off. When entrants break the tacit contract limiting the intensity of competition, it risks turning *coercive*: an invest-or-die game of price warfare and constrained investment then starts (Crotty 1993). Foreign competition, the historical record of the 1970s suggested, is one well-trodden road from the first regime to the second.

The claim of this thesis – announced in Chapter 1 and developed in Chapter 4, 5 and 6 – is that China’s catching up is walking that road. What Autor and Hanson (2025) have called

the ‘*China Shock 2.0*’ differs from the first in kind rather than in degree. While the first shock displaced U.S. labour in low-wage manufacturing, without threatening the position of U.S.-American *lead firms*, for whom it was mostly beneficial, as the Chinese competition was kept away from their core competence, the second is pitting U.S.-American *lead firms* against Chinese *national champions*, as well as against each other. The irony is therefore that China’s ascent now threatens the sustainability of the very model which it had previously enabled. Carried by the successful developmental policy of the Chinese state, Chinese firms have upgraded technologically, to the point that they are now able to enter into markets which had up until now exclusively been held by U.S.-American, European or advanced Asian corporations.

What is more, these *national champions* operate in conditions which favour sustained intense *coercive competition* – the condition which the Chinese leadership itself now diagnoses as *involution* – threatening to subject U.S.-American firms to the kind of competitive pressure which *corespective competition* normally avoids. This is the result of the fast accumulation of capital in China, of the Chinese state’s industrial policy which provides firms engaged in *coercive competition* access to the credit lines they need to continue the struggle. By lowering the cost of borrowing, that policy also effectively lowers the entry barriers into core sectors, destabilising *corespective* restraint. It can also be thought of as the contamination of the upper echelons of GVCs by the kind of competitive regimes that the leading firms had previously incentivised in the lower echelons. Because international competition is governed by absolute rather than comparative costs, the erosion of the barrier to entry into *core industries* for Chinese firms transmits the pressure quasi-directly to U.S.-American markups and profit rates. Even worse, these firms’ profitability cannot easily be supported by, for instance, erecting trade barriers: this is because the largest U.S.-American firms are most often also transnationals, which depend on low trade barriers to properly operate.

The empirical investigation of Chapter 6 gives this narrative measurable content. Mapping product-level Chinese import penetration onto a panel of large listed U.S. firms over 2000–2024, it finds four things. First, the association between exposure to Chinese imports and profitability has weakened and changed meaning: positive in 2000–2008, when offshoring to China still cut costs; insignificant in 2009–2016, when the two con-

figurations coexisted; and, in the robust weak form, no longer supportive of profitability in 2017–2024. Our NLP-based map suggests a negative and significant association in that last window, but this stronger reversal is sensitive to the concordance-based map; the robust claim is that exposure does not support profitability anymore. This finding of a mild compression of profits for firms in China-dominated sectors is consistent with the results of Milberg and Johnston (2025). Second, the dynamics of profit rates have shifted from a slowly decaying process, characteristic of low-intensity competition in the post-GFC period, towards fast, oscillatory mean-reversion, possibly indicating a more competitive environment. Third, investment still follows the pattern of a growth-safety trade-off, but its sensitivity to profitability has fallen significantly in the most recent time-window (2017–2024). We show a structural break in the investment function between the 2009–2016 and the 2017–2024 time windows. Fourth, and with low confidence, more exposed firms appear to invest *more*, as a process of coerced investment would suggest. In terms of the hypotheses stated in Chapter 5: H.1 holds in its weak form, since the profitable trade deficit model appears exhausted with respect to China; H.2 holds robustly, since profit rates have become less persistent; H.3 holds only in its weak form, since the growth-safety trade-off is weakened and structurally broken rather than abandoned; and H.4 is suggestive and dataset-sensitive. Taken together, these results do not signal a completed regime shift; they are more consistent with the hypothesis of a regime under strain, shifting partially and unevenly across firms and sectors.

It is against this background that the doctrinal shift of U.S. foreign economic policy should be read. If neoliberal globalisation was, as Burbaumer (2024) argues, a solution negotiated across classes, to the benefit of the transnational fraction of capital, as an answer to the profitability troubles of the 1970s, then the present turn towards tariffs, profit-sharing deals and industrial policy is best read symmetrically. Not as an aberration, the psychology of one man, but rather an attempt – perhaps unsuccessful – to renegotiate the terms of participation in a world market that is becoming less favourable to U.S.-American transnational capital. The continuity between the Biden and Trump administrations on China and on industrial intervention, and the pre-Trump pedigree of the rebalancing agenda, support reading the shift as the political surface of a deeper, material transformation rather than as its cause. It should be read as institutional *bricolage* rather than as an optimal restoration plan. What the new administration’s predatory

policy agenda – not only in economic, but also in military terms – signals, on this reading, is not strength but perhaps weakness.

The limits of the present work outline a path towards its continuation. Four points bear emphasis. First, the pricing channel – the most direct expression of coercive competition – has only been discussed from aggregate producer-price evidence rather than measured at the firm level. Securing access to the right dataset to perform a firm-level analysis of markup and price dynamics is a natural next step, which would add clarity to the exercise.

Second, the headline results are only partly robust to the alternative, concordance-based measure of exposure. The NLP pipeline used to construct the firm-product map is nevertheless part of the contribution of this thesis: it shows that publicly available SEC filings and low-cost open-source models can be used to build granular firm-level exposure indicators for empirical works. Clarifying how robust they are would require reproducing said results using perhaps a paid product-level database (for instance the *Panjiva* or *ImportGenius* database) as a ground truth against our NLP indicators; sadly, we did not have access to such a database at the time of the project. Furthermore, our novel indicator itself rests on simplifying choices – a time-invariant product map, equal weighting of products, trade data that count foreign-owned production in China as Chinese exports – which future work should attempt to relax.

Third, our dataset fails to reflect specific aspects of firm strategy and operation which we did take time to consider in our theoretical discussion. The two important aspects in that regard are, first, indicators of a firm’s transnationality – whether it operates across borders, its dependence on subcontractors or on foreign direct investment – and second, indicators of a firm’s investment into very high technology, its dependence on the sale of intangibles and its investment in R&D. It was not performed due to lack of time and resources, but is a natural extension of this work.

Fourth, we are only scratching the surface of the question which prompted this study, that of relating the economic and foreign policy of the United States to the situation of its businesses. It is far beyond the scope of this thesis to discuss the policy-making process and how it relates to the needs and interests of certain capital fractions in the

U.S., but it is a largely open and quite major question to be tackled. One which is only made more burning by what has been established here: that the profitability conditions of U.S.-American lead firms have deteriorated in the direction that the framework predicts.

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Appendix A

Mathematical Notation Reference

Accounting and balance-of-payments aggregates			
CAB	Current account balance	AFAB	Financial-account balance
S	Private savings	I	Private investment
T	Government revenue	G	Government spending
X	Aggregate exports	M	Aggregate imports
C	Aggregate consumption	A	Aggregate output
FYI	Primary income from abroad	FYO	Primary income paid abroad
SIB	Secondary income balance	OR	Stock of official reserves
KOF	Capital outflows	KIF	Capital inflows
NFK	Net international investment position		
Exchange rates, prices, and unit costs			
e	Nominal exchange rate	e_r	Real exchange rate
P	Price index	p	Unit-price index
v	Vertically integrated unit labour cost	τ	Relative price of tradables
a	Unit labour requirement	χ	intermediate input cost

ζ	Offshoring ratio		
Firm-level accounting, profitability			
Π	Profit level	π	Profit share
r	Rate of profit	K	Capital stock
δ	Depreciation rate	Z	Depreciation flow
κ	Potential output-capital ratio	u	Capacity-utilization rate
s	Sales	l	Required unit labour time
μ	Markup	w	Real hourly wage
f^*	Revenue from foreign investments	UNC	Unit normal cost
Firm investment and the growth-safety tradeoff			
α	Risk appetite	ϕ	Financial solidity
i	Interest rate	σ	Perceived uncertainty
Empirical panel and firm-level financial variables			
IP	Chinese import penetration	EXP	Firm exposure to Chinese competition
EBIT	Earnings before interest and taxes	AT	Total assets
DT	Total debt	PPENT	Property, plant, and equipment
CHE	Cash and cash equivalents	REVT	Total revenue
IR	Firm investment rate	$LR_{k,t}$	Leverage ratio
$CAPINT_{k,t}$	Firm capital intensity	$RELCHE_{k,t}$	Relative cash holdings
Indices, subscripts, and other conventions			
$\square_k$	Firm index	$\square_t$	Time index
$\square_c$	Consumption-basket subscript	$\square_r$	Real-variable subscript
$\square_0$	Baseline or no-offshoring subscript	$\square^*$	Foreign-country counterpart
$\square^+$	Positive effect	$\square^-$	Negative effect
$\square_{t-1}$	Lagged period	Δ	First difference operator

$\equiv$	Accounting identity	$\approx$	Approximate equality
$\propto$	Proportionality	$\in$	Set membership

Appendix B

Explaining Trade Imbalances

Figure 2.7 compares the effective and bilateral real exchange rates and the effective and bilateral real unit labour cost for the United States and China over the period 1995–2023 following Shaikh and Antonopoulos (2013)’s method. This short appendix section describes how the exchange rate and unit labour cost plots are constructed.

Bilateral real exchange rates (RER) are computed from the price level of GDP (`p1_gdpo`) in the Penn World Table 11 (Feenstra, Inklaar, and Timmer 2015) and from nominal exchange rates e_{it} (local currency per U.S. dollar) taken from the World Bank. Denoting the price level of country i at time t by P_{it} , the bilateral nominal exchange rate against partner j is $e_{ijt} = e_{it}/e_{jt}$. The bilateral real exchange rate is

$$\text{RER}_{ijt} = \frac{P_{it}}{e_{ijt} P_{jt}} = \frac{P_{it}}{\frac{e_{it}}{e_{jt}} P_{jt}}. \quad (\text{B.1})$$

Bilateral relative unit labour costs (RULC) use the labour share (`labsh`) from the same PWT release. Unit labour cost is proxied by the inverse of the labour share, $\text{ULC}_{it} = 1/\text{labsh}_{it}$, yielding

$$\text{RULC}_{ijt} = \frac{\text{ULC}_{it}}{e_{ijt} \text{ULC}_{jt}} = \frac{\text{labsh}_{it}^{-1}}{\frac{e_{it}}{e_{jt}} \text{labsh}_{jt}^{-1}}. \quad (\text{B.2})$$

The real effective exchange rate (REER) and the real effective unit labour cost

(REULC) aggregate these bilateral measures over a common set of 26 trading partners: Australia (AUS), Argentina (ARG), Bangladesh (BGD), Brazil (BRA), Canada (CAN), China (CHN), Egypt (EGY), France (FRA), Germany (DEU), India (IND), Indonesia (IDN), Italy (ITA), Japan (JPN), South Korea (KOR), Malaysia (MYS), Mexico (MEX), Netherlands (NLD), Pakistan (PAK), Poland (POL), Russia (RUS), Saudi Arabia (SAU), Spain (ESP), Thailand (THA), Turkey (TUR), the United Kingdom (GBR) and the United States (USA). For any reference country i , the partner set $\mathcal{J}_i$ contains all countries in this list except i itself.

Trade weights are constructed from bilateral import and export flows reported in UN Comtrade (United Nations 2026). Let M_{ijt} and X_{ijt} denote imports and exports between i and j as reported by country i . The total bilateral trade volume is $V_{ijt} = M_{ijt} + X_{ijt}$, and the time-varying weight of partner j is

$$\alpha_{ijt} = \frac{V_{ijt}}{\sum_{k \in \mathcal{J}_i} V_{ikt}}, \quad \text{with} \quad \sum_{j \in \mathcal{J}_i} \alpha_{ijt} = 1. \quad (\text{B.3})$$

The effective rates are obtained as geometric weighted averages of the bilateral series:

$$\text{REER}_{it} = \prod_{j \in \mathcal{J}_i} (\text{RER}_{ijt})^{\alpha_{ijt}}, \quad (\text{B.4})$$

$$\text{REULC}_{it} = \prod_{j \in \mathcal{J}_i} (\text{RULC}_{ijt})^{\alpha_{ijt}}. \quad (\text{B.5})$$

Missing bilateral observations are linearly interpolated before the geometric aggregation is performed. The World aggregate (W00) is dropped from the partner list, and all effective series are computed for the sample period 1995–2023.

Appendix C

Trade Imbalances, Profitability and Offshoring

This appendix simply serves to make explicit the derivation of the simple intermediate input import model which we discuss in Subsection 3.3.3.

Recall we build our model starting with the following price equation:

$$p = (1 + \mu)(wa + p_m a_m). \quad (\text{C.1})$$

This applies to one unit of goods produced; we define total output as Q (this is the number of commodities produced). The second step is to derive the domestic profit-share in a Kaleckian fashion:

$\Pi = \mu(wa + p_m a_m) \cdot Q$	Profits
$W = wa \cdot Q$	Wages
$\zeta = (a_m)/(a)$	Offshoring ratio
$z = (p_m a_m)/(wa) = \zeta \cdot (p_m/w)$	Intermediates import ratio
$\pi = \frac{\Pi}{W + \Pi} = \frac{\mu(wa + p_m a_m)}{\mu(wa + p_m a_m) + wa}$ $= \frac{\mu wa(1 + z)}{\mu wa(1 + z) + wa} = \frac{1}{1 + (\mu(1 + z))^{-1}}.$	

(C.2)

Aggregate demand, under the assumption of balanced government budget, is:

$$Y = C + I + X - M + (G - T) \quad \text{aggregate demand,} \quad (\text{C.3})$$

$$A = C + I \quad \text{domestic demand,}$$

$$0 = (G - T) \quad \text{balanced gov. budg.,}$$

$$Y = A + X - M. \quad (\text{C.4})$$

We then assume that the autonomous demand component (for demand leakage) and imports and exports are given by linear response functions. This gives the following equation for aggregate demand:

$$A = A_0 - \zeta \cdot \beta_A, \quad \beta_A > 0 \quad \text{dom. demand function}$$

$$X = X_0 + \zeta \cdot \beta_X, \quad \beta_X > 0 \quad \text{export function}$$

$$M = M_0 + \zeta \cdot \beta_M, \quad \beta_M > 0 \quad \text{import function}$$

$$Y = \overbrace{(A_0 + X_0 + M_0)}^{=Y_0} + \zeta \cdot (\beta_X - \beta_M - \beta_A). \quad (\text{C.5})$$

From here, the conditions we are looking for are those under which an increase in the *offshoring ratio* leads to an increase in aggregate profits *and* in the trade deficit. Expressed in formal terms, this is:

$$\frac{\partial}{\partial \zeta} (X - M) < 0 \quad \text{condition that an increase in offshoring increases the deficit,} \quad (\text{C.6})$$

$$\frac{\partial \Pi}{\partial \zeta} = \frac{\partial}{\partial \zeta} (\pi \cdot Y) > 0 \quad \text{condition that an increase in offshoring increases profits.} \quad (\text{C.7})$$

The effect of an increase in offshoring on the deficit is trivial to check:

$$\frac{\partial}{\partial \zeta} (X - M) = \frac{\partial}{\partial \zeta} (X_0 + \zeta \cdot \beta_X - M_0 - \zeta \cdot \beta_M) = \beta_X - \beta_M. \quad (\text{C.8})$$

By the assumption $\beta_X > 0, \beta_M > 0$ we get that $\frac{\partial}{\partial \zeta} (X - M)$ is always negative (so more offshoring always increases the trade deficit).

For the effect on profitability the computations are more involved since aggregate profit is given by:

$$\Pi = Y \cdot \pi, \quad (\text{C.9})$$

and both Y and π depend on ζ . So let us first compute these two derivatives separately, starting with differentiating π . Recall that:

$$\pi = \frac{1}{1 + (\mu(1+z))^{-1}} = \frac{\mu(1+z)}{1 + \mu(1+z)},$$

and also note that $z = \zeta \cdot (p_m/w)$. We will need to use the chain rule, so let us differentiate both our composed functions separately:

$$\frac{\partial}{\partial z} \left(\frac{\mu(1+z)}{1 + \mu(1+z)} \right) = \frac{\mu(1 + \mu(1+z)) - \mu(1+z)\mu}{(1 + \mu(1+z))^2} \quad (\text{C.10})$$

$$= \frac{\mu}{(1 + \mu(1+z))^2}. \quad (\text{C.11})$$

Trivially, $\frac{\partial z}{\partial \zeta} = (p_m/w)$. We now have all the terms to compute the derivative:

$$\frac{\partial \pi}{\partial \zeta} = \frac{\mu}{(1 + \mu(1+z))^2} \cdot \frac{p_m}{w} = \frac{\mu p_m/w}{(1 + \mu(1+z))^2} = \frac{\mu p_m/w}{(1 + \mu(1 + \zeta \frac{p_m}{w}))^2}. \quad (\text{C.12})$$

This is positive for any choice of parameters. The other derivative we need if we are to compute $\frac{\partial \Pi}{\partial \zeta}$ is that of Y which is much easier to get. Recalling the equation for Y we find that:

$$\frac{\partial Y}{\partial \zeta} = \frac{\partial}{\partial \zeta} (Y_0 + \zeta \cdot (\beta_X - \beta_M - \beta_A)) = \beta_X - \beta_M - \beta_A. \quad (\text{C.13})$$

This is an expression with an ambiguous sign.

We are now ready to compute the aggregate profitability response as:

$$\frac{\partial \Pi}{\partial \zeta} = \frac{\partial \pi}{\partial \zeta} \cdot Y + \frac{\partial Y}{\partial \zeta} \cdot \pi = Y \cdot \frac{\mu p_m/w}{(1 + \mu(1 + \zeta \frac{p_m}{w}))^2} + \pi(\beta_X - \beta_M - \beta_A). \quad (\text{C.14})$$

This is simply the product rule (of differentiation). This defines the set of parameters for which *profitable trade deficit* manifests itself.

Appendix D

Econometrics and Data Appendix

D.1 Reproducing the results

To ensure reproducibility of our results, we have published the source code required to reproduce our econometrics on two different `git` repositories. The product-to-firm data-mining pipeline is available at:

<https://github.com/RenardDesNeiges/HSCode-to-firm>

and can be run with an OpenRouter API Key.

The code to recompute our regressions as well as for generating every plot in this document is available at

https://github.com/RenardDesNeiges/USChina_thesis_reproduction

It requires access to the Compustat dataset as well as generating a product-to-firm map using the other repo.

D.2 Results using the Pierce and Schott-based exposition indicator

Table D.1 gives the regression coefficients for our profitability regression using the Pierce and Schott (2009) based dataset. Table D.2 gives the regression coefficients for our investment regression using the Pierce and Schott (2009) based dataset. We also include Figure D.1 which summarises the relationship between both of our indicators (the NLP-based indicator and the Pierce and Schott (2009) based indicator).

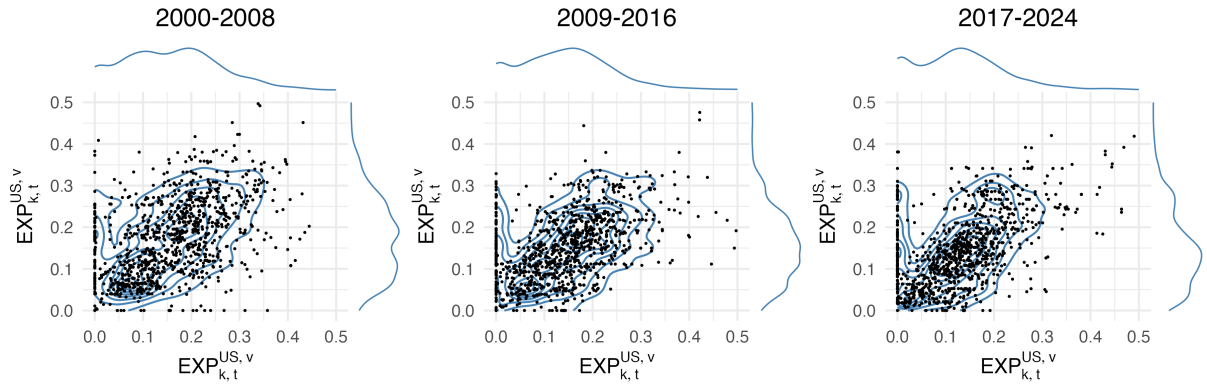


Figure D.1: Joint distribution plots of the NLP and the Pierce and Schott (2009) based exposition indicators. It is clear that they are strongly correlated. Notice that unlike the Pierce and Schott (2009) indicator, the NLP returns a significant number of firms with no exposition.

Table D.1: Profitability function: Difference-GMM across time windows.
Windmeijer-corrected robust standard errors (using the Pierce-hs-correspondance panel).

	2000- 2008	2009- 2016	2017- 2024
ROP _{<i>k,t-1</i>}	0.401* (0.242)	0.672*** (0.068)	0.549*** (0.069)
ROP _{<i>k,t-2</i>}	0.079 (0.101)	-0.001 (0.049)	-0.303*** (0.042)
EXP ^{US,v} _{<i>k,t</i>}	0.146*** (0.052)	-0.064 (0.080)	-0.032 (0.064)
EXP ^{US,v} _{<i>k,t-1</i>}	-0.072* (0.038)	0.032 (0.082)	-0.007 (0.049)
log(1 + LR _{<i>k,t-1</i>})	-0.051 (0.031)	-0.008 (0.045)	0.068 (0.045)
log(1 + RELCHE _{<i>k,t-1</i>})	-0.022 (0.040)	0.083*** (0.031)	0.029 (0.021)
log(CAPINT _{<i>k,t-1</i>})	0.049** (0.023)	0.020 (0.017)	0.034* (0.020)
K	531	560	522
T	9	8	8
Obs	4331	3882	3717
AR(1) p-value	0.034728	3.9725e-07	2.0629e-13
AR(2) p-value	0.7374	0.72628	0.34376
Sargan/Hansen p-value	0.086132	0.18944	0.093867
Instrument count	29	20	20

* p < 0.1, ** p < 0.05, *** p < 0.01

Table D.2: Investment function: Difference-GMM across time windows using the Pierce-
hs-correspondance panel.

Windmeijer-corrected robust standard errors.

	2000- 2008	2009- 2016	2017- 2024
$IR_{k,t-1}$	0.317*** (0.057)	0.618*** (0.106)	0.607*** (0.056)
$IR_{k,t-2}$	0.048* (0.028)	-0.002 (0.037)	-0.041 (0.042)
$EXP_{k,t}^{US,v}$	-0.018 (0.019)	-0.010 (0.033)	0.014 (0.012)
$EXP_{k,t-1}^{US,v}$	0.018 (0.023)	-0.019 (0.030)	0.007 (0.008)
$ROP_{k,t}$	0.079*** (0.020)	0.092*** (0.015)	0.030*** (0.010)
$ROP_{k,t-1}$	-0.005 (0.019)	-0.012 (0.012)	-0.008 (0.010)
$\log(1 + LR_{k,t-1})$	-0.053*** (0.012)	-0.048*** (0.011)	-0.033** (0.013)
$\log(1 + RELCHE_{k,t-1})$	0.007 (0.006)	-0.010 (0.009)	0.004 (0.005)
$\log(CAPINT_{k,t-1})$	-0.013*** (0.005)	-0.023*** (0.005)	-0.005 (0.003)
K	722	754	706
T	9	8	8
Obs	5849	5118	4925
AR(1) p-value	5.4875e-12	5.3486e-12	2.6051e-11
AR(2) p-value	0.59545	0.36976	0.2187
Sargan/Hansen p-value	0.85177	0.091654	0.96517
Instrument count	16	15	15

* p < 0.1, ** p < 0.05, *** p < 0.01